

TUNABLE NON-LINEAR MANIFOLD ROMs FOR EL-LIPTIC, PARABOLIC, AND HYPERBOLIC PDES VIA MATRIX-FREE PETROV–GALERKIN PROJECTION

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ABSTRACT

A reduced-order PDE solver should allow a practitioner to choose an accuracy and computational cost after training. We present a nonlinear manifold reduced-order model (NM-ROM) that provides this choice through nested correction directions added to a frozen decoder. Correction rank changes the trial space without retraining; solver tolerance and, where applicable, empirical quadrature control the cost of the reduced solve. The architecture combines a learned spatial bank with a small nonlinear coefficient map, exact boundary enforcement, and matrix-free weak residual projection. Linear operators are precomputed, while validated quadrature reduces the cost of nonlinear residual evaluation. Experiments on elliptic, parabolic and hyperbolic problems compare the resulting accuracy–cost tradeoff with neural operators, POD-based reduced models and named full-order solvers. On Burgers, a fixed-test-space correction ladder spans $2.44\times$ in same-grid error for $5.16\times$ in runtime; a separate scheduled ladder is tested across training seeds and a held-out cohort. Acceleration depends on the comparator and resolution: selected reduced solves are faster than iterative full-order methods, while neural operators and classical reduced models remain stronger in several cases. Linear-bank controls further identify when the nonlinear restriction adds cost without improving accuracy. These results establish deployment-time tunability and delimit its practical benefit, rather than a universal speed advantage.

1 INTRODUCTION

Neural operators for partial differential equations (PDEs) (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023; Li et al., 2024; Boullé and Townsend, 2024) provide fast amortised inference. At a fixed grid and inference procedure, a trained operator returns one accuracy and runtime point. A practitioner may instead need to spend more computation on a difficult query, or accept a less accurate solution when response time matters. Classical full-order methods (FOMs) expose numerical controls such as iteration tolerance, but their cost can grow substantially with mesh resolution. The question is whether a trained reduced model can provide a useful accuracy–cost tradeoff at deployment.

Projection-based reduced-order models address this question by solving the governing equations in a smaller space (Benner et al., 2015; Sirovich, 1987). A linear subspace is inexpensive to use, but can require many basis functions to represent moving fronts. Nonlinear-manifold ROMs learn a more compact representation (Lee and Carlberg, 2020; Kim et al., 2022); their online nonlinear solve can, however, offset the benefit of fewer unknowns. Representation accuracy and computational advantage therefore need to be measured separately.

We present an NM-ROM with a deployment-time accuracy–cost family from one trained decoder. The solution is represented by a learned spatial bank multiplied by a nonlinear map of latent variables. We augment this map with nested, precomputed correction directions. Activating more directions enlarges the trial space without changing the learned weights. The correction rank is the primary representation control; iteration limits and stopping tolerances govern the numerical solve, and empirical quadrature reduces residual-evaluation cost where its validation permits. All online coefficients are determined from the weak PDE residual, without access to the reference solution.

Our experiments ask whether this mechanism improves accuracy, what it costs, and when it is competitive with established solvers. We retain neural operators, POD models and the unrestricted learned bank as comparators. Full-order timings name both the algorithm and its settings: each linear-PDE speedup is measured against the displayed CG setting. The main section includes two- and three-dimensional experiments, with held-out final results distinguished from development comparisons.

Contributions.

1. **A tunable NM-ROM from one trained model.** Nested correction directions provide discrete deployment settings that change approximation capacity without retraining. We isolate correction rank at fixed residual test count and separately evaluate a prescribed test-count schedule across seeds and held-out cases.
2. **An architecture supporting the accuracy–cost tradeoff.** A learned spatial bank, a nonlinear head with a linear skip, and nested corrections support compact latent solves, boundary enforcement and node-local evaluation. Matched-dimension comparisons distinguish the benefit of the nonlinear representation from that of the bank itself.
3. **A practical weak-form implementation and paired evaluation.** Precomputed linear operators, matrix-free differentiation and validated empirical quadrature make the online solve measurable against neural operators and classical solvers. Paired accuracy and runtime measurements identify the PDEs, settings and resolutions where acceleration occurs.

2 RELATED WORK

Projection-based and nonlinear-manifold ROMs. Linear ROMs approximate solutions in a fixed subspace (Benner et al., 2015; Sirovich, 1987); moving fronts can require a large basis (Cohen and DeVore, 2015; Greif and Urban, 2019). Nonlinear-manifold ROMs combine learned representations with residual projection (Lee and Carlberg, 2020; Kim et al., 2022), including quadratic manifolds (Geelen et al., 2022; Barnett and Farhat, 2022), neural-field representations (Kim et al., 2024) and adaptive bases (Peherstorfer and Willcox, 2015; Carlberg, 2015). Our focus is a nested set of correction directions within one learned bank, selected after training. Least-squares Petrov–Galerkin projection (Carlberg et al., 2011) and preassembly of reduced operators (Ștefănescu and Sandu, 2014; Weder et al., 2024) are established components; the experiment isolates the effect of correction rank.

Neural operators and hyper-reduction. FNO (Li et al., 2021), DeepONet (Lu et al., 2021), U-Net (Ronneberger et al., 2015; Takamoto et al., 2022) and Transolver (Wu et al., 2024) learn solution maps. Their evaluation resolution or inference procedure can also change accuracy and cost; our mechanism instead changes the trial space used to solve the PDE. Empirical quadrature (Hernández et al., 2017; Yano and Patera, 2019) reduces nonlinear residual evaluation through non-negative weights. We retain that construction, testing its error on states reached by the solver and through independent re-draws. The main iterative linear-system comparator is conjugate gradient (Hestenes and Stiefel, 1952).

3 METHODOLOGY

We turn a learned manifold into a practical PDE solver through three components: a trial manifold whose decoder enforces Dirichlet conditions exactly and carries nested linear correction directions (§3.1); a least-squares Petrov–Galerkin projection of the discrete residual onto fixed weak tests (§3.2); and exact preassembly of every linear term with Empirical Quadrature for Burgers advection where validated (§3.3); §3.4 gives the architecture. Throughout, $u \in \mathbb{R}^n$ is the full-order state on a uniform grid of N intervals per axis, $A \in \mathbb{R}^{n \times n}$ the discrete negative Laplacian; R is the bank width, k the latent dimension, M the number of weak tests and m the number of quadrature nodes. The formulas below illustrate scalar Dirichlet problems; periodic vector fields and time-integrator details are specified per PDE in the appendix. Figure D.1 (Appendix D) shows the data flow; Appendix B gives the per-PDE derivations and exit codes.

3.1 TRIAL MANIFOLD WITH EXACT DIRICHLET ENFORCEMENT

We train a decoder $\mathcal{D} : \mathbb{R}^k \rightarrow \mathbb{R}^n$, $k \ll n$, with latent state z and no encoder in the deployed path, separable into a frozen spatial *bank* $G \in \mathbb{R}^{n \times R}$ times a small neural *head* $h_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^R$, $k \leq R \ll n$, the bank built once per mesh from a random-Fourier-feature coordinate network g_ϕ (Tancik et al., 2020),

$$G_{x,:} = \mu(x) g_\phi(x)^\top, \quad \mu(x) = 16 x_1(1 - x_1) x_2(1 - x_2). \quad (1)$$

To strictly enforce the homogeneous Dirichlet condition on the boundary Γ , we utilize the smooth vanishing factor μ , zero on Γ , folded into every column of the bank, so that $\partial u / \partial z = 0$ on Γ at every resolution (homogeneous data only).

Nested correction directions. The deployed model augments the head’s output with q fixed directions $C_q \in \mathbb{R}^{R \times q}$ and solves for the latent code and the correction coefficients together,

$$u(z, y) = G(h_\theta(z) + C_q y), \quad z \in \mathbb{R}^k, y \in \mathbb{R}^q, 0 \leq q \leq R. \quad (2)$$

The columns of C_q are chosen offline from training data; the Burgers construction uses principal components of the head’s coefficient errors. Per-PDE constructions are stated in Appendix B. The directions are ordered so that C_q is a prefix of $C_{q'}$ for $q < q'$; the ladder is nested and no rung needs retraining. At $q = 0$ the model is the frozen head; at $q = R$ the head is irrelevant and the representation is the full linear span of the bank. For linear PDEs its coefficients can be found by a reduced linear solve; for nonlinear PDEs the equations can remain nonlinear even in this full-bank representation; in between, q moves the reachable set from the head’s image towards the bank’s span, and the solved dimension from k to $k + q$. Two consequences follow: the bank’s projection error is a floor on the achievable error whatever the head or the solver does, so nonlinearity in h_θ buys a smaller *solved* dimension, not an escape from the span; and all x -dependence factors through G , which makes node sampling and exact preassembly available from one decoder.

3.2 PROJECTING THE PDE ONTO THE MANIFOLD

Each PDE we consider gives a discretised residual $r(u)$ that should vanish at the true solution. We substitute the trial manifold $u(z, y)$ into r and project onto a fixed, latent-independent test space. On the Dirichlet square, $P \in \mathbb{R}^{M \times n}$ collects low-frequency tensor-product sine vectors, eigenvectors of the five-point operator, $PA = \Lambda P$. The reduced problem solved online is

$$(z^*, y^*) = \arg \min_{z, y} \frac{1}{2} \|r_w(z, y)\|_2^2, \quad r_w(z, y) = \Lambda_\star^{-1} P r(u(z, y)) \in \mathbb{R}^M, \quad M > k + q, \quad (3)$$

with a diagonal row scaling $\Lambda_\star$ stated per PDE in Appendix B: a least-squares Petrov–Galerkin condition with an explicit test space (Carlberg et al., 2011; Lee and Carlberg, 2020), not tangent Galerkin, overdetermined rather than square, solved by a method appropriate to each PDE’s structure.

Elliptic (Poisson). The Poisson residual is $r(u) = Au - f$, and projection gives $r_w(z, y) = B_0(h_\theta(z) + C_q y) - b_0$, $B_0 = PG$ (Appendix B.1); the projector onto $\text{range}(B_0 C_q)$ does not depend on z , so y is eliminated exactly (Golub and Pereyra, 1973), the iteration stays k -dimensional at every q , and $q = R$ is a linear solve.

Parabolic (heat). The heat semi-discretisation $du/dt = -\kappa Au$ is advanced with Crank–Nicolson; the reduced step substitutes the manifold into the fully discrete equation *before* projecting and solves $z_{n+1} = \arg \min_z \|B_0 h_\theta(z) - D B_0 h_\theta(z_n)\|_2$ (Appendix B.2), a nonlinear least-squares problem, not a linear system; at $q = R$ the trajectory is the exact modal propagation of the bank coefficients.

Hyperbolic (Burgers). For $u_t + u(u_x + u_y) = \nu \Delta u$ with a sign-upwind stencil and backward Euler (Appendix B.3) the weak residual is nonlinear in the coefficients, so y is not eliminated in closed form; we damp the (z, y) blocks separately inside one Levenberg–Marquardt step (*block-damped* variable projection), which, in the recorded solver comparison, removes the joint LM configuration’s 6 budget exits (Table F.4).

Solver and exits. Each attempt solves the damped normal system $(H + \lambda \text{diag}(\text{diag } H)) \delta = -g$, $H = J^\top J$, $g = J^\top r_w$, accepting the step only if it strictly decreases the residual: damped Levenberg–Marquardt with a monotone test (Marquardt, 1963). Three exit families are recorded and never conflated: the scale-free stationarity measure $\eta(z, y) = \|J^\top r_w\|_2 / (\|J\|_F \|r_w\|_2) \leq \eta_{\text{tol}}$ (4), a residual threshold, and the stalls, reported as early-stopped, never as converged. No query uses the solution it predicts: the elliptic solve starts from the cached training code nearest the projected source, the Burgers query fits (z, y) to the supplied initial field (Appendix B.5).

3.3 HYPER-REDUCTION

The non-linear manifold reduces the DOF count from n to $k + q$, but a projected term such as $PN(Gc)$ still costs $O(n)$. For a fixed linear operator the tested residual is exactly precomputable, $P(Au - f) = B(h_\theta(z) + C_q y) - b$ with $B = \Lambda PG$ assembled offline by the corresponding discrete transforms, so linear terms are never approximated; the Burgers advection $PN(Gc)$ is the only term that resists preassembly, evaluated *densely*, $O(nR)$ per residual, or on an *empirical quadrature* rule of m nodes (Hernández et al., 2017; Yano and Patera, 2019), $O(mR)$. The rule is a non-negative *per-node* weight vector, *independent of the snapshot*, fitted by NNLS (Lawson and Hanson, 1974) to reproduce the projected advection term at n_{fit} stored codes with a hard cap of m nodes (Appendix B.4); changing m re-solves the fit, rules are not nested, and deployment selects among stored rules. *A rule is never accepted on its NNLS fit residual*; we score it by the held-out relative error ρ of the projected advection term (10) over states the solver actually reaches on trajectories disjoint from the fit and evaluation cases, against a primary bar $\rho_{\text{max}} \leq 0.116$ fixed before any certification job ran and a tight bar 0.06, and then re-draw the construction: a rule is *confirmed* only if every re-draw passes (§6.3).

3.4 MODEL ARCHITECTURE

Three properties of the problem motivate our architecture: the latent iteration is initialised cold, so the decoder Jacobian must retain a well-conditioned linear component, which motivates a *linear skip* in the head; hyper-reduction retains a sparse subset of mesh nodes, so the decoder must evaluate at any node in mesh-independent time, which motivates a *per-node bank*; and the family must reach from the head’s image to the bank’s whole span without retraining, which motivates the nested *correction directions*.

Linear skip, for cold-start convergence. The head is a two-layer SiLU MLP plus a linear skip, $h_\theta(z) = \varphi_\theta(z) + W^\top z$, so its Jacobian retains a latent-independent component along which the cold solve can descend; the elliptic solver checks the numerical rank of the projected Jacobian at every query. The skip is a design choice, not ablated here.

No encoder; per-node bank. No encoder is needed: the query fits (z, y) against the supplied input. The decoder restricted to a rule’s support is a cached block times the coefficients, and the network’s parameters do not depend on the mesh.

What is fixed, what is chosen, and how error is reported. Every operating point uses one frozen artefact per PDE; selected at run time are the rank q , the quadrature (dense or a stored rule) and the stopping tolerance and budget; M is held fixed along the headline ladder. The representation ablations distinguish three quantities: the *bank floor* (projection error onto range G), the *best-found* error (the smallest any point of the augmented manifold attains, from a multistart oracle) and the *solved* error the iteration returns.

4 IMPLEMENTATION

4.1 MATRIX-FREE PROJECTED OPERATORS VIA JAX

Every linear term is preassembled once per mesh as the $M \times R$ matrix B , so the online residual and Jacobian of a linear PDE are $B(h_\theta(z) + C_q y) - b$ and $B[Dh_\theta \ C_q]$, with Dh_θ from a forward-mode Jacobian-Vector Product (Bradbury et al., 2018); the sampled Burgers advection uses the cached

stencil block. The damped normal system is solved directly; no Krylov solve is applied to the projected operator.

4.2 TRAINING PROTOCOL

The bank and the head are trained in two stages, both as auto-decoders: the coordinate network g_ϕ is fitted to the training states through (1), then frozen, and the head is fitted with a code library Z on the bank-projected states; C_q is then constructed from training-only coefficient directions as specified for each PDE. Sizes, cohorts and offline cost are in Appendix C; every headline uses one checkpoint per PDE, named by hash in Table E.1.

5 EXPERIMENTAL SETUP AND BENCHMARK PROBLEMS

Problems and references. The completed two-dimensional study includes viscous Burgers, Poisson, heat, reflective waves and incompressible Navier–Stokes. Poisson is also tested on an L-shaped domain using a CG full-order comparator. Table F.1 gives meshes and cohorts; Table F.2 specifies the sampled families. Burgers provides the principal correction-rank study. Poisson, heat and waves test the benefit of nonlinear restriction for linear equations. Navier–Stokes and lower-viscosity Burgers test its limitations. The three-dimensional comparisons are included in the main results; source contracts and earlier development panels are retained in the appendix.

Accuracy. We report worst relative L^2 error over the stated cases and requested times, in percent. Poisson uses the steady reference norm; heat and wave displacement use the current reference norm. Burgers uses the initial-field norm and distinguishes all-times error, including input compression, from evolved-times error. Wave displacement alone does not establish velocity or energy accuracy. Same-grid error measures reduction error relative to the converged discrete solution. Error against a refined reference also includes discretisation error; the two are never interchanged. Development results and held-out tests are labelled separately. Settings for a held-out test are fixed before its cases are accessed.

Timing and speedup. Every accuracy–runtime pair comes from the same solver invocation. We use double precision, GPU burn-in, synchronised timing and medians of retained repetitions. GPU query time includes initialization, the solve or rollout, and requested device outputs. Complete-query time additionally includes host transfers and is labelled separately. Training and compilation are offline costs. Speedup is $S = T_{\text{FOM}}/T_{\text{method}}$, using measurements from the same allocation. Each table identifies the FOM algorithm, tolerance and error. Where a table selects the fastest tested passing FOM with no greater error than a method, that selection and the possibly differing denominators are explicit. A speedup is not an accuracy-equivalence claim. Nonconverged solves and failed accuracy targets remain visible; only settings satisfying the stated numerical checks enter a frontier.

Baselines. The main linear-PDE comparisons use CG, including the L-shaped domain. Burgers uses Newton–BiCGStab and Navier–Stokes uses the recorded CNAB2 time integrator; these nonlinear evolution problems are not relabelled as CG solves. We compare with FNO, U-Net, DeepONet and Transolver where matched measurements are available. Training budgets and transfer rules are recorded per model and need not be equal. POD is a classical reduced baseline. The linear ROM using the learned bank is an ablation that bypasses the nonlinear head; its gains are not NM-ROM speedups. Detailed solver diagnostics and historical alternative FOM comparisons remain in the appendix.

6 NUMERICAL EXPERIMENTS AND RESULTS

We first compare accuracy and runtime across problems, then examine settings from a single trained model and the controls that produce them. Errors are percentages and costs are median query times. Speedup $S = T_{\text{FOM}}/T_{\text{method}}$ is always against the named comparator measured in the same allocation. Tables identify final and development cohorts; a development result is not a final accuracy claim.

6.1 COMPARISON AGAINST NEURAL OPERATORS AND FULL-ORDER SOLVERS

Two-dimensional problems. Table 1 compares NM-ROM with CG on Poisson, heat and reflective waves. Increasing resolution can improve the speed ratio because the reduced computation grows more slowly than the full-grid iteration. Correction improves Poisson and wave displacement accuracy, but its runtime cost differs substantially between the two problems. The measured CG solutions remain more accurate than the corresponding NM-ROM settings. Wave displacement gains do not imply a passing velocity or energy error.

Table 1: Two-dimensional NM-ROM versus CG: collected development results. N is intervals per axis. Error is worst relative L^2 (%); time is median GPU query time (ms). Each row uses the fastest tested passing CG with no greater displayed error. Poisson and heat use relative tolerance 10^{-2} ; wave time steps and tolerances are listed in Appendix K. Heat uses an earlier audited checkpoint; wave NM-ROM fails its full-state target.

Problem	N	Method	Error (%)	GPU ms	CG error (%)	CG ms	S
Poisson	256	NM-ROM, $q = 0$	3.1567	6.7031	0.1527	22.2898	$3.33\times$
Poisson	256	NM-ROM, $q = 256$	0.9689	7.1213	0.1527	22.2898	$3.13\times$
Poisson	1024	NM-ROM, $q = 0$	3.1495	3.9406	0.0722	60.0170	$15.23\times$
Poisson	1024	NM-ROM, $q = 256$	0.9648	4.3188	0.0722	60.0170	$13.90\times$
Wave	256	NM-ROM, $q = 0$	6.1035	183.5155	0.4008	142.5111	$0.78\times$
Wave	256	NM-ROM, $q = 32$	2.8017	2520.1766	0.4008	142.5111	$0.06\times$
Wave	1024	NM-ROM, $q = 0$	6.1066	192.4624	5.1413	583.0259	$3.03\times$
Wave	1024	NM-ROM, $q = 32$	2.8094	2529.4450	0.4010	1043.3681	$0.41\times$
Heat	256	NM-ROM	4.5557	11.3414	0.9255	6.9389	$0.61\times$
Heat	1024	NM-ROM	4.5555	12.2062	0.7707	59.1780	$4.85\times$

On Burgers, Table 2 reports NM-ROM, POD and FNO against a common Newton–BiCGStab control. At this resolution the corrected model improves same-grid error, but does not beat the full-order solver in accuracy or runtime. On the separate matched operator cohort, U-Net and Transolver also outperform the ROM in accuracy (Table F.9); missing paired timings for that cohort are not filled from another run. The earlier 1024² paired study below also reports speed against a tight Newton control; relaxing that control substantially reduces the gain. These development solves permit stalls and do not establish stationarity. The newer high-resolution panels remain in Table L.1.

Table 2: Burgers2D development comparisons. Upper panel: 256². Worst evolved and all-times errors use the initial-field norm. Upper-panel speedups use the displayed Newton–BiCGStab control: nonlinear tolerance 10^{-3} , $\Delta t = 0.005$. Faster FOM settings remain in the full panel. The corrected EQ rule is single-draw; the uncorrected rule has repeated construction confirmation.

Method	Evolved error (%)	All-times error (%)	GPU ms	S
Newton–Krylov	0.0489	0.0489	31.788	$1.00\times$
NM-ROM $q = 0$	1.8891	2.5629	40.359	$0.79\times$
NM-ROM $q = 256$	0.5129	0.9053	746.020	$0.04\times$
POD-512	0.2184	0.6125	2790.828	$0.01\times$
FNO	7.4164	7.4164	7.183	$4.43\times$

Earlier 1024 ² development panel (separate allocation):			
Method	Error (%)	GPU ms	FOM/NM-ROM
NM-ROM	3.908	41.677	—
Tight Newton–BiCGStab	2.142	605.747	$14.53\times$
Relaxed Newton–BiCGStab	2.390	68.044	$1.63\times$

Lower panel: four cases, three repeats, job 3534502; initial-normalised worst error against a refined reference. Both FOMs use $\Delta t = 0.005$ and FFT preconditioning. Tight nonlinear/linear tolerances are $10^{-6}/10^{-8}$; relaxed are $10^{-2}/0.5$. Both methods meet the 5% accuracy target, with unequal measured errors; NM-ROM permits stalled exits. FOM/NM-ROM divides each FOM time by the NM-ROM time in this lower panel.

Three-dimensional problems. Tables 3 and 4 put the same comparisons in the main results. Each column block uses one allocation and one stated reference convention. The Poisson CG control uses the efficient solver without timed audit-history recording. Heat’s accepted error/runtime

panel is shown, with CG speedups left unreported until its paired CG experiment completes. All neural-operator variants include their stated reconstruction or projection cost.

Table 3: Three-dimensional linear PDEs, development: Poisson (left) and heat (right), 32 intervals per axis. Errors are worst same-grid relative L^2 percentages; heat uses current-normalised evolved error. Corrected ranks are $q = 96$ for both, with latent dimensions 16 and 32 respectively; POD uses rank 128. Poisson speedup is against CG at relative tolerance 10^{-2} . Heat CG is pending (—). DeepONet on Poisson uses the documented POD initialization.

Method	Poisson3D (development)			Heat3D (development)		
	Error (%)	GPU ms	S	Error (%)	GPU ms	S
NM-ROM, uncorrected	0.589	2.450	$1.06\times$	6.013	34.771	—
NM-ROM, corrected	0.161	2.632	$0.986\times$	1.379	114.343	—
POD	0.037	0.232	$11.2\times$	1.244	0.189	—
FNO	0.237	11.409	$0.227\times$	0.395	10.638	—
U-Net	0.193	4.099	$0.633\times$	0.318	4.373	—
DeepONet	1.807	2.599	$0.998\times$	21.769	3.889	—
Transolver	0.530	4.135	$0.627\times$	1.483	6.282	—
FOM	0.140	2.594	$1\times$	—	—	—

Table 4: Three-dimensional nonlinear PDEs: Burgers final (left, 33 nodes per axis) and Navier–Stokes development (right, 32 periodic points per axis). Errors are worst evolved same-grid relative L^2 percentages, normalised by the initial field. Corrected ranks are $q = 192$ and 256; POD ranks are 256 and 320. Burgers uses Newton–BiCGStab with $\Delta t = 0.01$, nonlinear tolerance 10^{-2} and inner tolerance 0.5; NS uses CNAB2 with $\Delta t = 0.01$. Every S uses that block’s displayed FOM.

Method	Burgers3D (final)			NS3D (development)		
	Error (%)	GPU ms	S	Error (%)	GPU ms	S
NM-ROM, uncorrected	16.263	219.161	$0.038\times$	13.535	1253.205	$0.0025\times$
NM-ROM, corrected	4.399	364.677	$0.0229\times$	12.534	4622.722	$0.000677\times$
POD	2.561	310.989	$0.0268\times$	23.732	30.460	$0.103\times$
FNO	2.378	6.072	$1.37\times$	0.470	8.903	$0.352\times$
U-Net	2.451	2.115	$3.94\times$	2.044	1.934	$1.62\times$
DeepONet	20.575	2.578	$3.23\times$	26.933	4.261	$0.735\times$
Transolver	2.017	3.489	$2.39\times$	2.691	3.361	$0.931\times$
FOM	2.312	8.337	$1\times$	0.628	3.130	$1\times$

Burgers3D is the frozen primary-seed result on reserved cases. Its correction ladder reduces error on every case for both evaluated initializations, but POD and several neural operators remain more accurate, and NM-ROM does not reach the measured cost–error frontier. The refined-reference checks fail the prospective physical-error budgets, so this result establishes same-grid accuracy only. The second initialization and all timing repetitions remain in Appendix M; in particular, Transolver’s seed variation is not hidden by reporting its better run. Navier–Stokes shows a smaller correction benefit and misses its accuracy target. Its displayed data are development measurements, pending final confirmation.

6.2 BEST CONFIGURATIONS PER PROBLEM AND RESOLUTION

The uncorrected and corrected rows use the same trained model within each problem. They represent different deployment settings, not separately trained fast and accurate networks. Architecture, correction ordering and numerical protocol are fixed before final testing. Table 5 isolates correction rank on Burgers2D with the residual test space held constant; increasing the rank spends more runtime to reduce error. A setting is called a passing fast setting only if it meets the declared numerical and accuracy requirements.

L-shaped Poisson provides a separate geometry comparison, using CG throughout Table 6. At the finer mesh, the NM-ROM is faster at larger error than CG; POD supplies another accuracy–cost choice. Complete-query timing here includes input and output transfer, so it is not mixed with the resident-device timings above.

Table 5: Burgers2D correction settings from one frozen model at 256^2 . The test count is fixed at $M = 1088$; residual evaluation is dense. Errors are worst evolved same-grid percentages on development cases, and times are paired median GPU query times. All rows meet the numerical stopping rule.

Correction rank q	Relative L^2 error (%)	GPU query time (ms)
0	1.2657	848.0
64	1.0593	1359.4
128	0.8711	1886.5
256	0.5194	4377.9

Table 6: L-shaped Poisson versus CG, development. Error is worst same-grid relative L^2 (%). Complete-query time includes host transfers. Each mesh uses its displayed CG tolerance and runtime as a common speedup denominator.

Mesh	Method	Error (%)	Complete ms	S
256^2	CG 0.01	0.637	11.644	$1.00\times$
256^2	NM-ROM $q = 0$	3.861	2.879	$4.04\times$
256^2	NM-ROM $q = 64$	2.131	3.028	$3.84\times$
256^2	POD-128	2.457	2.850	$4.09\times$
512^2	CG 0.01	0.385	29.127	$1.00\times$
512^2	NM-ROM $q = 0$	3.852	4.716	$6.18\times$
512^2	NM-ROM $q = 64$	2.123	4.801	$6.07\times$
512^2	POD-128	2.451	4.031	$7.23\times$

6.3 WHICH KNOB TO TURN

Correction rank changes representation capacity; tolerance and iteration limits control numerical work. EQ sample count controls residual-evaluation cost; Table 7 compares it with dense evaluation. The fixed-test-space Burgers2D ladder spans $2.44\times$ in error for $5.16\times$ in runtime. A smaller test space gives only $1.22\times$ error improvement, which is why test count is not silently varied with rank.

Table 7: Tabular comparison of dense and empirical-quadrature (EQ) residual evaluation on Burgers2D. Error is worst evolved relative L^2 (%); time is median GPU milliseconds from one allocation. The last column is dense time divided by EQ time. Dashes mean that no paired dense measurement was collected. Rules at $q = 0, 16, 32$ have repeated-construction confirmation; higher-rank constructions are marginal (Table F.24).

Correction rank q	Dense error (%)	Dense ms	EQ error (%)	EQ ms	Dense / EQ
0	1.8890	292.87	1.8891	59.07	$4.96\times$
16	—	—	1.4270	80.61	—
32	—	—	1.2493	97.70	—
64	1.0843	624.93	1.2275	120.63	$5.18\times$
128	0.8930	1190.49	0.8936	246.87	$4.82\times$
256	0.5194	3915.14	0.5389	722.21	$5.42\times$

The separate scheduled Burgers2D ladder was tested after freezing on a held-out cohort: all 4 checkpoints give monotone error reduction, with top-rank errors of 0.59–0.68%; 3 meet the full pre-registered criterion. The incumbent’s uncorrected solve reaches 10.1120% on a difficult held-out case, which remains in the reported maximum. Table F.30 retains every checkpoint.

Representation and numerical cost. Matched-dimension ablations identify a nonlinear representation benefit on Burgers, but not a universal runtime advantage (Tables F.17 and F.18). For linear PDEs, direct solution of the bank coefficients can be cheaper and more accurate; those are linear-ROM baseline results. EQ is distinct from correction rank: it approximates residual evaluation and is used only where validated. Repeated construction confirms the Burgers2D rules at $q = 0, 16, 32$; higher-rank rules retain their single-draw or marginal qualification. The current 3D results are dense and do not establish EQ acceleration. Tables 2, 5 and 7 separate method comparison, correction settings and quadrature cost; full construction diagnostics remain in the appendix.

Limitations. Some comparisons are developmental, cohorts are small, and operator training budgets differ. The fixed-test-space Burgers2D study has one checkpoint; its scheduled study and Burgers3D have separate seed checks. Nonlinear initialization can fail, and same-grid improvement does not remove discretisation error. Wave full-state accuracy, NS accuracy and Burgers3D physical refinement remain limitations. These results do not establish a universal speedup or convergence guarantee.

7 CONCLUSION AND FUTURE WORK

We presented a tunable NM-ROM whose nested corrections change approximation capacity after training. The learned bank and nonlinear head support a compact weak-form solve, while precomputation and validated quadrature control its evaluation cost. The experiments separate correction-rank accuracy gains from numerical stopping effects and measure runtime against explicitly named baselines. Burgers supplies the clearest tuning evidence; linear PDEs show when a direct reduced solve is preferable, and the remaining comparisons delimit the current method’s accuracy and cost. Future work concerns more reliable nonlinear initialization, resolved advection-dominated tests and extension beyond structured meshes.

REPRODUCIBILITY STATEMENT

Tables and numerical prose are generated from retained run records by the accompanying scripts. Source hashes, checkpoints, solver settings, cohorts, timing repetitions and audit results identify each comparison (Appendix E). The supplementary tables retain failed settings and distinguish development selection from held-out evaluation.

AI USE STATEMENT

Language-model assistants supported method and experiment development, implementation, result analysis, auditing, and manuscript drafting and editing. Numerical results were computed by the recorded solver runs; table generators reproduce the reported values from those records. The authors are responsible for verifying the methods, results and text.

REFERENCES

- J. Barnett and C. Farhat. Quadratic approximation manifold for mitigating the Kolmogorov barrier in nonlinear projection-based model order reduction. *Journal of Computational Physics*, 464:111348, 2022.
- P. Benner, S. Gugercin, and K. Willcox. A survey of projection-based model reduction methods for parametric dynamical systems. *SIAM Review*, 57(4):483–531, 2015.
- G. Berkooz, P. Holmes, and J. L. Lumley. The proper orthogonal decomposition in the analysis of turbulent flows. *Annual Review of Fluid Mechanics*, 25(1):539–575, 1993.
- S. Berrone, C. Canuto, M. Pintore, and N. Sukumar. Enforcing Dirichlet boundary conditions in physics-informed neural networks and variational physics-informed neural networks. *arXiv preprint arXiv:2210.14795*, 2022.
- N. Boullé and A. Townsend. A mathematical guide to operator learning. In *Handbook of Numerical Analysis*, volume 25, pages 83–125. Elsevier, 2024.
- J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs, 2018.
- R. Brecht, R. O. Popovych, A. Bihlo, and Y. Popovych. Improving physics-informed DeepONets with hard constraints. *arXiv preprint arXiv:2309.07899*, 2023.
- H. Cai, C. Gan, T. Wang, Z. Zhang, and S. Han. Once-for-All: Train one network and specialize it for efficient deployment. In *International Conference on Learning Representations*, 2020.

- S. Cao. Choose a Transformer: Fourier or Galerkin. In *Advances in Neural Information Processing Systems*, 2021.
- K. Carlberg, C. Bou-Mosleh, and C. Farhat. Efficient non-linear model reduction via a least-squares Petrov–Galerkin projection and compressive tensor approximations. *International Journal for Numerical Methods in Engineering*, 86(2):155–181, 2011.
- S. Chaturantabut and D. C. Sorensen. Nonlinear model reduction via discrete empirical interpolation. *SIAM Journal on Scientific Computing*, 32(5):2737–2764, 2010.
- W. Chen, J. Song, P. Ren, S. Subramanian, D. Morozov, and M. W. Mahoney. Data-efficient operator learning via unsupervised pretraining and in-context learning. In *Advances in Neural Information Processing Systems*, 2024.
- A. Cohen and R. DeVore. Approximation of high-dimensional parametric PDEs. *Acta Numerica*, 24:1–159, 2015.
- A. N. Diaz, Y. Choi, and M. Heinkenschloss. A fast and accurate domain-decomposition nonlinear manifold reduced order model. *Computer Methods in Applied Mechanics and Engineering*, 425:116943, 2024.
- M. S. Eshaghi, C. Anitescu, N. Valizadeh, Y. Wang, X. Zhuang, and T. Rabczuk. Nows: Neural operator warm starts for accelerating iterative solvers. *arXiv preprint arXiv:2511.02481*, 2025.
- C. Farhat, P. Avery, T. Chapman, and J. Cortial. Dimensional reduction of nonlinear finite element dynamic models with finite rotations and energy-based mesh sampling and weighting for computational efficiency. *International Journal for Numerical Methods in Engineering*, 98(9):625–662, 2014.
- S. Fresca and A. Manzoni. POD-DL-ROM: Enhancing deep learning-based reduced order models for nonlinear parametrized PDEs by proper orthogonal decomposition. *Computer Methods in Applied Mechanics and Engineering*, 388:114181, 2022.
- R. Geelen, S. Wright, and K. Willcox. Operator inference for non-intrusive model reduction with quadratic manifolds. *Computer Methods in Applied Mechanics and Engineering*, 403:115717, 2022.
- C. Greif and K. Urban. Decay of the Kolmogorov N -width for wave problems. *Applied Mathematics Letters*, 96:216–222, 2019.
- A. Griewank and A. Walther. *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*. SIAM, Philadelphia, 2nd edition, 2008.
- Z. Hao, Z. Wang, H. Su, C. Ying, Y. Dong, S. Liu, Z. Cheng, J. Song, and J. Zhu. GNOT: A general neural operator transformer for operator learning. In *International Conference on Machine Learning*, 2023.
- Z. Hao, C. Su, S. Liu, J. Berner, C. Ying, H. Su, A. Anandkumar, J. Song, and J. Zhu. DPOT: Auto-regressive denoising operator transformer for large-scale PDE pre-training. In *International Conference on Machine Learning*, 2024.
- M. Herde, B. Raonic, T. Rohner, R. Käppeli, R. Molinaro, E. de Bezenac, and S. Mishra. Poseidon: Efficient foundation models for PDEs. In *Advances in Neural Information Processing Systems*, 2024.
- J. A. Hernández, M. A. Caicedo, and A. Ferrer. Dimensional hyper-reduction of nonlinear finite element models via empirical cubature. *Computer Methods in Applied Mechanics and Engineering*, 313:687–722, 2017.
- M. R. Hestenes and E. Stiefel. Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards*, 49(6):409–436, 1952.
- J. S. Hesthaven, G. Rozza, and B. Stamm. *Certified Reduced Basis Methods for Parametrized Partial Differential Equations*. SpringerBriefs in Mathematics. Springer, 2016.

- M. Hirsch, F. Pichi, and J. S. Hesthaven. Neural empirical interpolation method for nonlinear model reduction. *arXiv preprint arXiv:2406.03562*, 2024.
- B. Holzsuh, Q. Liu, G. Kohl, and N. Thuerey. PDE-Transformer: Efficient and versatile transformers for physics simulations. In *International Conference on Machine Learning*, 2025.
- W. Hu, S. Liu, P. Qiao, Z. Sun, and Y. Dou. Transolver is a linear transformer: Revisiting physics-attention through the lens of linear attention. *arXiv preprint arXiv:2511.06294*, 2025.
- T. J. R. Hughes. *The Finite Element Method: Linear Static and Dynamic Finite Element Analysis*. Dover Publications, 2000.
- G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021.
- Y. Kim, Y. Choi, D. Widemann, and T. Zohdi. A fast and accurate physics-informed neural network reduced order model with shallow masked autoencoder. *Journal of Computational Physics*, 451:110841, 2022.
- M. Kim, T. Wen, K. Lee, and Y. Choi. Physics-informed reduced order model with conditional neural fields. *arXiv preprint arXiv:2412.05233*, 2024.
- D. Kochkov, J. A. Smith, A. Alieva, Q. Wang, M. P. Brenner, and S. Hoyer. Machine learning-accelerated computational fluid dynamics. *Proceedings of the National Academy of Sciences*, 118(21):e2101784118, 2021.
- A. Kopaničáková and G. E. Karniadakis. DeepOnet based preconditioning strategies for solving parametric linear systems of equations. *arXiv preprint arXiv:2401.02016*, 2024.
- N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- C. L. Lawson and R. J. Hanson. *Solving Least Squares Problems*. Prentice-Hall, Englewood Cliffs, NJ, 1974.
- K. Lee and K. T. Carlberg. Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *Journal of Computational Physics*, 404:108973, 2020.
- Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.
- Z. Li, D. Z. Huang, B. Liu, and A. Anandkumar. Fourier neural operator with learned deformations for PDEs on general geometries. *Journal of Machine Learning Research*, 24(388):1–26, 2023.
- Z. Li, K. Meidani, and A. B. Farimani. Transformer for partial differential equations’ operator learning. *Transactions on Machine Learning Research*, 2023.
- Z. Li, H. Zheng, N. Kovachki, D. Jin, H. Chen, B. Liu, K. Azizzadenesheli, and A. Anandkumar. Physics-informed neural operator for learning partial differential equations. *ACM/IMS Journal of Data Science*, 1(3):1–27, 2024.
- P. Lippe, B. Veeling, P. Perdikaris, R. E. Turner, and J. Brandstetter. PDE-Refiner: Achieving accurate long rollouts with neural PDE solvers. In *Advances in Neural Information Processing Systems*, 2023.
- L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.
- M. McCabe, B. Régald-Saint Blancard, L. H. Parker, R. Ohana, M. Cranmer, A. Bietti, M. Eickenberg, S. Golkar, G. Krawezik, F. Lanas, M. Pettee, T. Tesileanu, K. Cho, and S. Ho. Multiple physics pretraining for physical surrogate models. In *Advances in Neural Information Processing Systems*, 2024.

- M. A. Mirhoseini and M. J. Zahr. Accelerated solutions of convection-dominated partial differential equations using implicit feature tracking and empirical quadrature. *arXiv preprint arXiv:2305.15661*, 2023.
- F. Negri, A. Manzoni, and D. Amsallem. Efficient model reduction of parametrized systems by matrix discrete empirical interpolation. *Journal of Computational Physics*, 303:431–454, 2015.
- N. C. Nguyen. High-order empirical interpolation methods for real time solution of parametrized nonlinear PDEs. *arXiv preprint arXiv:2410.02100*, 2024.
- J. Nocedal and S. J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer, New York, 2nd edition, 2006.
- M. Ohlberger and S. Rave. Reduced basis methods: success, limitations and future challenges. In *Proceedings of the ALGORITHMY Conference*, 2016.
- A. Quarteroni, A. Manzoni, and F. Negri. *Reduced Basis Methods for Partial Differential Equations: An Introduction*. UNITEXT, Vol. 92. Springer, 2015.
- M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks. *Journal of Computational Physics*, 378:686–707, 2019.
- F. Romor, G. Stabile, and G. Rozza. Non-linear manifold reduced-order models with convolutional autoencoders and reduced over-collocation method. *Journal of Scientific Computing*, 94:74, 2023.
- Y. Saad. *Iterative Methods for Sparse Linear Systems*. SIAM, Philadelphia, 2nd edition, 2003.
- L. J. Shikhman. One operator to rule them all? On boundary-indexed operator families in neural PDE solvers. *arXiv preprint arXiv:2603.01406*, 2026.
- L. Sirovich. Turbulence and the dynamics of coherent structures, Parts I–III. *Quarterly of Applied Mathematics*, 45(3):561–590, 1987.
- N. Sukumar and A. Srivastava. Exact imposition of boundary conditions with distance functions in physics-informed deep neural networks. *Computer Methods in Applied Mechanics and Engineering*, 389:114333, 2022.
- A. Tran, A. Mathews, L. Xie, and C. S. Ong. Factorized Fourier neural operators. In *International Conference on Learning Representations*, 2023.
- K. Um, R. Brand, Y. Fei, P. Holl, and N. Thuerey. Solver-in-the-loop: Learning from differentiable physics to interact with iterative PDE-solvers. In *Advances in Neural Information Processing Systems*, 2020.
- S. Wen, A. Kumbhat, L. Lingsch, S. Mousavi, Y. Zhao, P. Chandrashekar, and S. Mishra. Geometry aware operator transformer as an efficient and accurate neural surrogate for PDEs on arbitrary domains. In *Advances in Neural Information Processing Systems*, 2025.
- T. Wang and C. Wang. Latent neural operator for solving forward and inverse PDE problems. In *Advances in Neural Information Processing Systems*, 2024.
- H. Wang, J. Li, A. Dwivedi, K. Hara, and T. Wu. BENO: Boundary-embedded neural operators for elliptic PDEs. In *International Conference on Learning Representations*, 2024.
- H. Wu, H. Luo, H. Wang, J. Wang, and M. Long. Transolver: A fast transformer solver for PDEs on general geometries. In *International Conference on Machine Learning*, 2024.
- M. Yano and A. T. Patera. An LP empirical quadrature procedure for reduced basis treatment of parametrized nonlinear PDEs. *Computer Methods in Applied Mechanics and Engineering*, 344:1104–1123, 2019.
- J. Yu, L. Yang, N. Xu, J. Yang, and T. Huang. Slimmable neural networks. In *International Conference on Learning Representations*, 2019.

- J. Barnett, C. Farhat, and Y. Maday. Neural-network-augmented projection-based model order reduction for mitigating the Kolmogorov barrier to reducibility. *Journal of Computational Physics*, 492:112420, 2023.
- K. Carlberg, C. Farhat, J. Cortial, and D. Amsallem. The GNAT method for nonlinear model reduction: Effective implementation and application to computational fluid dynamics and turbulent flows. *Journal of Computational Physics*, 242:623–647, 2013.
- K. Carlberg. Adaptive h -refinement for reduced-order models. *International Journal for Numerical Methods in Engineering*, 102(5):1192–1210, 2015.
- T. Chapman, P. Avery, P. Collins, and C. Farhat. Accelerated mesh sampling for the hyper reduction of nonlinear computational models. *International Journal for Numerical Methods in Engineering*, 109(12):1623–1654, 2017.
- G. H. Golub and V. Pereyra. The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate. *SIAM Journal on Numerical Analysis*, 10(2):413–432, 1973.
- D. W. Marquardt. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963.
- B. Peherstorfer and K. Willcox. Online adaptive model reduction for nonlinear systems via low-rank updates. *SIAM Journal on Scientific Computing*, 37(4):A2123–A2150, 2015.
- O. Ronneberger, P. Fischer, and T. Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015.
- R. Ștefănescu and A. Sandu. Efficient approximation of sparse Jacobians for time-implicit reduced order models. *arXiv preprint arXiv:1402.2018*, 2014.
- P. N. Swarztrauber. The methods of cyclic reduction, Fourier analysis and the FACR algorithm for the discrete solution of Poisson’s equation on a rectangle. *SIAM Review*, 19(3):490–501, 1977.
- M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 35, 2022.
- M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng. Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in Neural Information Processing Systems*, 33, 2020.
- J. Weder, P. Schwerdtner, and B. Peherstorfer. Online-efficient model reduction on quadratic manifolds with Neural Galerkin schemes. *arXiv preprint arXiv:2412.17695*, 2024.

A EXPERIMENTS AT A GLANCE

Table A.1 lists every experiment in the paper with the question it answers, the PDE and meshes, the number of cluster jobs and the lane, and where its full numbers are; every row’s jobs, GPUs and checkpoints are in Table E.1.

Table A.1: Experiments at a glance: one row per experiment, the question it answers, the PDE and meshes, the number of cluster jobs (distinct job ids in Table E.1) and the lane, and where the full numbers are.

experiment	question it answers	PDE, mesh	jobs (lane)	where the numbers are
Fixed- M correction ladder	With M held, does the rank q alone move the error, and by how much?	Burgers 256^2	5 (b-qxm)	Tables 5, F.5
Rank $\times$ test count	Which of q and M carries the error; where does M saturate?	Burgers 256^2	same (b-qxm)	Tables F.5, F.16
Three-seed retraining	Is the ladder a property of one checkpoint?	Burgers 256^2	3 (b-seeds)	Table F.29
Sealed cohort	Does the ladder hold on cases opened once, after every choice was frozen?	Burgers 256^2	1 (b-seeds)	Tables F.30, 5
Same-job panels	Is anything reduced on the frontier against POD-LSPG, an FNO, a Newton grid and the direct solve in one job?	Burgers 256^2 – 1024^2	3 (b-panel)	Tables L.1, F.13, F.15, F.14
Reference-error column	Does the knob move the physical error or only the same-grid error?	Burgers vs 4096^2	same (b-panel)	Tables 5, F.10, F.12, F.11
Mesh ladder	How do the cached solve and the complete query scale with mesh at a frozen checkpoint?	Burgers, Poisson 64^2 – 1024^2	2 (mesh-ladder)	Table F.26
Speed at parity	What does the fused implementation cost at bit-level agreement?	Burgers	3 (b-speed)	Table F.34
Solver knobs	Which solver knob moves accuracy and which moves cost?	Burgers	1 (tuning)	Table F.19
Quadrature certification, re-draw	Does the NNLS fit residual predict held-out error; do rules survive re-draws?	Burgers 256^2	3 (b-eqtop)	Tables F.21, F.25, F.22; Fig. F.1
Two rule sets, transfers	Do the two rule sets agree in one allocation; do rules transfer to 512^2 ?	Burgers 512^2	256^2 , same (b-panel)	Tables F.24, F.12
Neural operators on shared data	FNO ($\times 3$), U-Net, Transolver, one-variable controls, and the operators’ own knob	Burgers Poisson	256^2 ; 8 (no-second)	Tables F.9, F.33, F.32, F.31; App. G.1
Head ablation, matched dimension	Is the head better than the best linear, quadratic or POD map in the same bank?	Burgers Poisson 256^2 – 1024^2	2 (head-ablation)	Tables F.17, F.18, F.28; App. G.3
Three-layer decomposition	How much of the deployed error is the bank, the head, and the solver?	every cell	same	Table F.6
Linear PDEs	Poisson ladder, heat linear bank, waves: does the family collapse to a linear model?	Poisson, waves 1024^2	heat, $3+1+3$ (p-linear, heat, w-ladder)	Tables F.7, F.8, F.27; App. G.5
Navier–Stokes	Phase-2 gate in four settings; head-only data scaling; exploratory ladder	2D decaying NS 256^2	7 (ns2d)	Tables F.36, G.1, G.2, G.3
L-shaped Poisson	Where no fast transform applies, is a reduced solve cheaper than the cheapest full-order solve?	L-shape 512^2	64^2 –5 (lshape)	Tables 6, F.37, F.39, F.38
Low-viscosity Burgers	Does the manifold’s edge over linear reduction grow where the Kolmogorov width is worst? (under-resolved mesh)	Burgers 256^2	3 (b-lowvisc)	Tables G.6, G.7; App. G.7
Training study	Data density, objective, latent size and smoothness penalty of the head	Burgers 256^2	2 (b-head-train)	Table F.35; App. C
SMA-NM-ROM cold start	Not run under this protocol; a stated limitation	—	0	§6.3

B METHOD DETAILS

B.1 ELLIPTIC INSTANCE: POISSON

For $Au = f$ we take $\Lambda_\star = \Lambda$, so the weak residual is

$$r_w(z, y) = \Lambda^{-1}P(Au - f) = B_0(h_\theta(z) + C_q y) - b_0, \quad B_0 = PG, \quad b_0 = \Lambda^{-1}Pf. \quad (5)$$

Because the retained tests are orthonormal and $Au^\star = f$, this residual is exactly the tested error $P(u - u^\star)$, so the minimised objective is the squared discrete L^2 error of the reduced state in the retained modes; this holds only for the modes P retains.

Analytically eliminated corrections. Let $B_0 C_q = Q\mathcal{R}$ be a thin QR factorisation. Q does not depend on z , so the inner minimisation over y has the closed form $\mathcal{R}y = Q^\top(b_0 - B_0 h_\theta(z))$ and the

minimised value is

$$\min_y \|B_0(h_\theta(z) + C_q y) - b_0\| = \|B_\perp h_\theta(z) - b_\perp\|, \quad B_\perp = (I - QQ^\top)B_0, \quad b_\perp = (I - QQ^\top)b_0. \quad (6)$$

The LM iteration runs on z alone with the deflated operator, and y is recovered by one triangular solve. Because the projector is constant the elimination introduces no approximation and $B_\perp Dh_\theta$ is the exact Jacobian of the eliminated residual; the only dropped derivative anywhere in the elliptic solve is the head curvature. At $q = R$ the deflated operator is zero, the latent problem is degenerate by construction, and the model is the linear least-squares solve $\min_y \|B_0 y - b_0\|$; the timed top rung is that QR solve, not an LM iteration on a zero Jacobian. We report the reduced stationarity of the deflated problem, the augmented stationarity of the full problem, the backward error of the triangular recovery and the numerical rank of $B_0 C_q$.

B.2 PARABOLIC INSTANCE: HEAT

The semi-discrete heat equation $\dot{u} = -\kappa A u$ is advanced with Crank–Nicolson, $(I + \frac{\Delta t \kappa}{2} A)u^{n+1} = (I - \frac{\Delta t \kappa}{2} A)u^n$, and the reduced step substitutes the manifold into the fully discrete equation *before* projecting. With $u^n = u(z_n)$, projecting with P and row-scaling by $\Lambda_\star = I + \frac{\Delta t \kappa}{2} \Lambda$ gives

$$r_{w,n}(z) = B_0 h_\theta(z) - D B_0 h_\theta(z_n), \quad D = \text{diag}\left(\frac{1 - \frac{\Delta t \kappa}{2} \lambda_i}{1 + \frac{\Delta t \kappa}{2} \lambda_i}\right), \quad (7)$$

and $z_{n+1} = \arg \min_z \|r_{w,n}(z)\|_2$, warm-started at z_n . Two points are structural. The right-hand side is built from the *decoded current state* at every step: carrying the quantity the previous step already made small freezes the recursion and reproduces a one-step solution to round-off. And z_{n+1} enters through h_θ , so the step is a nonlinear least-squares problem, not a linear solve. At $q = R$ the step is linear in the coefficients and the whole trajectory is the exact modal propagation of the bank coefficients, with no head and no iteration; this is the “top rung” of the heat and wave ladders. The deployed path factors the damped normal matrix by Cholesky; a variant assembles the Gram matrix $S = B_0^\top B_0$ offline and forms $H = Dh_\theta^\top S Dh_\theta$ directly, which is exact and is reported as an algebraic ablation of the same step.

B.3 NONLINEAR ADVECTION: BURGERS

The full-order problem is $u_t + u(u_x + u_y) = \nu \Delta u$ with the non-conservative sign-upwind stencil

$$N(u)_{ij} = u_{ij} (\delta_x u + \delta_y u)_{ij}, \quad (\delta_x u)_{ij} = N \cdot \begin{cases} u_{ij} - u_{i-1,j}, & u_{ij} > 0, \\ u_{i+1,j} - u_{ij}, & u_{ij} \leq 0, \end{cases} \quad (8)$$

δ_y analogous and switching on the same centre value, ghost zeros on all four walls, and backward Euler in time, $r_n(u) = u - u^n + \Delta t (N(u) - \nu \Delta_h u)$. Projecting and row scaling by $\Lambda_\star = I + \Delta t \nu \Lambda$ gives, with $c = h_\theta(z) + C_q y$,

$$r_{w,n} = (I + \Delta t \nu \Lambda)^{-1} [B_0 c - B_0 c_n + \Delta t (\hat{N}(c) + \nu \Lambda B_0 c)], \quad (9)$$

where the diffusion term is exact and the row scaling is the diagonal of the linearised implicit operator in the modal basis, the modal form of the Helmholtz preconditioner the full-order Newton solver uses. Only $\hat{N}(c) \approx PN(Gc)$ requires a choice: dense evaluation at every interior node, the sampled rule $\hat{N}(c) = \sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(Gc)$ with the sign switch retained at the retained nodes, or the precomputed quadratic form $\frac{1}{2} c^\top Q c$ with Q the symmetrised tensor $T_{iab} = \sum_x P_{ix} G_{xa} (DG)_{xb}$, which is an identity only where $u > 0$ at every stencil node of every trial state the solver visits. The last is an algebraic route we verified in-job (residual and Jacobian parity below 10^{-11} relative on sign-satisfying states) and do not time in this paper. Time stepping uses a fixed substep count per output interval; each step is warm-started at the previous code, and the linear extrapolation is used instead when, and only when, its residual norm is smaller.

B.4 EMPIRICAL QUADRATURE: THE FIT AND THE COUNTS

A fitted rule with support $\mathcal{S}$ and weights w is scored by the held-out relative error of the projected advection term,

$$\rho(u) = \frac{\|\sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(u) - PN(u)\|_2}{\|PN(u)\|_2}, \quad \rho_{\max} \leq 0.116 \text{ (primary bar)}, \quad \rho_{\max} \leq 0.06 \text{ (tight)}, \quad (10)$$

over the held-out reachable states described below; the primary bar is the held-out ρ of the incumbent $q=0$ rule at the state carrying the first-interval penalty, measured in the q-diag cell and adopted as the primary bar in the q-ridge design before any certification job ran; the tight bar was declared before b-eqtop ran. A candidate node set $\mathcal{C}$ is drawn once with a fixed seed. For each of n_{fit} stored codes $c^{(\ell)}$ we evaluate the decoded state, form the per-node contributions $N_i(Gc^{(\ell)})$ and the exact full-grid target $PN(Gc^{(\ell)})$, and solve

$$\min_{w \geq 0} \|\mathcal{G} w - \beta\|_2, \quad \mathcal{G}_{(\ell,i),c} = P_{i,c} N_c(Gc^{(\ell)}), \quad \beta_{(\ell,i)} = [PN(Gc^{(\ell)})]_i, \quad (11)$$

rows normalised by their Euclidean norms, support grown greedily by the Lawson–Hanson criterion with an exact non-negative refit after each addition and a hard cap of m nodes, padded to the requested count if the support saturates. Three counts are separated: the requested node count m , the achieved support, and the number of fit states n_{fit} . In the certification cell the fit states are reachable states (states of the dense solver’s own converged per-step trajectory on fit trajectories), the held-out states are 512 reachable states per rung from certification trajectories disjoint from both the fit and the evaluation cases, and ρ of (10) is evaluated on those. A deployed rule is a stored triple: node indices, weights, and the cached $m \times 5 \times R$ bank block.

B.5 SOLVER CONSTANTS AND EXIT CODES

Damping starts at $\lambda_0 = 10^{-6}$ (Poisson, Burgers) or 10^{-4} (heat); $\lambda \leftarrow \max(\lambda/3, 10^{-12})$ on acceptance and $\min(10\lambda, \lambda_{\max})$ on rejection; the trust radius is taken from the spread of the training codes. Exit codes differ per cell and are reported per cell. Burgers: 4 stationarity, 1 residual, 2 tiny step, 3 rejected at $\lambda \geq 10^{14}$, 0 budget. Poisson: 6 stationarity, 2 relative residual, 1 stall, 3 damping limit, 5 non-finite initial value, 0 budget. Heat: 1 stationarity, 2 tiny step, 3 damping limit, 4 non-finite, 0 budget. The heat criterion divides by $\|J\|_F$ only, applied to a residual already normalised by the target norm; Poisson and Burgers use (4) directly. The completion rule that accepts an attained initial fit (§3.2) was pre-registered in the panel cell’s design before the job ran; the stricter flag is printed beside it in Table F.13.

B.6 WHAT THE METHOD IS, STATED ONCE

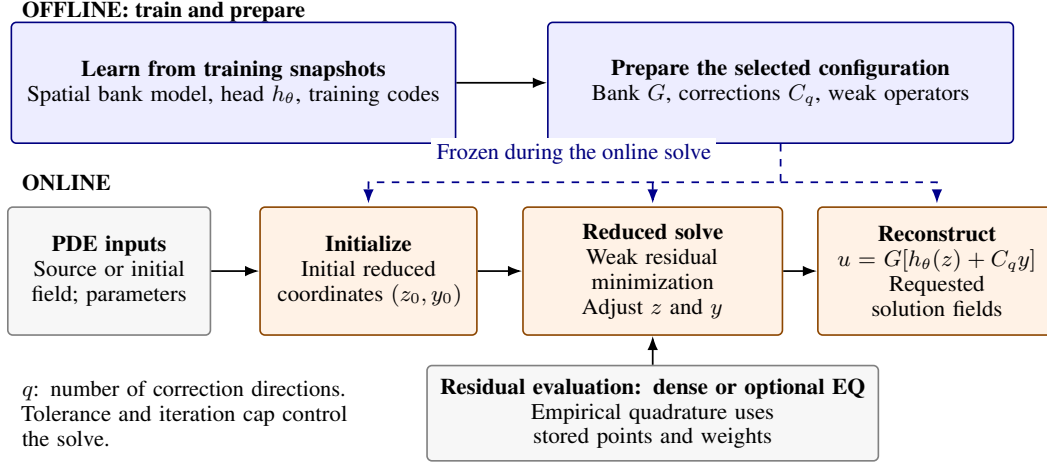
For a reader comparing this construction with related descriptions, the following hold for every result here. Dirichlet enforcement is a smooth polynomial factor folded into the bank, with zero lift, and covers homogeneous data only. The projection is least-squares Petrov–Galerkin with a fixed test space, not tangent Galerkin; the two differ even for a linear trial space, since $V^\top A^\top (AVz - b) = 0$ and $V^\top (AVz - b) = 0$ are different conditions. The heat scheme is Crank–Nicolson and the reduced step is a nonlinear least-squares problem, not a linear system in z_{n+1} ; no latent mass matrix is formed and no latent ODE is integrated. The quadrature fit targets the weak advection term with M rows per snapshot; the requested node count m , the achieved support and the fit-state count n_{fit} are three different numbers. No conjugate-gradient solve is applied to the projected operator. There is no encoder in the deployed path. The update is damped Levenberg–Marquardt, not undamped Gauss–Newton. The Poisson ground truth is the verified discrete solution, not an analytic field.

C PER-CELL TRAINING SCHEDULE

This appendix lists the per-cell configuration and training cost referenced in §4. The problem specification per cell (mesh, k , R , M , reference and cohorts) is Table F.1 and the sampled families Table F.2; the offline cost per stage, from the lane records, is Table F.3; the head-retraining arms on the frozen Burgers bank (training-set size, latent dimension and loss terms) are Table F.35. Every headline uses one checkpoint per PDE, named by hash in Table E.1.

D ARCHITECTURE CHOICES: EVIDENCE FROM SWEEPS

This appendix collects the evidence behind the architectural choices in §3.4: the matched-dimension head ablation on Burgers and Poisson (Tables F.17, F.18), the three error layers per cell (Table F.6), the head-capacity sweep on the frozen Poisson bank (Table F.28), the head-retraining arms (Table F.35), and the solver variants for the corrections on Burgers (Table F.4). The linear skip is a design choice adopted for cold-start convergence and is not ablated in this paper.



G : frozen spatial bank; h_θ : frozen neural head; C_q : fixed correction directions; z : nonlinear coordinates; y : correction coefficients. Solid arrows show online flow or an evaluation choice; dashed arrows show reuse of prepared components.

Figure D.1: NM-ROM from training to prediction. Blue components are prepared before the query and remain frozen; orange boxes compute the online solution. The inputs initialize reduced coordinates, which are adjusted to minimize the weak PDE residual before reconstructing the requested fields. Initialization and the reduced solver are PDE-specific; linear-PDE corrections can be eliminated analytically. Time-dependent problems repeat the reduced step, with reconstruction at requested output times. Empirical quadrature (EQ) is an optional residual evaluation, used in the Burgers2D EQ panels; the current 3D and wave panels do not use it. Correction rank changes the representation, whereas EQ changes residual evaluation. Baselines are evaluated independently and are not stages of this pipeline.

Figure D.1 draws the pipeline; Table D.1 lists its blocks, their sizes in the symbols of §3.1, and when each is fixed.

Table D.1: The blocks of Figure D.1. “Fixed when” distinguishes training, preparation and online work.

block	what it is	size	fixed when
coordinate net g_ϕ	random-Fourier-feature MLP, evaluated per node	R outputs per node	trained once
bank G	$\mu(x) g_\phi(x)^\top$ at the mesh nodes, (1)	$n \times R$	assembled once per mesh
head h_θ	two-layer SiLU MLP plus a linear skip	$k \rightarrow R$	trained once
code library Z	auto-decoder codes of the training states	k per state	trained once
tests P	the first M sine modes	$M \times n$	fixed per run
linear operator B	PAG	$M \times R$	assembled once per mesh
corrections C_q	PCA of the head residual on the training codes, nested in q	$R \times q$	once per checkpoint
quadrature rules	NNLS fit on reachable states, one rule per mesh and test space, validated by held-out ρ	m nodes	prepared offline
solve	damped LM on (z, y) (Burgers); y eliminated exactly (linear PDEs)	$k + q$ unknowns	inside the timed query
controls	rank q ; stored rule or dense; tolerance and budget	—	chosen at run time

Table E.1: Provenance of every result table: job id, GPU, source commit and checkpoint. The SHA256 of every machine-readable file the generator read is in `tables/provenance.json`. Full-order and reduced solver results require the GPU backend, float64 and highest matmul precision. The historical operator precision exception is disclosed in Appendix G.1.

table	lane	job id(s)	GPU	commit	checkpoint
T3, T5	b-panel (256 ² , bpn301)	3789570	NVIDIA A100 80GB PCIe	f3c5fbed8a21.../7a57f02e2bf4...	incumbent (gate checkpoint_unchanged; hash in T2, tuning row)
T3c, T5c	b-panel (512 ² , bpn401)	3805065	NVIDIA A100 80GB PCIe	f3c5fbed8a21.../7a57f02e2bf4...	incumbent (gate checkpoint_unchanged; hash in T2, tuning row)
T3b, T5b	b-panel (1024 ² , bpn203)	3789572	NVIDIA H200	f3c5fbed8a21.../7a57f02e2bf4...	incumbent (gate checkpoint_unchanged; hash in T2, tuning row)
T4	b-qxm	G1 = 3780175 (NVIDIA A100 80GB PCIe), G2 = 3780177 (NVIDIA A100-PCIE-40GB), S1 = 3780178 (NVIDIA A100-PCIE-40GB), E1 = 3783898 (NVIDIA A100-PCIE-40GB), E2 = 3783899 (NVIDIA A100 80GB PCIe) 3711424	per job	ed431edb498a76072bf42014	/ 18f0266ae6f04542...
T6a, T7	head-ablation (Burgers)	3711388	NVIDIA A100 80GB PCIe	2718bd320dce...	18f0266ae6f04542...
T6b, T7	head-ablation (Poisson)	3711736	NVIDIA A100 80GB PCIe	6759adcc9d85...	a128e7635c31...
T8	fixed-checkpoint tuning	3712269	NVIDIA A100 80GB PCIe	d2b93bd4f56d...	18f0266ae6f0...
T9	b-eqtop	bet101 = 3780164 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet201 = 3780165 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet301 = 3783811 (NVIDIA A100-PCIE-40GB, commit b2844c607290...) 3711388	see job list	see job list	18f0266ae6f04542...
T10	mesh-ladder (Burgers)	3711389	NVIDIA A100-PCIE-40GB	521cdced6f4d...	18f0266ae6f0...
T10	mesh-ladder (Poisson)	3711389	NVIDIA A100 80GB PCIe	521cdced6f4d...	a128e7635c31...
T21	p-linear + lshape (secondary: previous comparator, CG)	the p-linear and lshape jobs above	per job	per job	same checkpoints as T11b / T18c
T20, T20b	b-lowvisc (appendix; F4 under-resolution caveat)	panel 3817807; gate 3789639; training 3804337	NVIDIA A100-PCIE-40GB	gate 2cf5dc34; train 98aad39; 144f6b29	low-viscosity checkpoint hashed in the lane summary
T11e	ns2d phase 2 (K=16, K=32, family dimension 8, 2048 trajectories); no confirmatory ladder; exploratory ladder in T11g	3787319 (K=16, R=256), 3787320 (K=32, R=512), 3808498 (K=16, R=256, family dimension 8), 3808495 (K=16, R=256, 2048 trajectories); FOM 3780151	ns303 A100 (pax105); others per job	per job	checkpoints hashed in each result.json
T11f	ns2d ns301 (head-only data scaling on the frozen K=16 bank)	3808493	per job	per job	frozen K=16 bank; heads hashed in result.json
T11g, T11h	ns2d ns304 (exploratory after a failed phase-2 gate)	3808502	NVIDIA A100 80GB PCIe	31e0846f	ckpt_K32_R512 hashed in result.json
T11a	w-ladder	64 ² : job 3780447 (NVIDIA A100 80GB PCIe, commit 0bb3cc86); 256 ² : job 3783805 (NVIDIA A100-PCIE-40GB, commit 2655bb01); 1024 ² : job 3780450 (NVIDIA A100 80GB PCIe, commit 0bb3cc86) 3511417	per job	per job	frozen-math SHA asserted in job
T11d	heat linear bank (2026-09-10)	3511417	NVIDIA A100-PCIE-40GB	73fdaa88eb75...	expanded_seed790715 (frozen)
T11b, T11c	p-linear	256 ² : job 3780692 (NVIDIA A100-PCIE-40GB, commit b43a437d7360); 1024 ² : job 3783813 (NVIDIA H200, commit 3e411b5ac59d); head capacity job 3783883	per job	per job	R=512/K=32 checkpoint (pbh02 primary)
T11i	p-linear A8 + w-ladder A2 (literal/amended criteria)	same jobs as T11a and T11b	per job	pinned in tables/provenance.json	existing verdict records; no new solve
T14, T14c, T14d	no-second (5 of 8 jobs counted; two preamble deaths uncounted)	3702464, 3702709, 3710846, 3780138, 3780139, 3780625, 3783831, 3787189	A100 (per job)	per job	operator checkpoints hash-verified in job
T15	b-speed	3745655 (spd01), 3745656 (fine01), 3745913 (comp01)	A100 80GB PCIe	8fdffb08 / 94399dd6	18f0266ae6f04542...
T16	b-head-train	3745912 (training), 3749074 (evaluation)	A100-PCIE-40GB	0f0c56f7 / 2b9e7ce7	trained checkpoints hashed in archive
T18, T18c, T18d	lshape	training 3783786; solves 3784662, 3784663, 3789568; free rung 3784910	NVIDIA A100 80GB PCIe	1086ccefdcb5...	7 heads + bases Git-tracked
T12	b-seeds (development cohort)	seed1 = 3783776 (NVIDIA A100 80GB PCIe); seed2 = 3783777 (NVIDIA A100 80GB PCIe); seed3 = 3783778 (NVIDIA A100-PCIE-40GB) 3804465	per job	per job	three seed checkpoints hashed in summary
T13	b-seeds (sealed cohort)	3804465	NVIDIA A100-PCIE-40GB	be9415ab	four checkpoints (incumbent + three seeds) hashed in summary

Table E.2: Attempts that produced no reported number. A retracted attempt contributes no timed row, no gate and no oracle value; it is listed so that it is visibly retracted rather than absent.

lane	attempt	job	what it would have produced	why nothing is reported
b-panel	bpn201	3783817	1024 ² panel	config-parsing bug; no timed number produced
b-panel	bpn202	3787247	1024 ² panel	out of memory during autotuning inside an untimed diagnostic
lshape	lsh01	3780148	bank and head training	mis-scaled basis-orthonormality gate; rerun in full as lsh02
p-linear	plin1024	3780691	Poisson 1024 ² ladder	GPU out of memory in the untimed best-found oracle; rerun on an H200
w-ladder	wl256b	3780448	wave 256 ² ladder	retained-value gate failed on a tie-breaking difference; rerun as wl256c
no-second	pois01	3780224	Poisson U-Net screen	stager omitted a config directory; no training ran; rerun as pois02
ns2d	ns202	3783797	Navier-Stokes $K = 32$ head	pre- $\$A4$ attempt on a rank-capped bank; superseded by ns204

Table E.3: Attempts in flight when this version was prepared: none; every lane this version reads is closed.

lane	attempt	job	what it will add
—	—	—	none: every lane read by this version is closed

E PROVENANCE

F FULL TABLES

G EXTENDED RESULTS

G.1 NEURAL OPERATORS: CONTROLS AND THE RESOLUTION KNOB

Both pre-registered controls were run on the selected U-Net (Table F.32): its float64 twin lands at 1.6721 % and its second-seed twin at 1.5479 %, both below the ROM, so that comparison is not a seed or precision artefact — though the float64 margin is 0.1950 pp and, at equal wall clock, the float64 twins ran 0.25–0.42 $\times$ of their parents’ epochs, so precision is confounded with an epoch effect. The Transolver arm below the ROM has no twin and remains a single-seed float32 result, under the historical protocol’s explicit precision exception. These are achieved finite-budget errors, not bounds on attainable accuracy. For 7 of 8 of this lane’s arms and 4 of 4 FNO arms, the best validation checkpoint lies in the final five percent of training; this recency heuristic establishes neither convergence nor future improvement. Selection minimises mean case-maximum validation error, which need not minimise the worst case. The operators’ own knob, evaluation resolution, was tested under a pre-registered rule (Table F.31; speedups are within one model’s own curve in one job and carry no comparison to the ROM or the solver): `fno-large` is usable — one step, 256 $\rightarrow$ 128, buys the whole gain (2.12 $\times$ its own speed at 1.14 $\times$ its own error), below which the query is launch-bound (3.46, 3.44, 3.33 ms) while the error rises to 1.54 $\times$ and 2.17 $\times$ — while `tsol-refine`, `unet-refine` are not (the U-Net breaks at once, the Transolver is cost-flat). At rung 128 no family is within 32 $\times$ of the interpolation floor, so the off-resolution error is the network’s, not the grid’s.

G.2 QUADRATURE: THE FIT-STATE COUNT AND THE REPLICATION

Scored by (10) on reachable states, the top-rung blocker is the number of fit states, not the node count: 14- and 8-state rules at $q = 128$ and 256 fail at $m = 2048$ ($\rho_{\max} = 0.1908, 0.1678$) where 64-state rules pass in that draw (0.0669, 0.1074; Table F.21). Rebuilt with the cheapest rule passing the primary bar per rung and timed in one allocation, the EQ ladder runs 1.8891 $\rightarrow$ 0.5389 % at 59.1 $\rightarrow$ 722.2 ms (Table F.24; offline cost in Table F.3). In the replication (pool and fit-state subset, four draws at fixed q , m , fit-state count; Table F.25) the rules are confirmed at $q = 0, 16, 32$ and marginal at $q = 64(2/6), 128(4/5), 256(1/5)$ (draws passing / draws); at $q = 256$ the rule the ladder ran is the only one of 10f5 draws of its recipe to meet the bar (re-draws 0.130 – –0.242). “Every rung carries a certified rule” is withdrawn as a statement about the construction, and four earlier readings are withdrawn by the same data.

Table F.1: Problem specification. Cohort and reduced sizes are read from the run configurations where recorded. The Burgers sealed cohort has been opened and is reported in Table F.30; other rows describe their recorded development and validation cohorts.

PDE	equation, domain, boundary	meshes	time stepping	reduced sizes	reference	cohorts
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 (ladder 64–1024)	$\Delta t = 0.005$, backward Euler, sign- upwind	$K = 16$, $R = 512$	refined 4096^2 , $\Delta t = 0.00015625$	6 development cases; 32 held-out (tuning); sealed cohort opened once (job 3804465)
Poisson 2D	$-\Delta u = f$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 , 1024^2	none (elliptic)	$K = 16$, $R = 128$ (incumbent); $K = 32$, $R = 512$	exact discrete (DST); 2048 ² refinement	12 development sources
Heat 2D	$u_t = \kappa \Delta u$, $(0, 1)^2$	64^2 – 1024^2	Crank–Nicolson	$k = 8$, $R = 32$	exact modal	12 development cases (ear- lier cell, job 3511417)
Wave 2D (reflec- tive)	$u_{tt} = c^2 \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	64^2 , 256^2 , 1024^2	RK4 on the manifold; exact modal propaga- tion for the bank	$K = 32$, $R = 64$	direct DST	8 development cases
Poisson, L-shape	$-\Delta u = f$, $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	256^2 , 512^2	none	$K \in \{16, 32\}$, $R \in \{256, 512, 514\}$	sparse direct (Su- perLU)	3072 / 256 / 32 sources (train / selection / develop- ment)

Table F.2: Sampled problem families, transcribed from the generator sources named in the last column (code constants, not run outputs).

PDE	equation and boundary	sampled family (transcribed from the generator source)	source
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$u_0 = a \exp(- x - c ^2 / 2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim U(0.05, 0.20)$, $a \sim U(0.5, 2)$; $\nu \sim$ $\log U(0.01, 0.1)$; outputs $t \in \{0.05, \dots, 0.25\}$	experiments/ mr-burgers2d/ engines.py: params_draw
Poisson 2D	$-\Delta u = f$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$f = a \exp(- x - c ^2 / 2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim$ $\log U(0.02, 0.1)$, $a \sim U(0.5, 2)$	multistage- precision/ms_ parametric.py: sample_params
Poisson, L- shape	same source family on $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	as above; sources centred in the removed quadrant rejected	experiments/ lshape
Heat 2D	$u_t = \kappa \Delta u$ on $(0, 1)^2$, $\kappa = 0.02$ fixed	polynomial-boundary Gaussian family (sin- gle_bc_poly_gaussian_v1); Crank–Nicolson $\Delta t = 0.025$	heat linear-bank report, job 3511417
Wave 2D (re- flective)	$u_{tt} = c^2 \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	compact bump $\times$ Gaussian: half-widths $s_i \sim U(0.36, 0.42)$, centre $c_i \sim$ $U(s_i + 0.025, 1 - s_i - 0.025)$, amplitude $\sim U(0.7, 1.3)$, $\sigma_i \sim U(0.12, 0.16)$, advective velocity $v_i \sim$ $U(-0.5, 0.5)$ (zero every fourth case); speed $c \sim U(0.85, 1.15)$	experiments/ multiresolution- wave/audit_ dynamics.py: parameter_rows

G.3 THE HEAD AT MATCHED DIMENSION: METRIC DEPENDENCE AND THE THREE LAYERS

On Burgers, on the all-times metric no POD rank up to $k' = 128$ matches the head (10.1198 %), while on evolved times POD-128 reaches 1.9464 % against the $q = 0$ rung’s 1.8890 % (both on the development cohort; the rung’s sealed value is 10.1120 %, Table F.30). In three layers (Table F.6): floor 0.3918 %, best-found 2.5447 %, solved 2.5629 %. On Poisson a wider head moves the best-found/floor ratio only from 4.18 to 2.76 (Table F.28).

G.4 NAVIER–STOKES: THE HEAD-ONLY DATA-SCALING DIAGNOSIS

Job 3808493 retrained only the head on the frozen $K = 16$, $R = 256$ bank with 128, 256 and 512 training trajectories under three recipes and scored the held-out oracle at 256^2 (Table G.1). The pre-registered rule (ns2d/DESIGN.md §A9 (pre-registered data-scaling rule for the head-only diagnosis): least-squares log-log slope of the held-out oracle in the training-trajectory count; steeper than -0.25 = data-limited, shallower than -0.10 = head-limited, inside the band = ambiguous) returns “ambiguous” for every recipe; the decelerating slope and the inertness of regularisation lean towards a head-limited reading, which the lane does not upgrade beyond ambiguous. Caveat, pre-registered: the frozen bank was trained on all 512 trajectories, so the small- n arms are optimistic. The reported held-out/train ratios use different error normalisations and cannot quantify a generalisation gap or establish a capacity bottleneck.

Exploratory phase-3 ladder after the failed phase-2 gate. Every number in this paragraph and in Tables G.2 and G.3 is exploratory after a failed phase-2 gate: the $K = 32$, $R = 512$ manifold failed the held-out oracle bar (Table F.36), so the ladder answers a mechanism question only, not the pre-registered phase-3 question. Job 3808502 (NVIDIA A100 80GB PCIe) ran the correction

Table F.3: Offline cost per stage, from the lane records: head training on the retrained arms (the incumbent’s own training time was not recorded), the correction-direction fit, and the NNLS fit of each rule the primary EQ ladder ran. Every rule is refit when m changes; nothing is nested.

offline stage	seconds	amortised over	source
head training, like-for-like retrain of the incumbent recipe (4608 traj., 200k steps)	387	once per checkpoint	b-head-train job 3745912 (A100-PCIE-40GB)
head training, selected retrained arm	1459	once per checkpoint	b-head-train job 3745912 (A100-PCIE-40GB)
correction directions C_q (PCA of the head residual; flattened LM fit of the best-found codes)	1242	once per checkpoint, all q	cheap-corrections job 3734098
certified EQ rule, $q = 0$, $m = 1024$ (reachable, qrg304:reachable)	72	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 16$, $m = 1024$ (reachable, qrg304:reachable)	147	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 32$, $m = 1024$ (reachable, qrg304:reachable)	155	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 64$, $m = 1024$ (reachable, qrg304:reachable)	160	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 128$, $m = 2048$ (fs64, this_job:reachable)	2920	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 256$, $m = 2048$ (fs64, this_job:reachable)	2650	once per rung and rule	b-eqtop job 3780164

Table F.4: How the corrections are solved on Burgers, one job (job 3734098, NVIDIA A100 80GB PCIe): joint LM on (z, y) , block-damped, and plain variable projection at $q = 64$ and 128. Same error where all converge; the block-damped step is the one kept.

arm	M	worst all-times %	budget exits	all stationary	device ms
$q=64$, joint LM on (z, y)	320	2.1489	6	no	1119.7
$q=64$, block-damped	320	2.1489	0	yes	619.2
$q=64$, plain variable projection	320	2.1489	0	yes	9527.9
$q=128$, plain variable projection	256	1.8116	297	no	47566.9
$q=128$, block-damped	256	1.8116	0	yes	825.5

ladder $q \in \{0, 32, 64, 128, 256, 512\}$ at one fixed test count $M = 2176$ on 8 development cases with 3 timed repetitions, same-job POD-LSPG at every matched dimension $k' = K + q$, and a full-order Newton-tolerance ladder in the same job. The correction rank closes the manifold gap: the manifold layer (median over cases of the worst evolved time) falls from 16.4787 to 1.7434 %, the bank floor (1.7434 %), but reaches it only at $q = R = 512$, where the head no longer matters; the residual directions capture 26, 44, 68, 92, 100 % of the head-residual energy at $q = 32, 64, 128, 256, 512$. On the matched statistic (the same median-over-cases of the worst evolved time on both sides) the solve layer is 1.4–2.5 $\times$ the manifold layer, decreasing with q from 2.51 at $q = 0$ to 1.40 at $q = R$ (an earlier median over all 48 states, $t = 0$ included, gave 2.9–10.8 $\times$ and is superseded: it compared two different statistics). The loss accumulates along the trajectory: at $q = 0$ the per-time solved/manifold ratio runs 1.44, 1.66, 2.19, 2.89, 3.85 from $t = 0.2$ to $t = 1$. The ROM’s enstrophy over the reference at $t = 1$ is 1.043 (median) at $q = 0$, 0.990 at $q = 32$, a deficit of 0.95–0.97 at $q = 64$ –256 and 0.999 at $q = R$ (POD-32 control 1.026). Zero budget exits: worst evolved 68.7654 $\rightarrow$ 6.0140 % (11.4 $\times$ for 3.9 $\times$ in cost), monotone in the median (yes) but not in the worst (no; one inversion at the first rung). POD-LSPG at the same dimension is more accurate at every rung (yes) and cheaper at 5 of 6 rungs (all but $q = 512$, where both cost about the same); the non-dominated set on (ms, worst evolved) is fom_ntol0.0001, fom_ntol0.001, fom_ntol0.003, fom_ntol0.03, fom_ntol1e-06, fom_ntol1e-11, pod_k32 — 0 neural rungs, the falsification clause’s named outcome for this head. The loosest full-order setting that converges reaches 0.0041 % at 421.3 ms (converged reference 1177.7 ms).

G.5 THE LINEAR CELLS IN FULL

Poisson at 1024^2 ($R = 512$, $K = 32$; Table F.7): the middle rungs cost a flat 6.4–6.9 ms; the literal degenerate-curve criterion passes. At 256^2 its cost-span clause fails because the top rung is 3.17 $\times$ cheaper than the middle rungs. The literal falsification therefore fires there; the amended interpretation does not (Table G.4). Amendment A8 changed the deciding interpretation after observing the coarse-mesh results and before the fine-mesh job; that interpretation was not fixed before the campaign’s observations. The efficient endpoint uses direct QR, omitting redundant nonlinear iterations at full

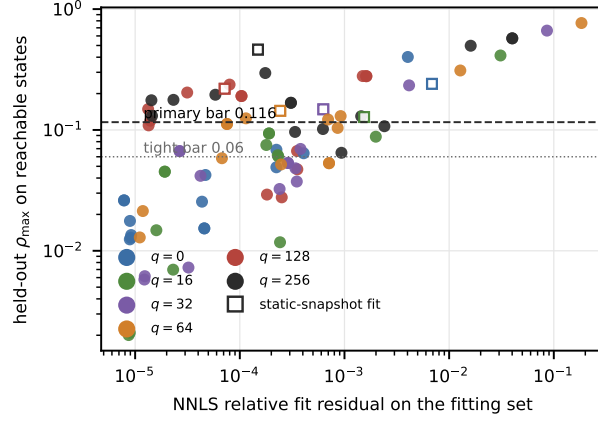


Figure F.1: Quadrature rules: fit residual against held-out error. Each point is one fitted rule; the horizontal axis is its NNLS fit residual on the states it was fitted to, the vertical axis its worst held-out ρ (10) on states the solver actually reaches, with the primary and tight bars drawn. Filled markers are rules fitted on reachable states, hollow squares rules fitted on static snapshots. Data: every rule of the b-eqtop lane, jobs 3780164, 3780165, 3783811 (Table F.22). What it shows: the fit residual does not predict the held-out error, and because the same construction re-drawn moves by $1.4\text{--}9.6\times$ (Table F.25), a point below the bar is a passing draw, not a certified construction.

Table F.5: Rank against test count (dense advection, budget 600, single seed). Both fixed- M ladders are shown; the $M = 256$ ladder fails the bar and the $M = 1088$ ladder passes it, and $M = 1088$ was chosen after the grid was run. The scheduled ladder’s cells span two jobs so its costs are not printed. Jobs G1 = 3780175 (NVIDIA A100 80GB PCIe), G2 = 3780177 (NVIDIA A100-PCI-E-40GB), S1 = 3780178 (NVIDIA A100-PCI-E-40GB), E1 = 3783898 (NVIDIA A100-PCI-E-40GB), E2 = 3783899 (NVIDIA A100 80GB PCIe); commit ed431edb498a / 76072bf42014; checkpoint 18f0266ae6f04542....

ladder	q	M	worst evolved %	device ms
fixed $M = 256$ (job 3780177)	0	256	1.2710	406.2
fixed $M = 256$ (job 3780177)	64	256	1.1255	645.6
fixed $M = 256$ (job 3780177)	128	256	1.0418	921.8
fixed $M = 1088$ (job 3780177)	0	1088	1.2657	848.0
fixed $M = 1088$ (job 3780177)	64	1088	1.0593	1359.4
fixed $M = 1088$ (job 3780177)	128	1088	0.8711	1886.5
fixed $M = 1088$ (job 3780177)	256	1088	0.5194	4377.9
scheduled $M = 4(K + q)$	0	64	1.8890	—
scheduled $M = 4(K + q)$	16	128	1.3985	—
scheduled $M = 4(K + q)$	32	192	1.2336	—
scheduled $M = 4(K + q)$	64	320	1.0843	—
scheduled $M = 4(K + q)$	128	576	0.8930	—
scheduled $M = 4(K + q)$	256	1088	0.5194	—

correction rank. The historical nonzero- q best-found columns remain retracted under amendment A10 because their projector was not orthonormal. A separate corrected full-cohort rerun is now independently confirmed in Appendix H. The historical solved ladder is unaffected; solver effort buys nothing at $q = 0$ (floor 0.7421, best-found 3.1139, solved 3.1495 % at $M = 129$). Heat at 1024^2 (an earlier cell of the same family, Table F.8): the bank evolved linearly reaches 1.68 % at 0.56 ms against the head’s 4.56 % at 12.3 ms, with the direct solve at 1.03 ms. The reflective wave (Table F.27): the top rung, the bank under exact modal propagation with no head, is not strictly the most accurate rung (strict reading: no) but matches the $q = 32$ rung within the amended integrator tie band at every mesh (256^2 : 0.020 pp, 5.121 against 5.104 %), and is $158\text{--}42\times$ cheaper than the cheapest head rung on device ms (on complete-query ms the dense input and output dominate both, Table F.27); the head ladder is monotone in q (yes), cost is not. On waves POD $k' = 64$ reaches 1.502 % against the bank’s 5.121 % at the same cost. Amendment A2 introduced the wave tie band after a local pilot showed a

Table F.6: Three layers of deployed error, worst over each cell’s development cases: what the bank can express, what the augmented manifold can reach (multistart oracle, an upper bound), what the deployed iteration returned. The p-linear $q > 0$ oracle was mis-scaled and is retracted; only its $q = 0$ oracle appears here. A separately sourced corrected rerun is reported in Table H.1.

cell	arm	bank floor %	best-found %	solved %	best/floor	solved – best (pp)	source
Burgers 256^2	neural head, $q=0$	0.3918	2.5447	2.5629	6.49	0.018	head-ablation
Poisson 1024^2 , $R=128$	neural head	2.3148	6.0926	6.0927	2.63	0.000	head-ablation
Poisson 256^2 , $R=512$	neural head $K=32$, $q=0$	0.7459	3.1211	3.1217	4.18	0.001	p-linear
Poisson 1024^2 , $R=512$	neural head $K=32$, $q=0$	0.7421	3.1139	3.1146	4.20	0.001	p-linear
wave 256^2	head, $q=0$	2.4571	8.0772	11.3314	3.29	3.254	w-ladder
wave 256^2	head, $q=32$	2.4571	2.4571	5.1038	1.00	2.647	w-ladder
L-shape 256^2	head_sdf_R512_K32	0.7766	3.1802	3.1814	4.09	0.001	lshape (untimed)
L-shape 256^2	head_sdf_R512_K16	0.7766	3.8607	3.8608	4.97	0.000	lshape (untimed)

Table F.7: Poisson ladder at 256^2 and 1024^2 (one job per mesh: 256^2 : job 3780692 (NVIDIA A100-PCIE-40GB, commit b43a437d7360); 1024^2 : job 3783813 (NVIDIA H200, commit 3e411b5ac59d); never compare costs across meshes; $q = 512$ via LM is degenerate by construction and its QR twin is the timed top rung).

mesh	subject	M	worst %	median %	complete-query ms	device ms	valid	non-dom. (all)	non-dom. (reduced)
256^2	$q=0$	129	3.1567	0.5080	9.214	6.703	36/36	no	no
256^2	$q=32$	257	2.4699	0.4437	9.537	6.893	36/36	no	no
256^2	$q=64$	384	2.0808	0.4127	9.819	7.347	36/36	no	no
256^2	$q=128$	640	1.5497	0.3195	10.219	7.536	36/36	no	no
256^2	$q=256$	1153	0.9689	0.1315	9.658	7.121	36/36	no	no
256^2	$q=512$	2177	0.7459	0.0580	117.854	114.990	0/36	no	no
256^2	linear top rung ($q=R$, QR)	2177	0.7459	0.0580	3.221	1.693	36/36	no	yes
256^2	free bank (LM)	2177	0.7789	0.0597	23.295	21.416	36/36	no	no
256^2	POD $k'=32$	129	9.9912	2.7371	3.766	2.094	36/36	no	no
256^2	POD $k'=64$	257	5.7375	0.6105	4.062	2.309	36/36	no	no
256^2	POD $k'=128$	512	2.5117	0.1060	4.825	3.231	36/36	no	no
256^2	POD $k'=256$	1025	0.8165	0.0134	6.171	4.566	36/36	no	no
256^2	POD $k'=512$	2048	0.1855	0.0014	10.002	8.340	36/36	no	yes
256^2	direct DST	—	0.0000	0.0000	2.526	0.227	36/36	yes	no
256^2	CG 10^{-2}	—	0.1527	0.0554	23.818	22.290	36/36	no	no
256^2	CG 10^{-4}	—	0.0012	0.0003	32.698	30.962	36/36	no	no
256^2	CG 10^{-6}	—	0.0000	0.0000	38.046	36.330	36/36	no	no
1024^2	$q=0$	129	3.1495	0.5079	6.394	3.941	36/36	no	no
1024^2	$q=32$	257	2.4641	0.4437	6.592	4.108	36/36	no	no
1024^2	$q=64$	384	2.0760	0.4128	6.745	4.240	36/36	no	no
1024^2	$q=128$	640	1.5459	0.3195	6.875	4.393	36/36	no	no
1024^2	$q=256$	1153	0.9648	0.1315	6.849	4.319	36/36	no	no
1024^2	$q=512$	2177	0.7421	0.0580	43.220	40.632	0/36	no	no
1024^2	linear top rung ($q=R$, QR)	2177	0.7421	0.0580	4.424	1.830	36/36	no	yes
1024^2	free bank (LM)	2177	0.7751	0.0596	13.021	10.662	36/36	no	no
1024^2	POD $k'=32$	129	9.9816	2.7357	3.696	1.318	36/36	no	no
1024^2	POD $k'=64$	257	5.7282	0.6099	3.970	1.571	36/36	no	no
1024^2	POD $k'=128$	512	2.5019	0.1058	4.397	2.095	36/36	no	no
1024^2	POD $k'=256$	1025	0.8119	0.0133	5.262	2.908	36/36	no	no
1024^2	POD $k'=512$	2048	0.1838	0.0014	6.972	4.572	36/36	no	yes
1024^2	direct DST	—	0.0000	0.0000	3.248	0.237	36/36	yes	no
1024^2	CG 10^{-2}	—	0.0722	0.0297	62.498	60.017	36/36	no	no
1024^2	CG 10^{-4}	—	0.0004	0.0001	85.988	83.437	36/36	no	no
1024^2	CG 10^{-6}	—	0.0000	0.0000	103.642	101.260	36/36	no	no

strict head advantage, before the cluster jobs. It is an empirical difference between worst errors, not a uniform bound on each trajectory’s integration error. The strict accuracy criterion fails at every mesh; the amended one passes (Table G.4). Conserving reduced energy does not certify trajectory accuracy.

On the L-shaped domain the sparse direct solve costs 1.28, 2.48, 8.39, 36.50 ms at 64^2 , 128^2 , 256^2 , 512^2 ; at 64^2 it dominates everything and at 128^2 only POD rungs join the non-dominated set (POD $k'=64$ at 7.701 %, $1.08\times$ the direct solve’s cost), cheapness rather than a trade; at 512^2 the best POD rung, POD $k'=128$, reaches 2.451 % at 4.031 ms. In a separate job at $M = 1024$, whose costs are not comparable with those above, the free rung $q = R$ is the one place the learned bank beats the linear baseline on both axes, at $1.003\times$ its own bank floor with no solver iterations (Table F.38).

G.6 AGAINST THE PREVIOUS SUBMISSION’S COMPARATOR: CONJUGATE GRADIENT

An earlier version of this work compared the reduced solve against unpreconditioned conjugate gradient (CG) at tolerance 10^{-6} . That comparison is retained here as a secondary one, same job only (Table G.5): on Poisson the reduced rungs are $3.7\text{--}23.4\times$ faster than CG 10^{-6} in the same job, but the direct transform (DST) solve is faster than every rung by $1.3\text{--}4.0\times$, which is why the paper

Table F.8: Heat 2D, an earlier cell of the same decoder family (job 3511417, NVIDIA A100-PCIE-40GB, commit 73fdaa88eb75...; $k = 8$, $R = 32$, twelve development cases, three repetitions; the finest three meshes of a five-rung ladder). The linear evolution of the full bank is the $q = R$ rung.

mesh	arm	worst physical %	worst same-grid %	device ms	complete-query ms
64^2	linear bank, exact modal evolution ($q=R$, no head)	1.675754	1.675754	0.116167	1.146627
64^2	nonlinear head ($k=8$)	4.559260	4.554246	12.091245	12.540239
64^2	same-grid direct solve	0.089749	0.000000	0.184825	0.557447
64^2	coarse (16^2) direct solve, interpolated	2.585355	2.623291	0.176126	0.518843
256^2	linear bank, exact modal evolution ($q=R$, no head)	1.675830	1.675830	0.137731	2.326213
256^2	nonlinear head ($k=8$)	4.555681	4.555390	11.888083	13.147866
256^2	same-grid direct solve	0.005602	0.000000	0.209627	1.438594
256^2	coarse (16^2) direct solve, interpolated	2.578601	2.581196	0.190585	1.400465
1024^2	linear bank, exact modal evolution ($q=R$, no head)	1.675830	1.675830	0.561659	17.681166
1024^2	nonlinear head ($k=8$)	4.555479	4.555461	12.318828	36.325427
1024^2	same-grid direct solve	0.000350	0.000000	1.034563	25.068664
1024^2	coarse (16^2) direct solve, interpolated	2.577910	2.578073	0.313116	24.364979

Table F.9: Neural operators on the shared Burgers data: capacity, epochs, checkpoint-recency flag (historically “still improving”), worst and median error on the matched eight-case cohort and on the 32 held-out cases. Accuracy is comparable across jobs; timing never is, and none is shown. Single seed, one mesh, one 3000 s budget per capacity.

arm	family	params	epochs	still improving	8-case worst %	val-32 worst %	val-32 median %	job
same_ntle-2_dt005	FOM	—	—	—	0.9978	—	—	3702709
same_ntle-6_dt005	FOM	—	—	—	1.3811	—	—	3702709
same_ntle-4_dt005	FOM	—	—	—	1.3811	—	—	3702709
unet-small	U-Net	4,368,389	2327	yes	1.4712	5.5937	1.3963	3780138
unet-medium	U-Net	7,763,461	1963	yes	1.5189	3.9622	1.3169	3780138
tsol-refine	Transolver	3,108,240	1628	no	1.5224	9.3183	1.3591	3780139
unet-refine	U-Net	7,763,461	1962	yes	1.7110	7.5176	1.0806	3780138
same_ntle-2_dt01	FOM	—	—	—	1.8642	—	—	3702709
rom	ROM	—	—	—	1.8671	—	—	3702709
coarse_half_dt005	FOM	—	—	—	1.8993	—	—	3702709
unet-large	U-Net	17,462,021	908	yes	2.3176	4.2340	1.3642	3780138
fno-large	FNO	17,877,317	692	yes	2.4829	6.3825	1.8054	3710846
tsol-small	Transolver	3,108,240	1628	yes	2.5273	6.8213	1.4522	3780139
tsol-medium	Transolver	6,952,208	586	yes	2.8498	9.5338	2.3846	3780139
fno-medium	FNO	5,780,213	1198	yes	2.8882	6.0423	1.7638	3710846
fno-small	FNO	1,193,125	1741	yes	2.9350	7.4812	1.9552	3710846
tsol-large	Transolver	12,322,960	882	yes	3.0988	6.3953	3.0690	3780139
fno-refine	FNO	17,877,317	688	yes	3.2660	7.1747	1.8810	3710846
coarse_quarter_dt01	FOM	—	—	—	3.6398	—	—	3702709

compares against the transform. On the L-shaped domain, where no transform applies, the lane timed CG at its own tolerances (tightest 10^{-10}) and an IC(0)-preconditioned CG on the CPU: the head is $2.9\text{--}14.9\times$ faster than the tightest CG across 64^2 , 128^2 , 256^2 , 512^2 , and the sparse direct solve is the competitive comparator (Table 6).

G.7 BURGERS AT TEN TIMES LOWER VISCOSITY (UNDER-RESOLVED; REDUCED-VERSUS-REDUCED ONLY)

A second Burgers cell (lane b-lowvisc; training job 3804337, gate job 3789639, panel job 3817807 on NVIDIA A100-PCIE-40GB) keeps the incumbent recipe and mesh and lowers the viscosity range ten-fold. The pre-registered under-resolution clause (F4) applies: the converged discrete operator itself sits $19.3\text{--}20.9\%$ from the 4096^2 reference ($5.17\times$ the incumbent cell’s discretisation error), so every number below is evidence about reduced models against each other on the same discrete system, not about the PDE. The source commits differ by stage: gate 2cf5dc34, training 98aad39, panel 144f6b29. The repaired provenance catalog retains hashes for historical incumbent training records whose source commit could not be recovered; those records do not inherit a later job’s source label. A POD orthogonality audit failed; the lane retained the bounded oblique-projection check as an explicit exception, not an all-checks-pass validation. Two things did happen. Linear reduction collapses: the POD-512 floor degrades $12.09\times$ and the trained linear bank’s floor $19.00\times$ ($0.3918 \rightarrow 7.4455\%$) against the incumbent cell, while the neural head’s solved $q = 0$ error degrades $3.27\times$ ($2.5629 \rightarrow 8.3933\%$) and its best-found $4.48\times$ (an upper bound here: solved over best-found is 0.74); the pre-registered F2 clause therefore does not fire (no). And every neural rung of the fixed- $M = 1088$ ladder, from $q = 0$, beats every POD-LSPG rank $k' = 16\text{--}512$ (best 11.2540% ; not monotone past $k' = 256$, the $k' = 512$ solve being solver-limited), and from $q = 128$ up

Table F.10: The correction ladder at 256^2 in one allocation (job 3789570, NVIDIA A100 80GB PCIe, checkpoint incumbent (gate checkpoint_unchanged; hash in T2, tuning row)), tolerance 10^{-6} : dense advection beside two quadrature rule sets, the rules the b-eqtop ladder used and the rules its replication selected (§6.3), with each rule’s status from that replication; worst error over evolved and over all times (%), median GPU ms, and the $t=0$ compression bounding the all-times metric. “Certified in one draw” is a single passing draw, not a certified construction. Every subject of the job, at both tolerances, is in Table F.13.

q	M	dense: evolved %	all %	ms	EQ, b-eqtop ladder rules: evolved %	all %	ms	rule status	EQ, replication-selected rules: evolved %	all %	ms	rule status	$t=0$ %
0	64	1.8890	2.5629	283.9	1.8891	2.5629	59.5	confirmed (3 of 3 re-draws)	1.8891	2.5629	58.6	confirmed (3 of 3 re-draws)	2.5629
16	128	1.3985	2.4806	360.8	1.4270	2.4806	81.7	confirmed (3 of 3 re-draws)	1.4270	2.4806	81.1	confirmed (3 of 3 re-draws)	2.4806
32	192	1.2336	2.3534	434.7	1.2493	2.3534	96.4	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.8	confirmed (2 of 2 re-draws)	2.3534
64	320	1.0843	2.1489	621.1	1.2275	2.1489	119.7	marginal (2 of 6 draws pass)	1.0840	2.1489	148.9	confirmed (2 of 2 re-draws)	2.1489
128	576	0.8930	1.8116	1186.9	0.8925	1.8116	246.5	marginal (4 of 5 draws pass)	0.8931	1.8116	273.5	single-draw	1.8116
256	1088	0.5194	0.9053	3939.8	1.0361	1.0361	697.6	marginal (1 of 5 draws pass)	0.5129	0.9053	746.0	single-draw	0.9053

Table F.11: The correction ladder at 1024^2 in one allocation (job 3789572, NVIDIA H200, same checkpoint), tolerance 10^{-6} : dense advection beside the quadrature rules transferred from 256^2 (rule status as at 256^2 ; the $q \geq 64$ rules miss the primary bar here, §6.1). Milliseconds are not comparable with Table F.10: different GPU, different job.

q	M	dense: evolved %	all %	ms	EQ, ladder rules transferred from 256^2 : evolved %	all %	ms	rule status	$t=0$ %
0	64	2.2886	3.8562	1551.7	2.2913	3.8562	40.3	transferred, primary at 1024^2 ; source confirmed (3 of 3 re-draws)	3.8562
16	128	1.5398	3.8071	1949.1	1.5273	3.8071	59.6	transferred, primary at 1024^2 ; source confirmed (3 of 3 re-draws)	3.8071
32	192	1.4482	3.7140	2382.9	1.4487	3.7140	72.0	transferred, primary at 1024^2 ; source confirmed (2 of 2 re-draws)	3.7140
64	320	1.2739	3.5773	3565.6	1.2762	3.5773	84.0	transferred, not primary at 1024^2 ; source marginal (2 of 6 draws pass)	3.5773
128	576	1.0444	3.2858	6895.2	1.0442	3.2858	163.7	transferred, not primary at 1024^2 ; source marginal (4 of 5 draws pass)	3.2858
256	1088	0.5861	2.7974	22053.9	1.8664	2.7974	502.2	transferred, not primary at 1024^2 ; source marginal (1 of 5 draws pass)	2.7974

also the free bank’s own solve (7.7940 % at 3174.6 ms): on the reduced-only frontier the neural rungs $q = 64, 128, 256$ are non-dominated. What did not happen: no reduced subject is non-dominated against the full-order grid (criterion P fails; the non-dominated set is nt1e-2_dt01, nt1e-3_dt005, dense_tight, fft_tight; the cheapest full-order setting beating every reduced subject is nt1e-2_dt01 at 14.0 ms and 3.90 %), the fixed- M ladder is monotone and converged but spans only $1.36\times$ in error for $3.25\times$ in cost ($9.0500 \rightarrow 6.6718$ %, $557.4 \rightarrow 1810.3$ ms), so the $2\times$ knob bar is not met, the $M = 256$ control is not monotone (test-starved at $q = 128$, as in the incumbent cell), and every reduced row’s error against the reference (17.98–19.44 %) is the discretisation error. 27 subjects, 0 budget exits.

Table F.12: The correction ladder at 512^2 in one allocation (job 3805065, NVIDIA A100 80GB PCIe, same checkpoint and GPU model as 256^2), tolerance 10^{-6} : dense advection beside both 256^2 rule sets transferred to this mesh (each transfer refits on its own fit-state draw); the status names whether the transfer passes the primary bar here and its source rule’s verdict at 256^2 . Milliseconds are not comparable with Table F.10: a different job.

ϵ	M	dense evolved %	all %	ms	EQ ladder rules transferred from 256^2	evolved %	all %	ms	rule status	EQ replication-selected rules transferred from 256^2	evolved %	all %	ms	rule status	$t \pm 0.5$
0	64	2.1376	3.2836	923.2		2.1400	3.2836	58.0	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)		2.1378	3.2836	58.1	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)	3.2836
18	128	1.5071	3.2248	1190.8		1.4014	3.2248	79.6	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)		1.4037	3.2248	79.2	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)	3.2248
32	192	1.3606	3.1172	1862.4		1.3072	3.1172	96.7	transferred, primary at 512^2 , source confirmed (2 of 2 re-draws)		1.4085	3.1172	93.9	transferred, not primary at 512^2 , source confirmed (2 of 2 re-draws)	3.1172
64	320	1.2015	2.9565	2231.7		1.2485	2.9565	120.8	transferred, not primary at 512^2 , source marginal (3 of 6 draws pass)		1.1994	2.9565	150.2	transferred, not primary at 512^2 , source confirmed (2 of 2 re-draws)	2.9565
128	576	0.8900	2.6066	4841.3		0.8084	2.6066	242.0	transferred, not primary at 512^2 , source marginal (1 of 6 draws pass)		0.9001	2.6066	262.4	transferred, primary at 512^2 , source single-draw	2.6066
256	1088	0.5625	2.0395	16219.9		1.8284	2.0395	686.6	transferred, not primary at 512^2 , source marginal (1 of 5 draws pass)		0.5510	2.0395	783.3	transferred, primary at 512^2 , source single-draw	2.0395

Table F.13: Every timed subject of the 256^2 same-allocation panel (job 3789570). “conv. (§5)” is the pre-registered completion rule as written — stationarity 10^{-6} at every step, or a residual-rule exit, with an attained initial fit accepted; “strict” is the earlier rule that does not accept it, tested against the arm’s own tolerance (so a 10^{-3} arm can be strict and still not converged under §5); “adm. (§5)” is admissibility for a frontier or a ratio (converged under that rule, and for EQ subjects a rule passing a bar). “adm. (own tol.)” is the superseded flag: it tested each arm against its own tolerance and so admitted the 12 arms run at 10^{-3} , which the rule as written does not (0 of 12 pass it). Those arms are kept here with all their numbers but are not admissible, which moves the admissible reduced count at this mesh from 39 to 27. The correction is the panel pre-registration’s §A13, and every panel number in this paper is read from that lane at commit 25434a27.

subject	family	q/k'	M	quad.	rule	evolved %	all %	$t=0$ %	vs ref %	device ms	complete-squary ms	conv. (§5)	strict	adm. (§5)	adm. (own tol.)
q0_M256_dense_glem06	rom	0	256	dense	none	1.2710	2.5629	2.5629	4.0637	341.5	343.4	yes	yes	yes	yes
q0_M64_dense_glem06	rom	0	64	dense	none	1.8890	2.5629	2.5629	4.5575	283.9	285.9	yes	yes	yes	yes
q0_M64_eqcert_g0p001	rom	0	64	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.8898	2.5628	2.5628	4.5521	43.6	45.8	no	yes	no	yes
q0_M64_eqcert_glem06	rom	0	64	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.8891	2.5629	2.5629	4.5529	59.5	61.7	yes	yes	yes	yes
q0_M64_eqtop_g0p001	rom	0	64	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.8898	2.5628	2.5628	4.5521	43.4	45.5	no	yes	no	yes
q0_M64_eqtop_glem06	rom	0	64	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.8891	2.5629	2.5629	4.5529	58.6	60.6	yes	yes	yes	yes
q16_M128_dense_glem06	rom	16	128	dense	none	1.3985	2.4806	2.4806	4.1065	360.8	362.8	yes	yes	yes	yes
q16_M128_eqcert_g0p001	rom	16	128	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.4272	2.4806	2.4806	4.1136	63.3	65.3	no	yes	no	yes
q16_M128_eqcert_glem06	rom	16	128	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.4270	2.4806	2.4806	4.1145	81.7	83.8	yes	yes	yes	yes
q16_M128_eqtop_g0p001	rom	16	128	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.4272	2.4806	2.4806	4.1136	62.1	64.2	no	yes	no	yes
q16_M128_eqtop_glem06	rom	16	128	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.4270	2.4806	2.4806	4.1145	81.1	83.3	yes	yes	yes	yes
q32_M192_dense_glem06	rom	32	192	dense	none	1.2336	2.3534	2.3534	4.0814	434.7	437.0	yes	yes	yes	yes
q32_M192_eqcert_g0p001	rom	32	192	eq	eqcert m=1024, confirmed (2 of 2 re-draws)	1.2497	2.3534	2.3534	4.0805	73.7	75.7	no	yes	no	yes
q32_M192_eqcert_glem06	rom	32	192	eq	eqcert m=1024, confirmed (2 of 2 re-draws)	1.2493	2.3534	2.3534	4.0818	96.4	98.4	yes	yes	yes	yes
q32_M192_eqtop_g0p001	rom	32	192	eq	eqtop m=1024, confirmed (2 of 2 re-draws)	1.2497	2.3534	2.3534	4.0805	74.1	76.2	no	yes	no	yes
q32_M192_eqtop_glem06	rom	32	192	eq	eqtop m=1024, confirmed (2 of 2 re-draws)	1.2493	2.3534	2.3534	4.0818	97.8	99.8	yes	yes	yes	yes
q64_M320_dense_glem06	rom	64	320	dense	none	1.0843	2.1489	2.1489	4.1013	621.1	623.0	yes	yes	yes	yes
q64_M320_eqcert_g0p001	rom	64	320	eq	eqcert m=1024, marginal (2 of 6 draws pass)	1.2278	2.1489	2.1489	4.0836	90.8	92.6	no	yes	no	yes
q64_M320_eqcert_glem06	rom	64	320	eq	eqcert m=1024, marginal (2 of 6 draws pass)	1.2275	2.1489	2.1489	4.0855	119.7	121.8	yes	yes	yes	yes
q64_M320_eqtop_g0p001	rom	64	320	eq	eqtop m=2048, confirmed (2 of 2 re-draws)	1.0841	2.1489	2.1489	4.1008	113.9	116.0	no	yes	no	yes
q64_M320_eqtop_glem06	rom	64	320	eq	eqtop m=2048, confirmed (2 of 2 re-draws)	1.0840	2.1489	2.1489	4.1027	148.9	151.0	yes	yes	yes	yes
q128_M576_dense_glem06	rom	128	576	dense	none	0.8930	1.8116	1.8116	4.0797	1186.9	1189.0	yes	yes	yes	yes
q128_M576_eqcert_g0p001	rom	128	576	eq	eqcert m=2048, marginal (4 of 5 draws pass)	0.8926	1.8116	1.8116	4.0791	189.3	191.5	no	yes	no	yes
q128_M576_eqcert_glem06	rom	128	576	eq	eqcert m=2048, marginal (4 of 5 draws pass)	0.8925	1.8116	1.8116	4.0809	246.5	248.6	yes	yes	yes	yes
q128_M576_eqtop_g0p001	rom	128	576	eq	eqtop m=2319, single-draw	0.8932	1.8116	1.8116	4.0787	205.1	207.2	no	yes	no	yes
q128_M576_eqtop_glem06	rom	128	576	eq	eqtop m=2319, single-draw	0.8931	1.8116	1.8116	4.0806	273.5	275.3	yes	yes	yes	yes
q256_M1088_dense_glem06	rom	256	1088	dense	none	0.5194	0.9053	0.9053	4.0391	3939.8	3942.0	yes	yes	yes	yes
q256_M1088_eqcert_g0p001	rom	256	1088	eq	eqcert m=2048, marginal (1 of 5 draws pass)	1.0324	1.0324	0.9053	4.0322	429.6	431.6	no	yes	no	yes
q256_M1088_eqcert_glem06	rom	256	1088	eq	eqcert m=2048, marginal (1 of 5 draws pass)	1.0361	1.0361	0.9053	4.0332	697.6	699.7	yes	yes	yes	yes
q256_M1088_eqtop_g0p001	rom	256	1088	eq	eqtop m=2560, single-draw	0.5129	0.9053	0.9053	4.0355	466.4	468.6	no	yes	no	yes
q256_M1088_eqtop_glem06	rom	256	1088	eq	eqtop m=2560, single-draw	0.5129	0.9053	0.9053	4.0365	746.0	748.4	yes	yes	yes	yes
q0_M64_eqcert_glem06_FastL4	fast	0	64	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.8891	2.5629	2.5629	4.5529	40.4	42.5	yes	yes	yes	yes
pod16_M64_dense	pod	16	64	dense	none	28.7250	61.6503	61.6503	61.6503	46.8	48.8	yes	yes	yes	yes
pod32_M128_dense	pod	32	128	dense	none	18.7995	47.0681	47.0681	47.0681	81.0	82.9	yes	yes	yes	yes
pod64_M256_dense	pod	64	256	dense	none	7.0835	19.8156	19.8156	19.8156	145.7	147.7	yes	yes	yes	yes
pod128_M512_dense	pod	128	512	dense	none	1.9464	10.1198	10.1198	10.1198	331.6	333.5	yes	yes	yes	yes
pod256_M1024_dense	pod	256	1024	dense	none	0.7109	3.7698	3.7698	4.0362	898.8	901.2	yes	yes	yes	yes
pod512_M2048_dense	pod	512	2048	dense	none	0.2184	0.6125	0.6125	4.0266	2790.8	2794.3	yes	yes	yes	yes
free512_M1024_dense	free	512	1024	dense	none	0.4471	0.6027	0.6027	4.0423	2439.6	2441.9	yes	yes	yes	yes
fno-large	fno	—	—	—	—	7.4164	7.4164	0.0000	5.7495	7.2	7.4	—	—	—	—
dense_1ight	dense	—	—	—	—	0.0000	0.0000	0.0000	4.0265	65.9	65.8	—	—	—	—
fft-1ight	fno	—	—	—	—	0.0000	0.0000	0.0000	4.0265	90.4	92.5	—	—	—	—
ntle-2_at005	fno	—	—	—	—	3.7127	3.7127	0.0000	2.4737	15.7	17.7	—	—	—	—
ntle-2_at01	fno	—	—	—	—	3.1999	3.1999	0.0000	3.6168	90.0	11.0	—	—	—	—
ntle-3_at005	fno	—	—	—	—	0.0489	0.0489	0.0000	4.0399	31.8	34.2	—	—	—	—
ntle-3_at01	fno	—	—	—	—	1.5179	1.5179	0.0000	5.1761	19.7	21.7	—	—	—	—
ntle-4_at005	fno	—	—	—	—	0.0338	0.0338	0.0000	4.0320	37.0	39.0	—	—	—	—
ntle-4_at01	fno	—	—	—	—	1.5109	1.5109	0.0000	5.1552	27.8	30.0	—	—	—	—

Table F.14: Every timed subject of the 1024² same-allocation panel (job 3789572, H200). Columns as in Table F.13; milliseconds are comparable only within this table. The §5 rule takes the admissible reduced count from 20 to 16 and the reduced subjects on the (GPU ms, worst evolved) frontier from 5 to 3 (q0_M64_eqxfer_glem06, q16_M128_eqxfer_glem06, q32_M192_eqxfer_glem06).

subject	family	q / k'	M	quad.	rule	evolved %	all %	t=0 %	vs ref %	device ms	complete-query ms	conv. (§5)	strict	adm. (§5)	adm. (own tol.)
q0_M64_dense_glem06	rom	0	64	dense	none	2.2886	3.8562	3.8562	3.8562	1551.7	1562.4	yes	yes	yes	yes
q0_M64_eqxfer_g0p001	rom	0	64	eq	eqxfer m=934, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	2.2925	3.8562	3.8562	3.8562	32.2	44.4	no	yes	no	yes
q0_M64_eqxfer_glem06	rom	0	64	eq	eqxfer m=934, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	2.2913	3.8562	3.8562	3.8562	40.3	51.4	yes	yes	yes	yes
q16_M128_dense_glem06	rom	16	128	dense	none	1.5398	3.8071	3.8071	3.8071	1949.1	1960.6	yes	yes	yes	yes
q16_M128_eqxfer_g0p001	rom	16	128	eq	eqxfer m=918, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	1.5277	3.8071	3.8071	3.8071	48.4	60.7	no	yes	no	yes
q16_M128_eqxfer_glem06	rom	16	128	eq	eqxfer m=918, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	1.5273	3.8071	3.8071	3.8071	59.6	70.5	yes	yes	yes	yes
q32_M192_dense_glem06	rom	32	192	dense	none	1.4482	3.7140	3.7140	3.7140	2382.9	2393.5	yes	yes	yes	yes
q32_M192_eqxfer_g0p001	rom	32	192	eq	eqxfer m=972, transferred, primary at 1024 ² ; source confirmed (2 of 2 re-draws)	1.4488	3.7140	3.7140	3.7140	57.6	69.2	no	yes	no	yes
q32_M192_eqxfer_glem06	rom	32	192	eq	eqxfer m=972, transferred, primary at 1024 ² ; source confirmed (2 of 2 re-draws)	1.4487	3.7140	3.7140	3.7140	72.0	83.0	yes	yes	yes	yes
q64_M320_dense_glem06	rom	64	320	dense	none	1.2739	3.5773	3.5773	3.5773	3565.6	3581.8	yes	yes	yes	yes
q64_M320_eqxfer_g0p001	rom	64	320	eq	eqxfer m=942, transferred, not primary at 1024 ² ; source marginal (2 of 6 draws pass)	1.2763	3.5773	3.5773	3.5773	64.4	76.5	no	yes	no	yes
q64_M320_eqxfer_glem06	rom	64	320	eq	eqxfer m=942, transferred, not primary at 1024 ² ; source marginal (2 of 6 draws pass)	1.2762	3.5773	3.5773	3.5773	84.0	94.5	yes	yes	no	no
q128_M576_dense_glem06	rom	128	576	dense	none	1.0444	3.2858	3.2858	3.2858	6895.2	6909.1	yes	yes	yes	yes
q128_M576_eqxfer_g0p001	rom	128	576	eq	eqxfer m=1713, transferred, not primary at 1024 ² ; source marginal (4 of 5 draws pass)	1.0443	3.2858	3.2858	3.2858	128.3	139.2	no	yes	no	yes
q128_M576_eqxfer_glem06	rom	128	576	eq	eqxfer m=1713, transferred, not primary at 1024 ² ; source marginal (4 of 5 draws pass)	1.0442	3.2858	3.2858	3.2858	163.7	175.4	yes	yes	yes	yes
q256_M1088_dense_glem06	rom	256	1088	dense	none	0.5861	2.7974	2.7974	2.7974	22053.9	22066.3	yes	yes	yes	yes
q256_M1088_eqxfer_g0p001	rom	256	1088	eq	eqxfer m=1658, transferred, not primary at 1024 ² ; source marginal (1 of 5 draws pass)	1.8655	2.7974	2.7974	2.7974	277.2	290.1	no	yes	no	yes
q256_M1088_eqxfer_glem06	rom	256	1088	eq	eqxfer m=1658, transferred, not primary at 1024 ² ; source marginal (1 of 5 draws pass)	1.8664	2.7974	2.7974	2.7974	502.2	515.1	yes	yes	no	no
pod16_M64_dense	pod	16	64	dense	none	29.2793	61.9285	61.9285	61.9285	198.2	209.6	yes	yes	yes	yes
pod32_M128_dense	pod	32	128	dense	none	19.4088	47.6410	47.6410	47.6410	391.2	402.5	yes	no	yes	yes
pod64_M256_dense	pod	64	256	dense	none	7.4077	20.5994	20.5994	20.5994	741.6	752.9	yes	no	yes	yes
pod128_M512_dense	pod	128	512	dense	none	2.1609	10.7339	10.7339	10.7339	1760.6	1773.0	yes	no	yes	yes
pod256_M1024_dense	pod	256	1024	dense	none	1.1017	4.1416	4.1416	4.1416	4897.6	4911.4	yes	no	yes	yes
free512_M1024_dense	free	512	1024	dense	none	0.5108	2.4466	2.4466	2.4466	13514.6	13529.0	yes	no	yes	yes
free-large	fno	—	—	—	—	6.2657	0.0000	0.0000	5.6472	58.9	71.0	—	—	—	—
fft-tight	fom	—	—	—	—	0.0000	0.0000	0.0000	2.1416	210.9	222.2	—	—	—	—
nt1e-2_dt005	fom	—	—	—	—	4.2628	4.2628	0.0000	2.3899	31.1	42.7	—	—	—	—
nt1e-2_dt01	fom	—	—	—	—	3.4582	3.4582	0.0000	2.8919	17.7	28.3	—	—	—	—
nt1e-4_dt005	fom	—	—	—	—	0.0343	0.0343	0.0000	2.1281	81.3	92.5	—	—	—	—
nt1e-4_dt01	fom	—	—	—	—	1.6287	1.6287	0.0000	3.7562	64.0	76.3	—	—	—	—

Table F.15: Every timed subject of the 512² same-allocation panel (job 3805065). Columns as in Table F.13; milliseconds are comparable only within this table. The \$5 rule takes the admissible reduced count from 31 to 23; nothing reduced is on the frontier under either flag.

subject	family	q / k'	M	quad.	rule	evolved %	all %	$t=0$ %	vs ref %	device ms	complete-query ms	conv. (\$)	strict	adm. (\$5)	adm. (own tol.)
q0_M256_dense_glem06	rom	0	256	dense	none	1.3761	3.2836	3.2836	3.2836	1175.7	1179.7	yes	yes	yes	yes
q0_M64_dense_glem06	rom	0	64	dense	none	2.1376	3.2836	3.2836	3.7578	923.2	927.3	yes	yes	yes	yes
q0_M64_eqtqpfer_g0p001	rom	0	64	eq	eqtqpfer $m=914$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1387	3.2836	3.2836	3.7448	43.8	47.5	no	yes	no	yes
q0_M64_eqtqpfer_glem06	rom	0	64	eq	eqtqpfer $m=914$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1378	3.2836	3.2836	3.7449	59.1	62.9	yes	yes	yes	yes
q0_M64_eqxfer_g0p001	rom	0	64	eq	eqxfer $m=922$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1409	3.2836	3.2836	3.7681	44.1	48.3	no	yes	no	yes
q0_M64_eqxfer_glem06	rom	0	64	eq	eqxfer $m=922$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1400	3.2836	3.2836	3.7681	58.0	61.9	yes	yes	yes	yes
q16_M128_dense_glem06	rom	16	128	dense	none	1.5071	3.2248	3.2248	3.2248	1190.8	1194.9	yes	yes	yes	yes
q16_M128_eqtqpfer_g0p001	rom	16	128	eq	eqtqpfer $m=879$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4538	3.2248	3.2248	3.2248	62.5	66.3	no	yes	no	yes
q16_M128_eqtqpfer_glem06	rom	16	128	eq	eqtqpfer $m=879$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4537	3.2248	3.2248	3.2248	79.2	83.1	yes	yes	yes	yes
q16_M128_eqxfer_g0p001	rom	16	128	eq	eqxfer $m=911$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4917	3.2248	3.2248	3.2248	62.5	66.7	no	yes	no	yes
q16_M128_eqxfer_glem06	rom	16	128	eq	eqxfer $m=911$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4914	3.2248	3.2248	3.2248	79.6	83.7	yes	yes	yes	yes
q32_M192_dense_glem06	rom	32	192	dense	none	1.3666	3.1172	3.1172	3.1172	1502.4	1506.1	yes	yes	yes	yes
q32_M192_eqtqpfer_g0p001	rom	32	192	eq	eqtqpfer $m=848$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.4060	3.1172	3.1172	3.1172	72.1	75.9	no	yes	no	yes
q32_M192_eqtqpfer_glem06	rom	32	192	eq	eqtqpfer $m=848$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.4055	3.1172	3.1172	3.1172	93.9	97.5	yes	yes	yes	yes
q32_M192_eqxfer_g0p001	rom	32	192	eq	eqxfer $m=973$, transferred, primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.3673	3.1172	3.1172	3.1172	74.4	78.6	no	yes	no	yes
q32_M192_eqxfer_glem06	rom	32	192	eq	eqxfer $m=973$, transferred, primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.3672	3.1172	3.1172	3.1172	96.7	100.8	yes	yes	yes	yes
q64_M320_dense_glem06	rom	64	320	dense	none	1.2015	2.9565	2.9565	3.0133	2221.7	2225.7	yes	yes	yes	yes
q64_M320_eqtqpfer_g0p001	rom	64	320	eq	eqtqpfer $m=1898$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.1995	2.9565	2.9565	3.0056	115.1	119.1	no	yes	no	no
q64_M320_eqtqpfer_glem06	rom	64	320	eq	eqtqpfer $m=1898$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.1994	2.9565	2.9565	3.0069	150.2	154.0	yes	yes	no	no
q64_M320_eqxfer_g0p001	rom	64	320	eq	eqxfer $m=937$, transferred, not primary at 512 ² ; source marginal (2 of 6 draws pass)	1.2488	2.9565	2.9565	3.0135	89.8	93.7	no	yes	no	no
q64_M320_eqxfer_glem06	rom	64	320	eq	eqxfer $m=937$, transferred, not primary at 512 ² ; source marginal (2 of 6 draws pass)	1.2485	2.9565	2.9565	3.0149	120.8	125.1	yes	yes	no	no
q128_M576_dense_glem06	rom	128	576	dense	none	0.9890	2.6066	2.6066	2.9236	4541.3	4545.3	yes	yes	yes	yes
q128_M576_eqtqpfer_g0p001	rom	128	576	eq	eqtqpfer $m=2152$, transferred, primary at 512 ² ; source single-draw	0.9902	2.6066	2.6066	2.9224	200.1	204.1	no	yes	no	yes
q128_M576_eqtqpfer_glem06	rom	128	576	eq	eqtqpfer $m=2152$, transferred, primary at 512 ² ; source single-draw	0.9901	2.6066	2.6066	2.9237	262.4	266.3	yes	yes	yes	yes
q128_M576_eqxfer_g0p001	rom	128	576	eq	eqxfer $m=1644$, transferred, not primary at 512 ² ; source marginal (4 of 5 draws pass)	0.9895	2.6066	2.6066	2.9188	188.4	192.4	no	yes	no	no
q128_M576_eqxfer_glem06	rom	128	576	eq	eqxfer $m=1644$, transferred, not primary at 512 ² ; source marginal (4 of 5 draws pass)	0.9894	2.6066	2.6066	2.9200	242.0	245.9	yes	yes	no	no
q256_M1088_dense_glem06	rom	256	1088	dense	none	0.5625	2.0195	2.0195	2.7611	16219.9	16224.4	yes	yes	yes	yes
q256_M1088_eqtqpfer_g0p001	rom	256	1088	eq	eqtqpfer $m=2438$, transferred, primary at 512 ² ; source single-draw	0.5510	2.0195	2.0195	2.7581	486.3	490.4	no	yes	no	yes
q256_M1088_eqtqpfer_glem06	rom	256	1088	eq	eqtqpfer $m=2438$, transferred, primary at 512 ² ; source single-draw	0.5510	2.0195	2.0195	2.7602	783.3	787.8	yes	yes	yes	yes
q256_M1088_eqxfer_g0p001	rom	256	1088	eq	eqxfer $m=1663$, transferred, not primary at 512 ² ; source marginal (1 of 5 draws pass)	1.8179	2.0195	2.0195	2.7920	395.7	399.9	no	yes	no	no
q256_M1088_eqxfer_glem06	rom	256	1088	eq	eqxfer $m=1663$, transferred, not primary at 512 ² ; source marginal (1 of 5 draws pass)	1.8204	2.0195	2.0195	2.7969	686.6	690.7	yes	yes	no	no
q0_M64_eqxfer_glem06_FastL4	fast	0	64	eq	eqxfer $m=922$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1400	3.2836	3.2836	3.7681	40.5	44.4	yes	yes	yes	yes
pod16_M64_dense	pod	16	64	dense	none	29.0916	61.8330	61.8330	61.8330	127.5	131.5	yes	yes	yes	yes
pod32_M128_dense	pod	32	128	dense	none	19.1988	47.4440	47.4440	47.4440	247.8	251.7	yes	yes	yes	yes
pod64_M256_dense	pod	64	256	dense	none	7.2900	20.3198	20.3198	20.3198	488.5	492.5	yes	yes	yes	yes
pod128_M512_dense	pod	128	512	dense	none	2.0372	10.5346	10.5346	10.5346	1206.7	1210.7	yes	yes	yes	yes
pod256_M1024_dense	pod	256	1024	dense	none	0.9578	4.0010	4.0010	4.0010	3488.2	3492.8	yes	yes	yes	yes
pod512_M2048_dense	pod	512	2048	dense	none	0.3328	0.6579	0.6579	0.7299	11933.9	11941.1	yes	yes	yes	yes
free512_M1024_dense	free	512	1024	dense	none	0.4912	1.8046	1.8046	2.7232	10019.3	10024.2	yes	yes	yes	yes
fno-large	fno	—	—	—	—	6.6169	6.6169	0.0000	5.6812	22.5	23.2	—	—	—	—
dense_tight	fom	—	—	—	—	0.0000	0.0000	0.0000	2.7025	147.4	151.4	—	—	—	—
fft_tight	fom	—	—	—	—	0.0000	0.0000	0.0000	2.7025	157.7	161.8	—	—	—	—
ntile-2_at005	fom	—	—	—	—	4.0614	4.0614	0.0000	2.0638	24.4	28.3	—	—	—	—
ntile-2_at01	fom	—	—	—	—	3.3550	3.3550	0.0000	3.3046	13.9	17.8	—	—	—	—
ntile-3_at005	fom	—	—	—	—	0.0516	0.0516	0.0000	2.7116	52.9	56.5	—	—	—	—
ntile-3_at01	fom	—	—	—	—	1.5893	1.5893	0.0000	4.1814	33.7	37.7	—	—	—	—
ntile-4_at005	fom	—	—	—	—	0.0341	0.0341	0.0000	2.6975	61.2	65.4	—	—	—	—
ntile-4_at01	fom	—	—	—	—	1.5869	1.5869	0.0000	4.1611	48.4	52.3	—	—	—	—

Table F.16: Fixed- q sweeps in the test count M (dense; errors comparable across the three b-qxm jobs, costs not shown). The last column restricts the span to at least two tests per unknown.

q	M sweep	worst evolved %	monotone	span	best M	span, $M \geq 2(K+q)$
0	64, 128, 256, 512, 1088, 2048, 4096	1.8890 / 1.4695 / 1.2710 / 1.2662 / 1.2657 / 1.2652 / 1.2649	yes	1.49×	4096	1.49×
16	128, 192, 256, 512, 1088	1.3985 / 1.2690 / 1.2305 / 1.2204 / 1.2204	yes	1.15×	1088	1.15×
32	192, 256, 320, 384, 768, 1088	1.2336 / 1.1869 / 1.1717 / 1.1665 / 1.1634 / 1.1629	yes	1.06×	1088	1.06×
64	128, 256, 320, 512, 576, 640, 1088, 1280, 2048, 4096	1.8032 / 1.1255 / 1.0843 / 1.0647 / 1.0633 / 1.0619 / 1.0593 / 1.0588 / 1.0594 / 1.0620	no	1.70×	1280	1.06×
128	256, 576, 1088, 1152	1.0418 / 0.8930 / 0.8711 / 0.8689	yes	1.20×	1152	1.03×
256	544, 1088, 2176, 3264, 4352, 6528	0.7566 / 0.5194 / 0.3736 / 0.3564 / 0.3523 / 0.3510	yes	2.16×	6528	2.16×

Table F.17: Head ablation, Burgers 256² (job 3711424, NVIDIA A100 80GB PCIe). Same-grid error against the converged full-order solve; “vs ref” against the 4096² reference.

arm	dim.	bank floor %	best-found %	solved same-grid %	vs ref %	device ms	stationary
(a) neural head, EQ	16	0.3918	2.5447	2.5629	4.5546	47.6	yes
(a) neural head, dense	16	0.3918	2.5447	2.5629	4.5575	281.7	yes
(b) linear map (fit to head outputs)	16	0.3918	56.9296	56.9296	56.9296	19.9	yes
(b) linear map (fit to truth)	16	0.3918	61.5226	61.5226	61.5226	19.7	yes
(c) quadratic map	16	0.3918	31.8276	31.8276	31.8276	28.0	yes
(e) POD-LSPG $k'=16$	16	61.6503	61.6503	61.6503	61.6503	46.0	yes
(e) POD-LSPG $k'=32$	32	47.0681	47.0681	47.0681	47.0681	80.7	yes
(e) POD-LSPG $k'=64$	64	19.8147	19.8147	19.8156	19.8156	144.1	no
(e) POD-LSPG $k'=128$	128	10.1189	10.1189	10.1198	10.1198	328.3	no
(d) unrestricted bank ($R=512$)	512	0.3918	0.3918	0.6027	4.0423	2417.6	no
FOM Newton, loose	—	—	—	3.7127	2.4737	15.4	—
FOM Newton, tight (reference)	—	—	—	0.0000	4.0265	88.7	—

Table F.18: Head ablation, Poisson 1024², $R = 128$ (job 3711736, NVIDIA A100 80GB PCIe).

arm	dim.	bank floor %	best-found %	solved %	query ms
(a) neural head	16	2.3148	6.0926	6.0927	5.843
(a+) head + 32 eliminated corrections	16	—	—	4.6670	7.317
(b) linear map (head outputs)	16	2.3148	17.2966	17.2966	5.163
(b) linear map (truth)	16	2.3148	21.1648	21.1648	5.329
(c) quadratic map	16	2.3148	16.6302	16.6303	5.774
(e) POD-LSPG $k'=8$	8	22.3196	22.3196	22.3196	4.633
(e) POD-LSPG $k'=16$	16	20.0548	20.0548	20.0548	4.659
(e) POD-LSPG $k'=32$	32	11.8589	11.8589	11.8589	5.050
(e) POD-LSPG $k'=64$	64	7.7397	7.7397	7.7398	5.170
(e) POD-LSPG $k'=128$	128	4.3213	4.3213	4.3452	5.930
(d) unrestricted bank ($R=128$)	128	2.3148	2.3148	2.3274	5.995
direct DST	—	—	—	0.0007	4.341

Table F.19: Solver knobs at a fixed Burgers checkpoint on 32 held-out cases (job 3712269, NVIDIA A100 80GB PCIe, checkpoint 18f0266ae6f0. . .). Loosening the tolerance from 10^{-8} to 10^{-3} removes 35.6 % of the query for a change from 6.712 to 6.701 % worst error; a cap of 2 is a cliff (90.3 %). The full-order control at tolerance 10^{-4} beats the best ROM setting on both axes.

setting	worst % (32 held-out)	device ms	early-stopped
EQ $m=256$, tol 10^{-6} , cap 180	7.245	51.0	0/96
EQ $m=512$, tol 10^{-8}	6.712	64.7	0/96
EQ $m=512$, tol 10^{-3}	6.701	41.6	0/96
EQ $m=512$, cap 2 (early-stopped)	90.324	28.4	96/96
FOM Newton 10^{-2} , $\Delta t=0.01$	6.807	9.9	0/96
FOM Newton 10^{-2} , $\Delta t=0.005$	35.357	17.4	0/96
FOM Newton 10^{-4} , $\Delta t=0.005$	6.171	22.4	0/96
FOM Newton 10^{-6} , $\Delta t=0.005$	6.172	88.3	0/96
FOM 128^2 , $\Delta t=0.005$	9.849	21.2	0/96
FOM 64^2 , $\Delta t=0.01$	16.095	14.0	0/96

Table F.20: Cold-start iteration-cap sweep on Poisson at 1024 intervals (local GB10 diagnostic; only ratios transfer). From the nearest training code the converged answer is reached at a small cap; from $z = 0$ the cap traces a monotone curve that looks like an accuracy–cost frontier, every point of which is dominated by starting well.

start	LM cap	worst %	median %	median iters	ms (GB10, ratios only)
nearest	1	13.55	4.29	1	20.18
nearest	2	6.80	1.29	2	25.68
nearest	4	6.09	1.24	4	33.83
nearest	6	6.09	1.24	5	40.58
nearest	8	6.09	1.24	5	38.67
nearest	10	6.09	1.24	5	38.23
nearest	12	6.09	1.24	5	38.92
nearest	20	6.09	1.24	5	38.47
nearest	300	6.09	1.24	5	70.17
zero	1	91.88	74.88	1	20.60
zero	2	91.88	74.88	2	25.46
zero	4	91.88	74.88	4	33.43
zero	6	91.88	54.37	6	45.09
zero	8	30.03	5.96	8	52.23
zero	10	8.19	2.10	10	61.98
zero	12	6.09	1.46	12	71.31
zero	20	6.09	1.24	14	85.48
zero	300	6.09	1.24	14	80.89

Table F.21: Cheapest rule passing each bar per rung in its own draw, with its fit-state count, and the parent lane’s rule re-scored on the same held-out states. Jobs bet101 = 3780164 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet201 = 3780165 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet301 = 3783811 (NVIDIA A100-PCI-E-40GB, commit b2844c607290...). Construction status is in Table F.25.

q	cheapest rule passing the primary bar (its draw)	cheapest rule passing the tight bar (its draw)	best parent-lane rule, re-scored: primary?
0	this_job/std, $m=512$, 64 states, $\rho_{\max}=0.0641$	this_job/rhow64, $m=622$, 64 states, $\rho_{\max}=0.0491$	$m=1024$, $\rho_{\max}=0.0153$, yes
16	this_job/std, $m=512$, 64 states, $\rho_{\max}=0.0879$	this_job/std, $m=1024$, 64 states, $\rho_{\max}=0.0118$	$m=2048$, $\rho_{\max}=0.0452$, yes
32	qrg304/reachable, $m=1024$, 42 states, $\rho_{\max}=0.0533$	qrg304/reachable, $m=1024$, 42 states, $\rho_{\max}=0.0533$	$m=1024$, $\rho_{\max}=0.0533$, yes
64	qrg304/reachable, $m=1024$, 25 states, $\rho_{\max}=0.0531$	qrg304/reachable, $m=1024$, 25 states, $\rho_{\max}=0.0531$	$m=1024$, $\rho_{\max}=0.0531$, yes
128	this_job/fs64, $m=2048$, 64 states, $\rho_{\max}=0.0669$	this_job/rhow64, $m=2048$, 64 states, $\rho_{\max}=0.0472$	$m=2048$, $\rho_{\max}=0.1908$, no
256	this_job/fs64, $m=2048$, 64 states, $\rho_{\max}=0.1074$	—	$m=2048$, $\rho_{\max}=0.1678$, no

Table F.22: Every quadrature rule at $q \leq 32$: NNLS fit residual beside the held-out ρ , so the anti-correlation stays visible. Bars: primary $\rho_{\max} \leq 0.116$, tight $\rho_{\max} \leq 0.06$ (source: b-eqtop/DESIGN.md §5 (byte-identical from commit 33e8bcd, 2026-09-17 01:11, through the pinned 8542c604); origin q-ridge/DESIGN.md amendment A3, commit a1822ae6 (2026-09-16); every b-eqtop job post-dates it). Continued in Table F.23.

q	fit arm	m	population	NNLS rel. fit	$\rho_{\max}$	ρ_{95}	primary	job
0	reachable	1024	qrg304:reachable	4.58e-05	0.0153	0.0088	yes	3780164
0	reachable	1024	qrg304:reachable	4.58e-05	0.0153	0.0088	yes	3780165
0	reachable	1024	qrg304:reachable	4.58e-05	0.0153	0.0088	yes	3783811
0	reachable	1521	qrg304:reachable	7.83e-06	0.0261	0.0083	yes	3780164
0	reachable	1521	qrg304:reachable	7.83e-06	0.0261	0.0083	yes	3780165
0	reachable	1521	qrg304:reachable	7.83e-06	0.0261	0.0083	yes	3783811
0	static	256	qrg304:static	6.85e-03	0.2407	0.2073	no	3780164
0	static	256	qrg304:static	6.85e-03	0.2407	0.2073	no	3780165
0	static	256	qrg304:static	6.85e-03	0.2407	0.2073	no	3783811
0	fs64	1024	this_job:reachable	4.70e-05	0.0424	0.0154	yes	3780165
0	fs64	1546	this_job:reachable	8.92e-06	0.0177	0.0091	yes	3780165
0	rhow64	609	this_job:reachable	2.24e-04	0.0686	0.0550	yes	3780165
0	rhow64	622	this_job:reachable	2.24e-04	0.0491	0.0328	yes	3780165
0	std	256	this_job:reachable	4.03e-03	0.4018	0.3674	no	3780165
0	std	512	this_job:reachable	4.07e-04	0.0641	0.0545	yes	3780165
0	std	1024	this_job:reachable	4.35e-05	0.0255	0.0087	yes	3780165
0	std	1526	this_job:reachable	9.17e-06	0.0135	0.0032	yes	3780165
0	std	1528	this_job:reachable	8.90e-06	0.0124	0.0038	yes	3780165
0	std	1530	this_job:reachable	9.07e-06	0.0128	0.0040	yes	3780165
16	reachable	1024	qrg304:reachable	1.90e-04	0.0935	0.0273	yes	3780164
16	reachable	1024	qrg304:reachable	1.90e-04	0.0935	0.0273	yes	3780165
16	reachable	1024	qrg304:reachable	1.90e-04	0.0935	0.0273	yes	3783811
16	reachable	2048	qrg304:reachable	1.92e-05	0.0452	0.0041	yes	3780164
16	reachable	2048	qrg304:reachable	1.92e-05	0.0452	0.0041	yes	3780165
16	reachable	2048	qrg304:reachable	1.92e-05	0.0452	0.0041	yes	3783811
16	static	512	qrg304:static	1.54e-03	0.1279	0.0523	no	3780164
16	static	512	qrg304:static	1.54e-03	0.1279	0.0523	no	3780165
16	static	512	qrg304:static	1.54e-03	0.1279	0.0523	no	3783811
16	fs64	1024	this_job:reachable	1.79e-04	0.0751	0.0232	yes	3780165
16	fs64	2048	this_job:reachable	1.59e-05	0.0148	0.0043	yes	3780165
16	rhow64	946	this_job:reachable	2.30e-04	0.0618	0.0174	yes	3780165
16	rhow64	972	this_job:reachable	2.33e-04	0.0603	0.0098	yes	3780165
16	std	256	this_job:reachable	3.09e-02	0.4133	0.3926	no	3780165
16	std	512	this_job:reachable	1.99e-03	0.0879	0.0723	yes	3780165
16	std	1024	this_job:reachable	2.42e-04	0.0118	0.0108	yes	3780165
16	std	2048	this_job:reachable	2.31e-05	0.0070	0.0031	yes	3780165
16	std	2556	this_job:reachable	8.89e-06	0.0021	0.0015	yes	3780165
16	std	2581	this_job:reachable	8.67e-06	0.0020	0.0014	yes	3780165
32	reachable	1024	qrg304:reachable	2.89e-04	0.0533	0.0226	yes	3780164
32	reachable	1024	qrg304:reachable	2.89e-04	0.0533	0.0226	yes	3780165
32	reachable	1024	qrg304:reachable	2.89e-04	0.0533	0.0226	yes	3783811
32	reachable	2048	qrg304:reachable	2.66e-05	0.0668	0.0120	yes	3780164
32	reachable	2048	qrg304:reachable	2.66e-05	0.0668	0.0120	yes	3780165
32	reachable	2048	qrg304:reachable	2.66e-05	0.0668	0.0120	yes	3783811
32	static	768	qrg304:static	6.27e-04	0.1482	0.0214	no	3780164
32	static	768	qrg304:static	6.27e-04	0.1482	0.0214	no	3780165
32	static	768	qrg304:static	6.27e-04	0.1482	0.0214	no	3783811
32	fs64	1024	this_job:reachable	3.80e-04	0.0696	0.0324	yes	3780165
32	fs64	2048	this_job:reachable	4.21e-05	0.0417	0.0036	yes	3780165
32	rhow64	1024	this_job:reachable	3.50e-04	0.0374	0.0308	yes	3780165
32	rhow64	1208	this_job:reachable	2.40e-04	0.0325	0.0280	yes	3780165
32	std	256	this_job:reachable	8.60e-02	0.6639	0.6101	no	3780165
32	std	512	this_job:reachable	4.14e-03	0.2339	0.1944	no	3780165
32	std	1024	this_job:reachable	3.40e-04	0.0481	0.0329	yes	3780165
32	std	2048	this_job:reachable	3.23e-05	0.0073	0.0049	yes	3780165
32	std	2628	this_job:reachable	1.23e-05	0.0058	0.0029	yes	3780165
32	std	2675	this_job:reachable	1.24e-05	0.0062	0.0021	yes	3780165

Table F.23: Every quadrature rule at $q \geq 64$ (continuation of Table F.22; same columns and bars).

q	fit arm	m	population	NNLS rel. fit	$\rho_{\max}$	ρ_{95}	primary	job
64	reachable	1024	qrg304:reachable	7.12e-04	0.0531	0.0230	yes	3780164
64	reachable	1024	qrg304:reachable	7.12e-04	0.0531	0.0230	yes	3780165
64	reachable	1024	qrg304:reachable	7.12e-04	0.0531	0.0230	yes	3783811
64	reachable	2048	qrg304:reachable	7.54e-05	0.1120	0.0172	yes	3780164
64	reachable	2048	qrg304:reachable	7.54e-05	0.1120	0.0172	yes	3780165
64	reachable	2048	qrg304:reachable	7.54e-05	0.1120	0.0172	yes	3783811
64	static	1280	qrg304:static	2.43e-04	0.1439	0.0235	no	3780164
64	static	1280	qrg304:static	2.43e-04	0.1439	0.0235	no	3780165
64	static	1280	qrg304:static	2.43e-04	0.1439	0.0235	no	3783811
64	fs64	1024	this_job:reachable	9.19e-04	0.1303	0.0332	no	3780165
64	fs64	2048	this_job:reachable	1.14e-04	0.1248	0.0032	no	3780165
64	repro64m2048s1	2048	this_job:reachable	1.26e-04	0.0459	0.0088	yes	3783811
64	repro64m2048s2	2048	this_job:reachable	1.18e-04	0.1036	0.0090	yes	3783811
64	repro64m2048s3	2048	this_job:reachable	1.27e-04	0.0134	0.0042	yes	3783811
64	repro64m2048s4	2048	this_job:reachable	1.36e-04	0.0108	0.0052	yes	3783811
64	reprovincumbentm1024s1	1024	this_job:reachable	4.85e-04	0.1420	0.0608	no	3783811
64	reprovincumbentm1024s2	1024	this_job:reachable	4.66e-04	0.1509	0.0526	no	3783811
64	reprovincumbentm1024s3	1024	this_job:reachable	6.86e-04	0.1587	0.0924	no	3783811
64	reprovincumbentm1024s4	1024	this_job:reachable	6.26e-04	0.0670	0.0309	yes	3783811
64	rhov64	1024	this_job:reachable	8.60e-04	0.1039	0.0364	yes	3780165
64	rhov64	1563	this_job:reachable	2.48e-04	0.0518	0.0238	yes	3780165
64	std	256	this_job:reachable	1.84e-01	0.7670	0.7632	no	3780165
64	std	512	this_job:reachable	1.28e-02	0.3106	0.2780	no	3780165
64	std	1024	this_job:reachable	6.97e-04	0.1220	0.0262	no	3780165
64	std	2048	this_job:reachable	6.75e-05	0.0583	0.0074	yes	3780165
64	std	3054	this_job:reachable	1.18e-05	0.0213	0.0022	yes	3780165
64	std	3128	this_job:reachable	1.11e-05	0.0129	0.0023	yes	3780165
128	reachable	1024	qrg304:reachable	1.62e-03	0.2786	0.2246	no	3780164
128	reachable	1024	qrg304:reachable	1.62e-03	0.2786	0.2246	no	3780165
128	reachable	1024	qrg304:reachable	1.62e-03	0.2786	0.2246	no	3783811
128	reachable	2048	qrg304:reachable	1.04e-04	0.1908	0.0518	no	3780164
128	reachable	2048	qrg304:reachable	1.04e-04	0.1908	0.0518	no	3780165
128	reachable	2048	qrg304:reachable	1.04e-04	0.1908	0.0518	no	3783811
128	static	2048	qrg304:static	7.16e-05	0.2194	0.0677	no	3780164
128	static	2048	qrg304:static	7.16e-05	0.2194	0.0677	no	3780165
128	static	2048	qrg304:static	7.16e-05	0.2194	0.0677	no	3783811
128	fs64	2048	this_job:reachable	3.52e-04	0.0669	0.0214	yes	3780164
128	fs64	2560	this_job:reachable	1.82e-04	0.0292	0.0111	yes	3780164
128	repro64m2048s1	2048	this_job:reachable	4.12e-04	0.0497	0.0372	yes	3783811
128	repro64m2048s2	2048	this_job:reachable	4.31e-04	0.0426	0.0298	yes	3783811
128	repro64m2048s3	2048	this_job:reachable	3.25e-04	0.0788	0.0234	yes	3783811
128	repro64m2048s4	2048	this_job:reachable	3.66e-04	0.2040	0.0506	no	3783811
128	reprovincumbentm1024s1	1024	this_job:reachable	1.62e-03	0.3523	0.3057	no	3783811
128	reprovincumbentm1024s2	1024	this_job:reachable	1.70e-03	0.2098	0.1380	no	3783811
128	reprovincumbentm1024s3	1024	this_job:reachable	1.01e-03	0.3357	0.2786	no	3783811
128	reprovincumbentm1024s4	1024	this_job:reachable	1.36e-03	0.4609	0.4217	no	3783811
128	rhov64	2048	this_job:reachable	3.54e-04	0.0472	0.0176	yes	3780164
128	rhov64	2319	this_job:reachable	2.52e-04	0.0277	0.0102	yes	3780164
128	std	1024	this_job:reachable	1.49e-03	0.2793	0.1615	no	3780164
128	std	2048	this_job:reachable	7.98e-05	0.2373	0.0520	no	3780164
128	std	2560	this_job:reachable	3.14e-05	0.2040	0.0308	no	3780164
128	std	3055	this_job:reachable	1.40e-05	0.1193	0.0197	no	3780164
128	std	3088	this_job:reachable	1.35e-05	0.1091	0.0183	yes	3780164
128	std	3128	this_job:reachable	1.33e-05	0.1487	0.0161	no	3780164
256	reachable	1024	qrg304:reachable	4.01e-02	0.5731	0.5370	no	3780164
256	reachable	1024	qrg304:reachable	4.01e-02	0.5731	0.5370	no	3780165
256	reachable	1024	qrg304:reachable	4.01e-02	0.5731	0.5370	no	3783811
256	reachable	2048	qrg304:reachable	3.08e-04	0.1678	0.0957	no	3780164
256	reachable	2048	qrg304:reachable	3.08e-04	0.1678	0.0957	no	3780165
256	reachable	2048	qrg304:reachable	3.08e-04	0.1678	0.0957	no	3783811
256	static	2048	qrg304:static	1.49e-04	0.4621	0.4275	no	3780164
256	static	2048	qrg304:static	1.49e-04	0.4621	0.4275	no	3780165
256	static	2048	qrg304:static	1.49e-04	0.4621	0.4275	no	3783811
256	fs64	2048	this_job:reachable	2.41e-03	0.1074	0.0878	yes	3780164
256	fs64	2560	this_job:reachable	9.35e-04	0.0647	0.0314	yes	3780164
256	repro64m2048s1	2048	this_job:reachable	1.34e-03	0.1820	0.1299	no	3783811
256	repro64m2048s2	2048	this_job:reachable	1.64e-03	0.1899	0.1209	no	3783811
256	repro64m2048s3	2048	this_job:reachable	1.41e-03	0.1299	0.0909	no	3783811
256	repro64m2048s4	2048	this_job:reachable	1.33e-03	0.2421	0.1413	no	3783811
256	reprovincumbentm1024s1	1024	this_job:reachable	1.21e-02	0.6743	0.4499	no	3783811
256	reprovincumbentm1024s2	1024	this_job:reachable	2.26e-02	0.5737	0.5502	no	3783811
256	reprovincumbentm1024s3	1024	this_job:reachable	3.35e-02	0.7319	0.7216	no	3783811
256	reprovincumbentm1024s4	1024	this_job:reachable	1.71e-02	0.5075	0.4916	no	3783811
256	rhov64	2048	this_job:reachable	1.44e-03	0.1299	0.0836	no	3780164
256	rhov64	2560	this_job:reachable	6.23e-04	0.1018	0.0380	yes	3780164
256	rhov64	3072	this_job:reachable	3.37e-04	0.0963	0.0229	yes	3780164
256	std	1024	this_job:reachable	1.61e-02	0.4977	0.4450	no	3780164
256	std	2048	this_job:reachable	1.75e-04	0.2958	0.1822	no	3780164
256	std	2560	this_job:reachable	5.84e-05	0.1956	0.1052	no	3780164
256	std	3072	this_job:reachable	2.32e-05	0.1774	0.0722	no	3780164
256	std	3370	this_job:reachable	1.45e-05	0.1308	0.0685	no	3780164
256	std	3410	this_job:reachable	1.43e-05	0.1761	0.0836	no	3780164

Table F.24: The EQ ladder with the cheapest rule passing the primary bar in its draw per rung, timed in one allocation (job 3780164, A100 80 GB), with its same-job dense twins, and the construction status from the four-draw replication (job 3783811): confirmed = every re-draw passes, marginal = only the listed fraction does. Errors are as measured with these rules (SHA256-identified); they are not properties of the construction above $q = 32$. Ladder rows read from the lane summary.json (final).

q	m	fit states	$\rho_{\max}$ (this draw)	construction status	EQ evolved %	EQ all %	EQ ms	dense evolved %	dense ms	EQ/dense cost
0	1024	64	0.0153	confirmed (3 of 3 re-draws)	1.8891	2.5629	59.1	1.8890	292.9	0.202
16	1024	64	0.0935	confirmed (3 of 3 re-draws)	1.4270	2.4806	80.6	—	—	—
32	1024	42	0.0533	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.7	—	—	—
64	1024	25	0.0531	marginal (2 of 6 draws pass)	1.2275	2.1489	120.6	1.0843	624.9	0.193
128	2048	64	0.0669	marginal (4 of 5 draws pass)	0.8936	1.8116	246.9	0.8930	1190.5	0.207
256	2048	64	0.1074	marginal (1 of 5 draws pass)	0.5389	0.9053	722.2	0.5194	3915.1	0.184

Table F.25: The pre-registered draw replication (job 3783811): four independent draws of candidate pool and fit-state subset at fixed q , m , fit-state count and scaling, with the number of draws passing each bar and the construction status over every draw of that construction in all three jobs. Withdrawn by these data: the parent lane’s extrapolated crossing node count; the inherited fit-state convention for $q \geq 128$ (0 of 6 draws pass); “64 states is uniformly better” (1 of 5 at $q = 256$); and the interim rule export at $q \geq 64$, superseded by rules chosen by moving up in m , a post-hoc choice the lane discloses.

q	m	fit states	four draws: $\rho_{\max}$	min / median / max	spread	primary	tight	construction	status (all draws)
64	1024	25	0.1420, 0.1509, 0.1587, 0.0670	0.0670 / 0.1464 / 0.1587	2.37	1/4	0/4	marginal (2 of 6 draws pass)	marginal (2 of 6 draws pass)
64	2048	64	0.0459, 0.1036, 0.0134, 0.0108	0.0108 / 0.0297 / 0.1036	9.57	4/4	3/4	marginal (4 of 5 draws pass)	marginal (4 of 5 draws pass)
128	1024	14	0.3523, 0.2098, 0.3357, 0.4609	0.2098 / 0.3440 / 0.4609	2.20	0/4	0/4	marginal (0 of 6 draws pass)	marginal (0 of 6 draws pass)
128	2048	64	0.0497, 0.0426, 0.0788, 0.2040	0.0426 / 0.0642 / 0.2040	4.79	3/4	2/4	marginal (4 of 5 draws pass)	marginal (4 of 5 draws pass)
256	1024	8	0.6743, 0.5737, 0.7319, 0.5075	0.5075 / 0.6240 / 0.7319	1.44	0/4	0/4	marginal (0 of 6 draws pass)	marginal (0 of 6 draws pass)
256	2048	64	0.1820, 0.1899, 0.1299, 0.2421	0.1299 / 0.1860 / 0.2421	1.86	0/4	0/4	marginal (1 of 5 draws pass)	marginal (1 of 5 draws pass)

Table F.26: Frozen-checkpoint mesh ladder (Burgers job 3711388on NVIDIA A100-PCIE-40GB, checkpoint 18f0266ae6f0...; Poisson job 3711389on NVIDIA A100 80GB PCIe, checkpoint a128e7635c31...). “efficient FOM” is the cheapest same-job full-order arm meeting the 5 % target on that mesh.

PDE	intervals	unknowns	ROM cached ms	ROM device ms	ROM complete-query ms	ROM worst vs ref %	ROM vs same-grid FOM %	efficient FOM	FOM ms	FOM/ROM	ROM meets 5%
Burgers	64	3,969	44.2	48.5	50.4	10.8570	5.5018	fom_same_ntile-2	12.9	0.265	no
Burgers	128	16,129	44.7	46.5	48.1	6.5431	2.8939	fom_same_ntile-2	14.9	0.320	no
Burgers	256	65,025	44.7	48.2	50.6	4.5546	2.5629	fom_same_ntile-2	15.7	0.325	yes
Burgers	512	261,121	45.1	47.4	53.7	3.7168	3.2836	fom_same_ntile-2	23.5	0.495	yes
Burgers	1024	1,046,529	43.6	50.1	76.8	3.8847	3.8362	fom_coarsest_ntile-4	22.2	0.443	yes
Poisson	64	3,969	2.1	2.3	3.2	6.1119	6.1282	dst	0.1	0.063	no
Poisson	128	16,129	2.1	2.2	3.2	6.1107	6.1149	dst	0.1	0.065	no
Poisson	256	65,025	2.1	2.2	3.3	6.1106	6.1116	dst	0.2	0.067	no
Poisson	512	261,121	1.9	2.3	3.8	6.1106	6.1108	dst	0.2	0.079	no
Poisson	1024	1,046,529	2.0	3.0	6.6	6.1106	6.1106	dst	0.3	0.112	no

Table F.27: Reflective 2D wave, one job per mesh (64^2 : job 3780447 (NVIDIA A100 80GB PCIe, commit 0bb3cc86); 256^2 : job 3783805 (NVIDIA A100-PCI-E-40GB, commit 2655bb01); 1024^2 : job 3780450 (NVIDIA A100 80GB PCIe, commit 0bb3cc86)); costs comparable only within a mesh. The top rung is the learned bank evolved by exact modal propagation with no head.

mesh	arm	q / k'	worst energy-state %	$t=0$ %	device ms	complete-query ms
64^2	head_q0	0	11.2055	5.5314	184.535	185.575
64^2	trained_nested40	8	6.8128	4.7478	200.232	201.072
64^2	nested_q8	8	5.9379	3.7715	199.289	200.375
64^2	nested_q16	16	5.0053	2.5994	1977.402	1978.383
64^2	nested_q32	32	4.9161	2.0345	2500.060	2501.038
64^2	linear_bank64	$64 (= R)$	4.9323	2.0345	1.168	1.986
64^2	pod_k40	$k'=40$	4.8825	4.2685	1.190	1.989
64^2	pod_k64	$k'=64$	1.4893	1.4709	1.202	2.018
64^2	pod_k128	$k'=128$	0.8668	0.4932	1.252	2.052
64^2	dst	—	0.0000	0.0000	3.391	4.286
64^2	rk4_fom	—	0.0612	0.0000	5.511	6.865
64^2	cg_1e-06	—	0.5745	0.0000	109.836	111.350
256^2	head_q0	0	11.3314	5.5585	183.516	198.619
256^2	trained_nested40	8	6.8764	4.7840	197.587	213.401
256^2	nested_q8	8	6.0627	3.8045	197.389	212.596
256^2	nested_q16	16	5.1661	2.6270	1989.463	2004.700
256^2	nested_q32	32	5.1038	2.0731	2520.177	2536.035
256^2	linear_bank64	$64 (= R)$	5.1208	2.0731	1.428	16.659
256^2	pod_k40	$k'=40$	4.9963	4.2844	1.365	16.639
256^2	pod_k64	$k'=64$	1.5018	1.4817	1.399	16.657
256^2	pod_k128	$k'=128$	0.9873	0.5122	1.482	16.773
256^2	dst	—	0.0000	0.0000	4.356	19.703
256^2	rk4_fom	—	0.0016	0.0000	24.346	40.083
256^2	cg_1e-06	—	0.6018	0.0000	142.511	158.115
1024^2	head_q0	0	11.3388	5.5602	192.462	451.994
1024^2	trained_nested40	8	6.8789	4.7863	207.832	467.228
1024^2	nested_q8	8	6.0692	3.8066	206.632	468.808
1024^2	nested_q16	16	5.1763	2.6287	2003.425	2264.489
1024^2	nested_q32	32	5.1164	2.0755	2529.445	2789.801
1024^2	linear_bank64	$64 (= R)$	5.1335	2.0755	4.562	265.785
1024^2	pod_k40	$k'=40$	5.0030	4.2854	4.183	263.883
1024^2	pod_k64	$k'=64$	1.5010	1.4824	4.232	264.860
1024^2	pod_k128	$k'=128$	0.9888	0.5135	5.879	265.409
1024^2	dst	—	0.0000	0.0000	17.103	277.324
1024^2	rk4_fom	—	0.0000	0.0000	958.007	1174.433
1024^2	cg_1e-06	—	0.6079	0.0000	1043.368	1278.837

Table F.28: Poisson head capacity on the frozen $R = 512$ bank (job 3783883, NVIDIA A100-PCI-E-40GB): width helps, depth alone does not, and no arm reaches a $2\times$ ratio to the floor. The re-run of the primary recipe reproduces it exactly.

head	best-found dev. %	best-found / floor	solved 1024^2 %	complete-query ms
K32_w128_L2	3.1212	4.18	3.1146	12.93
K32_w256_L2	2.2652	3.04	2.2600	12.84
K32_w128_L3	3.0625	4.11	3.0558	13.23
K32_w256_L3	2.3614	3.17	2.3571	13.39
K64_w128_L2	2.5625	3.44	2.5587	13.72
K64_w256_L3	2.0553	2.76	2.0549	14.12
K32_w128_L2_x3	2.6428	3.54	2.6360	12.97

Table F.29: Three training seeds on the development cohort (lane b-seeds, jobs `seed1` = 3783776 (NVIDIA A100 80GB PCIe); `seed2` = 3783777 (NVIDIA A100 80GB PCIe); `seed3` = 3783778 (NVIDIA A100-PCIE-40GB)). Top: per rung of the dense $M = 4(K + q)$ ladder, worst error over the six development cases as mean $\pm$ sample standard deviation over the 3 seed checkpoints, beside the incumbent checkpoint re-run in each seed’s own job (its value is identical in all three jobs). Bottom: the per-seed ladder verdicts. Costs are never averaged across jobs. This is the development cohort; the sealed cohort (Table F.30) is the headline.

q	M	evolved %, seeds	evolved %, incumbent	all-times %, seeds	all-times %, incumbent	best-found %, seeds	converged
0	64	2.0002 $\pm$ 0.4986	1.8890	2.6659 $\pm$ 0.1715	2.5629	2.6552 $\pm$ 0.1781	3 of 3
16	128	1.5988 $\pm$ 0.0995	1.3985	2.5842 $\pm$ 0.1534	2.4806	2.5731 $\pm$ 0.1576	3 of 3
32	192	1.3727 $\pm$ 0.1314	1.2336	2.4971 $\pm$ 0.1344	2.3534	2.4928 $\pm$ 0.1359	3 of 3
64	320	1.2130 $\pm$ 0.0580	1.0843	2.2759 $\pm$ 0.0790	2.1489	2.2632 $\pm$ 0.0946	3 of 3
128	576	0.9562 $\pm$ 0.0264	0.8930	1.8806 $\pm$ 0.0704	1.8116	1.8793 $\pm$ 0.0704	3 of 3
256	1088	0.4987 $\pm$ 0.1164	0.5194	0.9538 $\pm$ 0.0381	0.9053	0.9431 $\pm$ 0.0398	2 of 3

seed	job	GPU	monotone (evolved)	monotone (all-times)	every rung converged	error span	cost span	knob bar
seed1	3783776	NVIDIA A100 80GB PCIe	yes	yes	yes	4.07 $\times$	16.26 $\times$	yes
seed2	3783777	NVIDIA A100 80GB PCIe	yes	yes	yes	4.01 $\times$	10.34 $\times$	yes
seed3	3783778	NVIDIA A100-PCIE-40GB	yes	yes	no	3.93 $\times$	20.97 $\times$	no

Table F.30: The sealed cohort (lane b-seeds, job 3804465, NVIDIA A100-PCIE-40GB), opened once after every choice was frozen. Top: per rung of the dense $M = 4(K + q)$ ladder, the incumbent’s sealed worst evolved error and device time, the seed mean $\pm$ sample standard deviation on the sealed and the development cohort, the lane’s sealed-over-development ratio of seed means (raw and difficulty-normalised) and the converged count over the 4 checkpoints. Bottom: per-checkpoint verdicts. The incumbent’s $q = 0$ rung is a single sealed case converged to a wrong branch (gradient exit, zero budget exits); the pre-registered ratio criterion fails at $q = 0$ for the incumbent alone (5.35) and the convergence criterion fails on `seed2` at $q = 64$ (3 budget exits).

q	M	incumbent sealed %	ms	seeds sealed %	seeds development %	sealed / dev	normalised	converged
0	64	10.1120	375.1	2.8023 $\pm$ 0.6357	2.0002 $\pm$ 0.4986	1.40	1.41	4 of 4
16	128	1.4617	480.8	1.7305 $\pm$ 0.1297	1.5988 $\pm$ 0.0995	1.08	1.09	4 of 4
32	192	1.4458	540.5	1.5044 $\pm$ 0.1422	1.3727 $\pm$ 0.1314	1.10	1.14	4 of 4
64	320	1.3252	754.5	1.3528 $\pm$ 0.1099	1.2130 $\pm$ 0.0580	1.12	1.13	3 of 4
128	576	1.0964	1460.4	1.0461 $\pm$ 0.0619	0.9562 $\pm$ 0.0264	1.09	1.17	4 of 4
256	1088	0.6789	6725.5	0.6188 $\pm$ 0.0345	0.4987 $\pm$ 0.1164	1.24	1.20	4 of 4

checkpoint	job	monotone (evolved)	monotone (all-times)	every rung converged	error span	cost span	knob bar
incumbent	3804465	yes	yes	yes	14.89 $\times$	17.93 $\times$	yes
seed1	3804465	yes	yes	yes	3.52 $\times$	13.25 $\times$	yes
seed2	3804465	yes	yes	no	4.81 $\times$	13.17 $\times$	no
seed3	3804465	yes	yes	yes	5.22 $\times$	14.62 $\times$	yes

Table F.31: The operators’ own inference-time knob: each validation-selected checkpoint evaluated at coarser grids and prolonged back (job 3787189, NVIDIA A100 80GB PCIe), with the interpolation floor (a perfect operator’s error at that rung) and the pre-registered gates ($\geq 1.5\times$ own speed at $\leq 2\times$ own error). Speedups are within one model’s own curve, same job.

operator	rung	worst evolved %	interp. floor %	device ms	speed vs 256	error vs 256	both gates
fno-large	256	6.38	0.00	7.35	—	—	ref.
fno-large	128	7.30	0.23	3.46	2.12	1.14	yes
fno-large	64	9.82	0.92	3.44	2.14	1.54	yes
fno-large	32	13.85	4.02	3.33	2.21	2.17	no
tsol-refine	256	9.32	0.00	11.20	—	—	ref.
tsol-refine	128	15.29	0.23	8.15	1.37	1.64	no
tsol-refine	64	35.18	0.92	8.09	1.38	3.78	no
tsol-refine	32	75.13	4.02	8.04	1.39	8.06	no
unet-refine	256	7.52	0.00	5.90	—	—	ref.
unet-refine	128	38.13	0.23	2.35	2.51	5.07	no
unet-refine	64	60.18	0.92	2.31	2.55	8.01	no
unet-refine	32	67.65	4.02	2.30	2.56	9.00	no

Table F.32: One-variable controls on the validation-selected capacities (job 3783831): each repeats a screen arm under the identical protocol and budget with exactly one variable changed, and is never eligible for capacity selection. “vs ROM” is the matched eight-case worst against the ROM’s, from job 3702709; at equal wall clock a float64 twin is also a fewer-epochs twin.

control	twin	one variable changed	epochs vs twin	control worst %	twin worst %	control vs ROM	twin vs ROM
ctrl-medium-f64	unet-medium	network dtype float64	500 vs 1963 (0.25 $\times$)	1.6721	1.5189	below (0.1950 pp)	below
ctrl-medium-seed2	unet-medium	training seed	1969 vs 1963 (1.00 $\times$)	1.5479	1.5189	below (0.3192 pp)	below
ctrl-tsol-small-f64	tsol-small	network dtype float64	682 vs 1628 (0.42 $\times$)	3.1329	2.5273	above (1.2658 pp)	above

Table F.33: U-Net against FNO on the Poisson operator-screen dataset (physical reference). The U-Net arms hit the epoch cap with recent best checkpoints; the FNO early-stopped.

arm	family	params	median %	worst %	job
fno-large	FNO	17,876,673	2.03	29.37	3702464
fno-medium	FNO	5,779,729	2.35	28.28	3702464
fno-small	FNO	1,192,801	2.64	32.47	3702464
unet-large	U-Net	17,461,393	1.95	8.31	3780625
unet-medium	U-Net	7,763,041	1.68	6.02	3780625
unet-small	U-Net	4,368,073	2.40	12.80	3780625

Table F.34: Speed at bit-level parity (jobs 3745655 (spd01), 3745656 (fine01), 3745913 (comp01)): fastest arm whose fields agree with the incumbent to 10^{-12} with identical iteration counts and exit reasons. The pre-registered $2\times$ target fails at every mesh.

attempt	intervals	arm	incumbent ms	arm ms	GPU speedup	incl. host	parity	$2\times$ target
spd01	256	L4	46.856	30.824	1.520x	1.478x	2.565e-13	FAIL
fine01	512	L4	46.928	31.106	1.509x	1.454x	5.053e-13	FAIL
fine01	1024	L4	49.283	33.544	1.469x	1.310x	6.951e-13	FAIL
comp01	256	C1	46.627	30.723	1.518x	1.482x	2.562e-13	FAIL
comp01	1024	C1	49.804	33.282	1.496x	1.408x	7.129e-13	FAIL

Table F.35: Training study on the Burgers head (jobs 3745912 (training), 3749074 (evaluation)): data density, objective, latent dimension and a joint bank arm, each evaluated through the unchanged head-ablation machinery. 0 of 26 arms pass the pre-registered success criterion. The like-for-like retrain does not reproduce the incumbent, so the density verdict is measured below baseline.

arm (EQ query)	K	bank floor %	best-found %	solved all %	solved evolved %	device ms	conv.
incumbent (4608 traj., $K=16$)	16	0.3918	2.5447	2.5629	1.9002	48.7	yes
128 traj., $K=16$	16	0.3918	12.6496	12.6496	7.2529	45.4	yes
512 traj.	16	0.3918	6.8202	6.8204	3.7153	39.3	yes
2048 traj.	16	0.3918	3.5570	3.5574	1.8752	46.6	yes
4608 traj. (like-for-like retrain)	16	0.3918	3.9616	3.9710	1.9522	59.1	yes
128 traj., $K=32$	32	0.3918	8.8489	8.8496	6.2351	80.7	yes
2048 traj., $K=32$	32	0.3918	3.1132	3.1137	2.4882	54.5	yes
128 traj., weak-residual term	16	0.3918	14.2164	14.2165	7.5935	43.9	yes
128 traj., trajectory term	16	0.3918	12.8237	12.8238	7.8543	42.8	yes
128 traj., code-smoothness term	16	0.3918	13.6497	13.6505	13.4135	36.8	yes
selected: 2048 traj., $K=32$, weak term	32	0.3918	2.8289	3.1275	3.1275	55.1	yes
joint bank+head, $R=512$	32	2.8517	5.0383	5.0471	2.4399	46.3	yes

Table F.36: Navier–Stokes 2D, phase-2 gates in four settings: the $K = 16$ head on the full-rank $R = 256$ bank, the $K = 32$ head on the full-rank $R = 512$ bank, the same $K = 16$ recipe on a family of intrinsic dimension 8 instead of 14 (3 modes; job 3808498, A100 (pax105)), and the $K = 16$ recipe at $4\times$ the training data (2048 trajectories: 512 gated plus 1536 extra, same family; last row, job 3808495, A100 (pax106)) (jobs 3787319 ($K=16$, $R=256$), 3787320 ($K=32$, $R=512$), 3808498 ($K=16$, $R=256$, family dimension 8), 3808495 ($K=16$, $R=256$, 2048 trajectories); full-order solver and dataset certified in job 3780151). Every gate passes except the held-out oracle, whose pre-registered bar was a $2.0\times$ margin over POD at matched dimension (ns2d/DESIGN.md H-ORACLE gate: oracle median $\leq 1/2 \times$ POD-K held-out, i.e. ratio ≥ 2.0 ; pre-registered at commit 1281ff73 (2026-09-17 00:55), before jobs 3783796/3787319/3787320); the ratio is 1.15–1.46 across the 4 settings. No confirmatory ladder on a passing head was run; the exploratory phase-3 ladder after this failed gate is in Table G.2. With $4\times$ the data the training reconstruction rises from 5.0345 to 12.3875 % and the held-out/training ratio falls from 4.0 to 1.3. These are mixed-normalisation ratios: training and held-out errors use different denominators, so the change does not establish a shift from generalisation to capacity limits. The held-out POD/oracle ratios remain comparable (0.52 at $t = 0$, 1.39 on evolved times). The $4\times$ -data arm’s B-DATA gate passed by the pre-registered value path (a hash mismatch on its node, at 10^{-16} on the archived trajectories). On the lower-dimensional family every error falls $3.3\text{--}4.4\times$ (POD-16 3.3, oracle 3.8, bank floor 4.4) while the ratio stays at 1.39 (1.83 at $t = 0$, 1.44 on evolved times) and the held-out/training ratio at 3.8 against 4.0. Oracle values are upper bounds: 12 ($K=16$), 18 ($K=32$), 4 (family dimension 8) and 6 (2048 trajectories) held-out fits at 256^2 hit the LM budget, which biases the ratio downward, so the result establishes failure to clear the gate with the available fits, not a proof that the globally best manifold fit cannot clear it.

mesh	bank rank	B-ORTH	bank worst %	POD- R worst %	B-FLOOR	oracle median %	POD- K median %	POD- K / oracle	H-ORACLE (≥ 2.0)
64^2	256	yes	14.0487	13.0669	yes	20.3442	24.1747	1.19	no
128^2	256	yes	14.0932	13.0156	yes	20.2574	24.0400	1.19	no
256^2	256	yes	14.0838	13.0322	yes	20.2417	24.0219	1.19	no
64^2 ($K=32$)	512	yes	7.4103	6.0761	yes	12.1757	14.0697	1.16	no
128^2 ($K=32$)	512	yes	7.5545	5.9861	yes	12.1115	13.9568	1.15	no
256^2 ($K=32$)	512	yes	7.5643	5.9983	yes	12.0948	13.9455	1.15	no
256^2 (family dim. 8)	256	yes	6.4018	6.2295	yes	5.2823	7.3651	1.39	no
256^2 (2048 traj.)	256	yes	13.8371	11.5733	yes	16.6598	24.2492	1.46	no

Table F.37: L-shaped Poisson, solve layer at $M = 257$ (jobs 3784662, 3784663, 3789568, one per mesh): the non-dominated set on (complete-query ms, worst same-grid error) at each mesh, over every reduced and full-order subject. The sparse direct solve is exact to round-off. The 512^2 job’s first attempt (Table E.2) failed a residual gate that is unreachable in float64 at that mesh, where the achievable residual floor grows like N^2 ; the gate now takes the larger of the two bounds, which changed no mesh already run, and the re-run differs from that attempt only by the gate.

mesh	subject	family	worst same-grid %	complete-query ms
64^2	sparse direct (SuperLU)	fom	0.0000	1.282
128^2	POD $k'=16$	pod	34.6279	2.086
128^2	POD $k'=32$	pod	19.0053	2.233
128^2	POD $k'=64$	pod	7.7013	2.294
128^2	sparse direct (SuperLU)	fom	0.0000	2.475
256^2	POD $k'=16$	pod	34.5764	2.240
256^2	POD $k'=32$	pod	18.9497	2.341
256^2	POD $k'=64$	pod	7.6645	2.483
256^2	head $q=0$ (head_smooth_R256_K16)	neural	3.4928	2.848
256^2	POD $k'=128$	pod	2.4575	2.850
256^2	head $q=64$ (head_sdf_R512_K16)	neural+linear	2.1305	3.028
256^2	sparse direct (SuperLU)	fom	0.0000	8.386
512^2	POD $k'=16$	pod	34.5639	3.217
512^2	POD $k'=32$	pod	18.9358	3.390
512^2	POD $k'=64$	pod	7.6554	3.512
512^2	POD $k'=128$	pod	2.4511	4.031
512^2	head $q=128$ (head_sdf_R512_K16)	neural+linear	2.1977	4.748
512^2	head $q=64$ (head_sdf_R512_K16)	neural+linear	2.1233	4.801
512^2	CG 10^{-2}	fom	0.3845	29.127
512^2	sparse direct (SuperLU)	fom	0.0000	36.505

Table F.38: L-shaped Poisson, free rung at $M = 1024$ (job 3784910). **These costs are not comparable with Table F.37:** the dense projection is charged inside every reduced query here. Non-dominated set only.

subject	family	worst same-grid %	complete-query ms
POD $k'=16$	pod	34.5764	2.451
POD $k'=32$	pod	18.9497	2.458
free rung $q=R$ (head_sdf_R512_K32)	free	0.7791	2.582
free rung $q=R$ (head_sdf_R512_K16)	free	0.7791	2.583
sparse direct (SuperLU)	fom	0.0000	8.351

Table F.39: L-shaped Poisson (job 3783786, NVIDIA A100 80GB PCIe): bank floors by boundary factor and rank, and the head layer at 256^2 . The solve layer is in Table F.37 (64^2 – 512^2 , one job per mesh).

bank	floor, dev. 256^2 %	floor, dev. 512^2 %	floor, common 256^2 %
smooth_R256	1.5113	1.5057	3.1933
smooth_R512	0.7123	0.7093	1.9344
sdf_R256	1.5697	1.5626	3.2101
sdf_R512	0.7766	0.7698	1.6155
enrich_R512	0.6676	0.6650	1.7028
smooth_ff128s2_R512	2.5438	2.5388	3.8411

head arm	best-found %	weak solve % (untimed)	best-found / bank floor
head_enrich_R512_K16	3.8508	3.8510	5.77
head_sdf_R256_K16	3.5177	3.5181	2.24
head_sdf_R512_K16	3.8607	3.8608	4.97
head_sdf_R512_K32	3.1802	3.1814	4.09
head_smooth_R256_K16	3.4919	3.4928	2.31
head_smooth_R512_K16	3.5924	3.5931	5.04
head_smooth_ff128s2_R512_K16	6.7961	6.7970	2.67

Table G.1: Navier–Stokes head-only data scaling (job 3808493): held-out oracle median per recipe and training-set size, POD-16 at matched dimension, the held-out/train ratio, the selected training step, and the pre-registered log-log slope with its verdict (band -0.25 to -0.10). The held-out/train column is a historical mixed-normalisation ratio, not a generalisation-gap estimate.

recipe	training trajectories	held-out oracle median %	POD-16 median %	POD / oracle	held-out / train	selected step
plain recipe	128	27.4938	23.5757	0.86	16.5	100000
plain recipe	256	20.6889	22.9610	1.11	6.9	100000
plain recipe	512	19.8279	22.7726	1.15	4.1	100000
plain recipe (all n)	—	slope -0.236	—	—	—	ambiguous
weight decay + early stopping	128	26.1095	23.5757	0.90	4.6	5000
weight decay + early stopping	256	21.0204	22.9610	1.09	3.2	10000
weight decay + early stopping	512	19.3892	22.7726	1.17	3.1	30000
weight decay + early stopping (all n)	—	slope -0.215	—	—	—	ambiguous
weight decay, head $4\times$ smaller	128	24.6583	23.5757	0.96	3.4	95000
weight decay, head $4\times$ smaller	256	21.3431	22.9610	1.08	1.8	10000
weight decay, head $4\times$ smaller	512	20.0874	22.7726	1.13	1.6	45000
weight decay, head $4\times$ smaller (all n)	—	slope -0.148	—	—	—	ambiguous

Table G.2: Navier–Stokes, exploratory after a failed phase-2 gate (job 3808502): the correction ladder on the $K = 32$, $R = 512$ manifold at fixed $M = 2176$, its three-layer decomposition (manifold layer and bank floor as the median over cases of the worst evolved time, and the solved/manifold ratio on that matched statistic), the share of head-residual energy the q directions capture, and same-job POD-LSPG at the matched dimension $k' = K + q$; costs are median ms inside this job. “Non-dom.” marks membership of the non-dominated set on (ms, worst evolved error) over every subject in the job.

q	$k' = K + q$	neural worst %	median %	ms	budget exits	manifold layer %	bank floor %	solved / manifold	energy %	POD- k' worst %	median %	ms	neural non-dom.	POD non-dom.
0	32	68.7654	41.3922	13387.5	0	16.4787	1.7434	2.51	0	54.6331	33.4692	212.1	—	yes
32	64	70.7701	36.8058	16979.1	0	15.9678	1.7434	2.31	26	35.5092	19.0979	431.0	—	—
64	96	61.5127	32.7278	21289.9	0	14.8960	1.7434	2.20	44	27.7285	13.2380	832.7	—	—
128	160	56.5456	25.6655	29011.5	0	13.4634	1.7434	1.91	68	17.1958	8.0974	2398.2	—	—
256	288	33.0536	13.8903	46267.9	0	6.8226	1.7434	2.04	92	9.5434	3.9876	9963.9	—	—
512	544	6.0140	2.4377	52592.3	0	1.7434	1.7434	1.40	100	4.1664	1.4871	53902.5	—	—

Table G.3: Navier–Stokes, **exploratory after a failed phase-2 gate** (job 3808502): the full-order Newton-tolerance ladder timed in the same job as Table G.2; the last row is the converged reference.

Newton tol.	worst evolved %	median %	ms	non-dominated
0.03	106.4413	95.1088	46.7	yes
0.01	106.4413	95.1088	47.2	—
0.003	25.5895	7.3378	263.2	yes
0.001	0.0041	0.0030	421.3	yes
0.0001	0.0033	0.0006	425.6	yes
1e-06	0.0033	0.0007	477.8	yes
1e-11	0.0000	0.0000	1177.7	yes

Table G.4: Literal and amended linear-PDE criteria. Wave rows ask whether the linear endpoint is at least as accurate as every head rung; the amended test permits the listed tie band. Poisson rows ask whether the collapse claim is falsified; the observed quantity is the full ladder’s maximum/minimum query-cost ratio. Amendments followed earlier observations, as described in the text; “yes” has the meaning of the criterion in its row.

PDE	mesh	criterion	literal	amended	observed quantity
wave	64^2	top-rung accuracy (D1)	no	yes	0.019 pp tie band
wave	256^2	top-rung accuracy (D1)	no	yes	0.020 pp tie band
wave	1024^2	top-rung accuracy (D1)	no	yes	0.020 pp tie band
Poisson	256^2	falsification fires	yes	no	3.17 cost span
Poisson	1024^2	falsification fires	no	no	1.55 cost span

Table G.5: Secondary comparison against the previous submission’s comparator, unpreconditioned CG, beside the competitive one, all inside one job per mesh: the reduced rungs (and the linear top rung and POD) with their complete-query ms, the CG solve at the tolerance shown, the direct solve (DST on the square, SuperLU on the L-shape) and the ratios. Poisson from the p-linear jobs, L-shape from the lshape jobs (CG at its own tolerances, the tightest shown; the IC(0)-PCG arm runs on the CPU). Burgers has no CG comparator (its full-order solver is Newton). Heat is omitted: its committed CG record is per-invocation raw timing of an earlier decoder family and is not re-aggregated here.

mesh	subject	error %	ms	CG tol.	CG ms	CG / ROM	direct	direct ms	direct / ROM	IC(0)-PCG (CPU) ms	PCG / ROM
256 ² (job 3780692)	q=0	3.1567	9.214	10 ⁻⁶	38.05	4.1	DST	2.526	0.27	—	—
256 ² (job 3780692)	q=32	2.4699	9.537	10 ⁻⁶	38.05	4.0	DST	2.526	0.26	—	—
256 ² (job 3780692)	q=64	2.0808	9.819	10 ⁻⁶	38.05	3.9	DST	2.526	0.26	—	—
256 ² (job 3780692)	q=128	1.5497	10.219	10 ⁻⁶	38.05	3.7	DST	2.526	0.25	—	—
256 ² (job 3780692)	q=256	0.9689	9.658	10 ⁻⁶	38.05	3.9	DST	2.526	0.26	—	—
256 ² (job 3780692)	linear top rung (q=R)	0.7459	3.221	10 ⁻⁶	38.05	11.8	DST	2.526	0.78	—	—
256 ² (job 3780692)	POD k'=512	0.1855	10.002	10 ⁻⁶	38.05	3.8	DST	2.526	0.25	—	—
1024 ² (job 3783813)	q=0	3.1495	6.394	10 ⁻⁶	103.64	16.2	DST	3.248	0.51	—	—
1024 ² (job 3783813)	q=32	2.4641	6.592	10 ⁻⁶	103.64	15.7	DST	3.248	0.49	—	—
1024 ² (job 3783813)	q=64	2.0760	6.745	10 ⁻⁶	103.64	15.4	DST	3.248	0.48	—	—
1024 ² (job 3783813)	q=128	1.5459	6.875	10 ⁻⁶	103.64	15.1	DST	3.248	0.47	—	—
1024 ² (job 3783813)	q=256	0.9648	6.849	10 ⁻⁶	103.64	15.1	DST	3.248	0.47	—	—
1024 ² (job 3783813)	linear top rung (q=R)	0.7421	4.424	10 ⁻⁶	103.64	23.4	DST	3.248	0.73	—	—
1024 ² (job 3783813)	POD k'=512	0.1838	6.972	10 ⁻⁶	103.64	14.9	DST	3.248	0.47	—	—
64 ² (job 3784662)	head q=0 (head_sdf_R512_K16)	4.0451	2.693	10 ⁻¹⁰	8.25	3.1	SuperLU	1.282	0.48	10.22	3.8
64 ² (job 3784662)	head q=32 (head_sdf_R512_K16)	2.7644	2.832	10 ⁻¹⁰	8.25	2.9	SuperLU	1.282	0.45	10.22	3.6
64 ² (job 3784662)	head q=64 (head_sdf_R512_K16)	2.3004	2.846	10 ⁻¹⁰	8.25	2.9	SuperLU	1.282	0.45	10.22	3.6
64 ² (job 3784662)	head q=128 (head_sdf_R512_K16)	2.3580	2.796	10 ⁻¹⁰	8.25	2.9	SuperLU	1.282	0.46	10.22	3.7
64 ² (job 3784662)	POD k'=128	2.5913	2.711	10 ⁻¹⁰	8.25	3.0	SuperLU	1.282	0.47	10.22	3.8
128 ² (job 3784662)	head q=0 (head_sdf_R512_K16)	3.8965	2.673	10 ⁻¹⁰	14.96	5.6	SuperLU	2.475	0.93	66.51	24.9
128 ² (job 3784662)	head q=32 (head_sdf_R512_K16)	2.6261	2.789	10 ⁻¹⁰	14.96	5.4	SuperLU	2.475	0.89	66.51	23.8
128 ² (job 3784662)	head q=64 (head_sdf_R512_K16)	2.1643	2.806	10 ⁻¹⁰	14.96	5.3	SuperLU	2.475	0.88	66.51	23.7
128 ² (job 3784662)	head q=128 (head_sdf_R512_K16)	2.2217	2.868	10 ⁻¹⁰	14.96	5.2	SuperLU	2.475	0.86	66.51	23.2
128 ² (job 3784662)	POD k'=128	2.4833	2.764	10 ⁻¹⁰	14.96	5.4	SuperLU	2.475	0.90	66.51	24.1
256 ² (job 3784663)	head q=0 (head_sdf_R512_K16)	3.8608	2.879	10 ⁻¹⁰	28.08	9.8	SuperLU	8.386	2.91	484.34	168.2
256 ² (job 3784663)	head q=32 (head_sdf_R512_K16)	2.5950	2.963	10 ⁻¹⁰	28.08	9.5	SuperLU	8.386	2.83	484.34	163.5
256 ² (job 3784663)	head q=64 (head_sdf_R512_K16)	2.1305	3.028	10 ⁻¹⁰	28.08	9.3	SuperLU	8.386	2.77	484.34	159.9
256 ² (job 3784663)	head q=128 (head_sdf_R512_K16)	2.2055	3.042	10 ⁻¹⁰	28.08	9.2	SuperLU	8.386	2.76	484.34	159.2
256 ² (job 3784663)	POD k'=128	2.4575	2.850	10 ⁻¹⁰	28.08	9.9	SuperLU	8.386	2.94	484.34	169.9
512 ² (job 3789568)	head q=0 (head_sdf_R512_K16)	3.8521	4.716	10 ⁻¹⁰	70.12	14.9	SuperLU	36.505	7.74	4672.61	990.8
512 ² (job 3789568)	head q=32 (head_sdf_R512_K16)	2.5875	4.812	10 ⁻¹⁰	70.12	14.6	SuperLU	36.505	7.59	4672.61	971.1
512 ² (job 3789568)	head q=64 (head_sdf_R512_K16)	2.1233	4.801	10 ⁻¹⁰	70.12	14.6	SuperLU	36.505	7.60	4672.61	973.3
512 ² (job 3789568)	head q=128 (head_sdf_R512_K16)	2.1977	4.748	10 ⁻¹⁰	70.12	14.8	SuperLU	36.505	7.69	4672.61	984.1
512 ² (job 3789568)	POD k'=128	2.4511	4.031	10 ⁻¹⁰	70.12	17.4	SuperLU	36.505	9.06	4672.61	1159.2

Table G.6: Burgers at ten times lower viscosity, panel job 3817807 (NVIDIA A100-PCIE-40GB): the neural ladder at fixed $M = 1088$, the $M = 256$ control and the (256, 2176) subject. **F4 applies:** the mesh is under-resolved (the converged discrete operator is 19.3–20.9 % from the 4096^2 reference), so “vs ref” is the discretisation error and every comparison is reduced-versus-reduced on the same discrete system. Costs are median device ms inside this job; “non-dom.” against all subjects and against reduced subjects only.

subject	worst evolved %	all times %	vs ref %	best-found %	device ms	converged	budget exits	non-dom. (all)	non-dom. (reduced only)
$q = 0, M = 1088$	9.0500	9.0500	17.98	11.3934	557.4	yes	0	no	no
$q = 16, M = 1088$	8.7562	8.7562	18.03	11.3462	681.4	yes	0	no	no
$q = 32, M = 1088$	8.6360	8.6360	18.08	11.2022	772.2	yes	0	no	no
$q = 64, M = 1088$	8.3165	8.3165	18.36	10.9966	913.2	yes	0	no	yes
$q = 128, M = 1088$	7.6276	7.6276	18.70	10.6478	1220.7	yes	0	no	yes
$q = 256, M = 1088$	6.6718	6.6718	19.44	10.0347	1810.3	yes	0	no	yes
$q = 0, M = 256$	8.3933	8.3933	18.00	11.3934	300.3	yes	0	no	yes
$q = 16, M = 256$	8.4298	8.4298	17.90	11.3462	379.8	yes	0	no	no
$q = 32, M = 256$	8.7191	8.7191	17.75	11.2022	422.7	yes	0	no	no
$q = 64, M = 256$	8.9538	8.9538	18.14	10.9966	500.6	yes	0	no	no
$q = 128, M = 256$	10.2614	10.2614	19.15	10.6478	658.6	yes	0	no	no
$q = 256, M = 2176$	6.1869	6.1869	18.94	10.0347	2864.2	yes	0	no	yes

Table G.7: The same job as Table G.6: POD-LSPG at matched dimension, the free $R = 512$ bank and the full-order tolerance/step grid. **F4 applies** (under-resolved mesh); the non-dominated set against all subjects is ntle-2_dt01, ntle-3_dt005, dense_tight, fft_tight.

subject	worst evolved %	all times %	vs ref %	best-found %	device ms	converged	budget exits	non-dom. (all)	non-dom. (reduced only)
POD-LSPG $k' = 16, M = 64$	57.9837	67.3429	60.09	67.3429	46.5	yes	0	no	yes
POD-LSPG $k' = 32, M = 128$	45.6825	54.4540	47.84	54.4539	80.2	yes	0	no	yes
POD-LSPG $k' = 64, M = 256$	26.2116	35.1340	28.35	35.1339	150.1	yes	0	no	yes
POD-LSPG $k' = 128, M = 512$	16.4288	23.2032	23.21	23.2019	331.2	yes	0	no	no
POD-LSPG $k' = 256, M = 1024$	11.2540	15.2899	21.96	15.2670	881.2	yes	0	no	no
POD-LSPG $k' = 512, M = 2048$	17.7782	17.7782	24.50	10.2797	2735.3	yes	0	no	no
free bank $R = 512, M = 1024$	7.7940	7.7940	20.36	7.4455	3174.6	yes	0	no	no
ntle-2_dt01	3.8950	3.8950	19.33	—	14.0	—	—	yes	no
ntle-2_dt005	5.2542	5.2542	20.59	—	17.8	—	—	no	no
ntle-3_dt01	3.2798	3.2798	19.72	—	63.3	—	—	no	no
ntle-3_dt005	0.1971	0.1971	20.86	—	54.9	—	—	yes	no
ntle-4_dt01	3.2798	3.2798	19.77	—	88.7	—	—	no	no
ntle-4_dt005	0.0571	0.0571	20.81	—	123.0	—	—	no	no
dense_tight	0.0000	0.0000	20.82	—	112.3	—	—	yes	no
fft_tight	0.0000	0.0000	20.82	—	171.2	—	—	yes	no

H CORRECTED POISSON REPRESENTATION DIAGNOSTIC

The nonzero-correction representation fits originally reported by the Poisson lane remain retracted under amendment A10. A separate rerun on the complete original development cohort now confirms the corrected query-metric projection (job 3996320, scientific source 5a8d7bc8dea9, retained evidence pinned at aa9b55ee7e17). All 240 case/rank fits meet the recorded stationarity criterion; the independent NumPy field and derivative audit passes 1595 checks. These are untimed, truth-informed representation fits on already opened cases. The historical timed results, their source pins and the original retraction are unchanged.

Let R_G be the triangular factor of the evaluated bank in the query mesh’s field norm, and let Q_q have orthonormal columns spanning $R_G C_q$. Eliminating the free correction coefficients leaves

$$r_q(z) = (I - Q_q Q_q^\top)(R_G h_\theta(z) - T). \quad (12)$$

Here T is the reference field projected into these coordinates. The original implementation omitted the required normalization of the projector. The corrected finite best-found error is an upper bound on the unknown global minimum, while the free-bank floor is a lower bound. At $q = R$, direct free-coefficient projection attains that floor and removes the redundant nonlinear head.

Table H.1: Separately confirmed Poisson representation errors (%) on the original development sources. Original starts use the unchanged nearest-training-code multistart prescription with the repaired projector. Best found additionally permits previously retained solved states and the preceding rank’s best fit as feasible starts. Median summarizes the safeguarded errors. Original starts and best found report worst-case errors; bank floor is the worst free-bank projection error. Stationary counts cases satisfying the normalized-gradient rule, with the direct endpoint stationary by construction. None is a new inference timing or sealed-test result.

Intervals	R	K	q	Original starts	Best found	Median	Bank floor	Stationary
256	512	32	0	3.1211	3.1211	0.5050	0.7459	12/12
256	512	32	32	2.4554	2.4554	0.4420	0.7459	12/12
256	512	32	64	2.0787	2.0787	0.4126	0.7459	12/12
256	512	32	128	1.5491	1.5491	0.3194	0.7459	12/12
256	512	32	256	0.9688	0.9688	0.1315	0.7459	12/12
256	512	32	512	direct	0.7459	0.0580	0.7459	12/12
1024	512	32	0	3.1139	3.1139	0.5049	0.7421	12/12
1024	512	32	32	2.4497	2.4497	0.4421	0.7421	12/12
1024	512	32	64	2.0738	2.0738	0.4127	0.7421	12/12
1024	512	32	128	1.5453	1.5453	0.3194	0.7421	12/12
1024	512	32	256	0.9648	0.9648	0.1315	0.7421	12/12
1024	512	32	512	direct	0.7421	0.0580	0.7421	12/12
256	128	16	0	6.1013	6.1013	1.2423	2.3226	12/12
256	128	16	32	4.6715	4.6715	0.9365	2.3226	12/12
256	128	16	64	3.4336	3.4336	0.7863	2.3226	12/12
256	128	16	128	direct	2.3226	0.3684	2.3226	12/12
1024	128	16	0	6.0930	6.0930	1.2421	2.3152	12/12
1024	128	16	32	4.6644	4.6644	0.9362	2.3152	12/12
1024	128	16	64	3.4271	3.4271	0.7860	2.3152	12/12
1024	128	16	128	direct	2.3152	0.3682	2.3152	12/12

I THREE-DIMENSIONAL DEVELOPMENT COMPARISONS

Provisional evidence, excluded from the abstract and main claims. The following panels are audited development comparisons during an ongoing tuning campaign. Their inputs were available for checkpoint or configuration selection. These tables preserve the recorded development snapshot; later frozen final evaluations are not included here. Independent-seed confirmation and final evaluation must be distinguished from checkpoint selection. Failure to reach an accuracy target is retained as a result, and a finite training budget does not establish a model family’s attainable error floor. No cross-PDE or cross-allocation runtime ratio is formed.

Each table pairs accuracy with the cost of the same solver invocation. GPU warm-up precedes measurement, parameters use double precision and matrix multiplications use the highest-precision setting. Query costs include the recorded initialization, solve and requested field output with inputs and outputs resident on the GPU; unmeasured host transfers are not assigned a cost. Timing summaries retain all repetitions. Operator variants include any stated interpolation or incompressibility projection in the measured query. Failed direct mesh-transfer variants are displayed separately from native-grid prediction followed by interpolation; their failure is not an intrinsic impossibility result for the operator family.

Stationarity columns report each run’s recorded stopping criterion; they do not prove a global minimum. The selected Poisson panel has independently audited fields and references, but its nonlinear stopping summary lacks saved selected latent states for a separate gradient reconstruction. The selected Navier–Stokes panel retains histories for sampled independent weak-gradient checks. Its new paired measurements pass the numerical-validity audit, but a separate strict cross-run frozen-field replay gate failed and remains failed. This supports neither cross-run equivalence nor an implementation-speed ratio. A zero recorded stopping-failure count does not remove these qualifications.

The weak NM-ROM residual is projected against smooth test functions. The learned bank, compressed head and correction rank have sizes R , K and q ; M is the actual number of weak tests. The Burgers basis completes the degenerate Laplacian eigenvalue shell at its requested cutoff, preserving coordinate symmetry. Dense full-grid operations remain charged where quadrature has not met its certificate. Full-bank Galerkin controls and a full-bank weak least-squares endpoint are distinct reduced equations. On the native grid, POD uses the matched training states. On finer heat and Poisson meshes, POD is rebuilt offline from the same training members; these mesh-adapted controls are distinct from transferring frozen neural weights. Efficient classical controls remain in the comparison even when they are both faster and more accurate.

Table I.1: Provisional development panel provenance. Each row names a separate GPU allocation. Scientific source identifiers refer to staged file contents verified against repository history; snapshot hashes and upstream result/audit hashes are in `tables/campaign-provenance.json`.

PDE	Attempt	Job	GPU	Scientific source
Burgers 3D	b3d004	3995688	A100 80GB PCIe	c026d56b8390
Heat 3D	extra03	3995709	A100-PCIE-40GB	4ac8b16455f5
Poisson 3D	extra03	3995104	A100 80GB PCIe	a50ce0977373
Navier–Stokes 3D	extra03	3995695	A100 80GB PCIe	8ba0a11ee83a

Table I.2: Provisional training and query contracts from saved run configurations. Grid counts are per spatial axis. Fields are full spatial states, including the initial state when requested; vector states include all channels. The number of fields is not a count of statistically independent examples. Heads lists saved trained candidates; the following panels explicitly identify the displayed candidate. All model families share the lane’s training membership.

PDE	Training grid	Train cases	Fields/case	Outputs	Channels	R	K	M
Burgers 3D	33 nodes	512	8	51	1	256	32	642
Heat 3D	32 intervals	128	6	6	1	128	8, 16, 32	512
Poisson 3D	32 intervals	512	1	1	1	128	8, 16	512
Navier–Stokes 3D	32 periodic points	512	6	6	3	512	16	1024

Table I.3: Provisional operator training records. Budget, done and selected are requested, completed and checkpoint-selected optimizer updates; batch is examples per update and LR is the initial learning rate. Seed records the initializer. Exit records why training stopped. Selection uses the recorded development criterion; an update limit is not a convergence certificate. Reused training checkpoints may originate in an earlier allocation. Compute budgets are recorded, not claimed equal.

PDE	Operator	Parameters	Budget	Done	Selected	Batch	LR	Seed	Exit
Burgers 3D	fno	1771431	20000	20000	16100	2	0.001	920310	steps
Burgers 3D	unet	89431	20000	20000	17500	2	0.001	920311	steps
Burgers 3D	deeponet	1861959	30000	30000	30000	2	0.001	920312	steps
Burgers 3D	transolver	564392	30000	30000	29400	2	0.001	920313	steps
Heat 3D	fno_w16_m6	1771349	20000	20000	19500	4	0.001	920321	steps
Heat 3D	unet_w8	89197	20000	20000	20000	4	0.001	920322	steps
Heat 3D	deeponet_r128_w16	857749	20000	20000	20000	4	0.001	920323	steps
Heat 3D	transolver_w48_s32	562840	20000	20000	20000	4	0.001	920324	steps
Poisson 3D	fno_w24_m8	9440953	30000	30000	26750	2	0.0005	920450	steps
Poisson 3D	unet_w16	354577	30000	30000	28750	2	0.0005	920451	steps
Poisson 3D	deeponet_r128_w16	791697	30000	30000	29000	2	0.001	920452	steps
Poisson 3D	transolver_w48_s32	561272	30000	30000	29750	2	0.001	920453	steps
Navier–Stokes 3D	deeponet	6790319	32000	32000	23200	2	0.001	202609208	steps
Navier–Stokes 3D	transolver	569064	32000	32000	32000	2	0.001	202609209	steps
Navier–Stokes 3D	fno	4196511	8000	8000	7100	2	0.001	202609205	steps
Navier–Stokes 3D	unet	89287	8000	8000	7900	2	0.001	202609206	steps

FNO is a Fourier neural operator; U-Net is a multiscale convolutional model; DeepONet uses an input branch and coordinate trunk; Transolver uses learned physics-attention slices. The pinned machine-readable training records retain each architecture, normalization, selected checkpoint hash and training curve metadata. Burgers operators return their trained time knots, pass through the supplied initial state and interpolate to the common requested output times. The Navier–Stokes case is a periodic interacting three-component flow family; the reported panel does not claim turbulent-flow coverage. Each selected panel now includes paired measurements for all four operator families.

Table I.4: Provisional NM-ROM architecture and training records, derived from saved checkpoints and their original training metadata. MLP widths list input, hidden and output sizes of each multilayer perceptron. Head parameter counts include its linear skip; optimized training codes and fixed Fourier, scale and whitening assets are excluded. Budget, done and selected are requested, completed and checkpoint-selected updates. Base seed initializes the recipe; stage-specific key splits and sampling seeds are retained in the pinned configuration. Selected updates follow the recorded development criterion or the final optimizer iterate, as specified in the machine-readable protocol. Update limits do not certify convergence. Reused checkpoints retain their original training record rather than inheriting the query allocation’s budget.

PDE	Part	MLP widths	Parameters	Budget	Done	Selected	Base seed
Burgers 3D	bank	128 → 384 → 384 → 256	295936	200000	200000	200000	920201
Burgers 3D	head K32	32 → 256 → 256 → 256	148224	100000	100000	100000	920201
Heat 3D	bank	64 → 256 → 256 → 128	115328	100000	100000	100000	920312
Heat 3D	head K8	8 → 256 → 256 → 128	102016	80000	80000	80000	920312
Heat 3D	head K16	16 → 256 → 256 → 128	105088	80000	80000	80000	920312
Heat 3D	head K32	32 → 256 → 256 → 128	111232	80000	80000	45000	920312
Poisson 3D	bank	128 → 256 → 256 → 128	131712	150000	150000	150000	920412
Poisson 3D	head K8	8 → 256 → 256 → 128	102016	100000	100000	100000	920412
Poisson 3D	head K16	16 → 256 → 256 → 128	105088	100000	100000	100000	920412
Navier–Stokes 3D	bank	512 → 512 → 512 → 1536	1313280	12000	12000	12000	202609204
Navier–Stokes 3D	head K16	16 → 512 → 512 → 512	542208	30000	30000	30000	202609204

The coordinate network supplies the spatial bank, which is frozen while the compressed coefficient head is trained. The head’s optimized training codes are offline fitting variables, not an encoder or online inputs. The vector bank’s output groups all velocity components; its terminal width therefore differs from the number of bank functions. Correction directions and query-mesh solver assets are prepared offline. Changing correction rank within a reported ladder leaves all neural weights frozen. The pinned protocol records checkpoint content hashes, full training configurations and the source of reused training metadata.

I.1 BURGERS 3D: PROVISIONAL DEVELOPMENT PANEL

Provisional: Audited development data with all four operator families and the same training membership. Errors use the initial-field norm and exclude time zero. Operator predictions interpolate trained time knots to the common output times. The newer operator schedules are still under evaluation. Physical refinement covers only two cases; this panel supports same-grid comparisons only.

Table I.5: 33 nodes per axis; 8 development cases per method. Initial-field-normalized evolved error; same-grid reference. Each accuracy/cost pair comes from the same invocation in job 3995688. NF/NS counts nonfinite/nonstationary cases; outliers count calls above 1.5 times the method median.

Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers/calls
NM-ROM K32, q0	5.7071	10.8780	186.364	230.973	0/0	0/24
NM-ROM K32, q64	4.9359	8.9108	267.817	325.616	0/0	0/24
NM-ROM K32, q128	3.3258	6.6491	353.542	428.614	0/0	0/24
NM-ROM K32, q192	1.3600	3.4908	418.712	620.808	0/0	0/24
Full learned bank R256	0.8437	2.4345	316.964	317.629	0/0	0/24
POD32	9.5224	22.6947	95.738	96.211	0/0	0/24
POD96	1.9855	5.5785	151.273	152.382	0/0	0/24
POD160	0.7675	2.5057	224.346	225.287	0/0	0/24
POD224	0.3988	1.2100	284.540	285.347	0/0	0/24
POD256	0.3292	1.0363	316.567	317.512	0/0	0/24
FNO, time interpolation	1.6413	1.9979	5.862	6.687	0/0	0/24
U-Net, time interpolation	1.6239	1.9129	2.212	2.633	0/0	0/24
DeepONet, time interpolation	7.9145	21.1988	2.534	3.009	0/0	0/24
Transolver, time interpolation	1.8700	3.7391	3.545	3.888	0/0	0/24
FOM n1e-02, 15e-01	1.2739	1.6185	16.068	16.725	0/0	0/24
FOM n1e-02, 15e-03	0.0382	0.6543	17.940	20.571	0/0	0/24
FOM n1e-03, 15e-01	0.4692	0.5248	19.753	21.495	0/0	0/24
FOM n1e-04, 15e-01	0.0267	0.0303	32.494	34.086	0/0	0/24
FOM n1e-04, 11e-06	0.0188	0.0207	37.394	41.505	0/0	0/24
FOM n1e-06, 11e-01	0.0002	0.0003	48.156	50.230	0/0	0/24

All timed arms in the main panel are included. The separately measured larger-head candidate on the same allocation is retained in the complete report, alongside representation and limited refinement diagnostics. FOM labels identify nonlinear (n) and inner linear (l) tolerances.

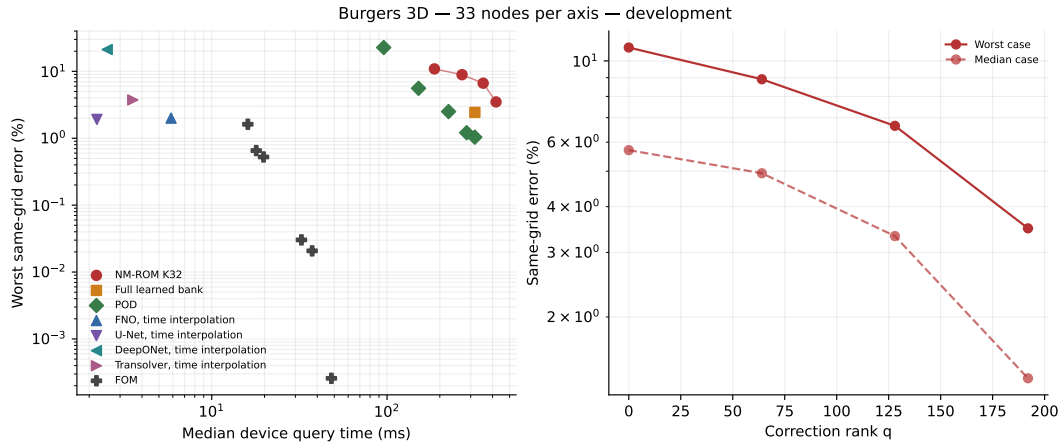


Figure I.1: Provisional paired development measurements from Table I.5. The right panel isolates the fixed-head correction ladder; the left retains classical controls and operator variants. Zero-error direct solves are identified separately rather than clipped onto the logarithmic error axis.

I.2 HEAT 3D: PROVISIONAL DEVELOPMENT PANEL

Provisional: Audited development data with all four operator families. Errors use the current reference-field norm and exclude time zero. All recorded fits and evolved solves are stationary. The larger head improves the high-correction endpoint but retains a generalization gap; improved initialization and longer operator schedules are under evaluation. Failed direct transfer variants are retained; native-grid interpolation is charged separately. Finer-grid POD is rebuilt offline from the same training members; it is a mesh-adapted classical control, not frozen-weight transfer.

Table I.6: 64 intervals per axis; 16 development cases per method. Current-reference-normalized evolved error; same-grid reference. Each accuracy/cost pair comes from the same invocation in job 3995709. NF/NS counts nonfinite/nonstationary cases; outliers count calls above 1.5 times the method median.

Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers/calls
NM-ROM K32, q0	1.1494	5.9683	37.143	48.919	0/0	0/48
NM-ROM K32, q8	1.0223	5.1921	40.126	80.283	0/0	5/48
NM-ROM K32, q32	0.8224	4.4151	40.445	56.685	0/0	1/48
NM-ROM K32, q64	0.5591	3.1815	52.322	92.258	0/0	5/48
NM-ROM K32, q96	0.3903	1.3555	113.590	151.651	0/0	1/48
Full bank, exact Galerkin	0.2563	1.3166	0.564	0.766	0/0	1/48
Full bank, exact weak	0.2542	1.3113	1.495	1.694	0/0	0/48
POD32	3.0811	9.3730	0.311	0.647	0/0	9/48
POD128	0.1414	1.2338	0.565	0.616	0/0	0/48
FNO, direct transfer	15.1234	17.6641	16.246	17.031	0/0	0/48
FNO, native + interpolation	0.7669	0.9891	10.681	11.592	0/0	0/48
U-Net, direct transfer	75.7115	92.6756	10.960	11.808	0/0	0/48
U-Net, native + interpolation	0.7512	0.9468	4.240	4.764	0/0	0/48
DeepONet, direct transfer	33.8829	52.5118	11.270	12.017	0/0	0/48
DeepONet, native + interpolation	7.0168	21.7903	4.029	4.545	0/0	0/48
Transolver, direct transfer	19.8616	25.2954	20.828	22.118	0/0	0/48
Transolver, native + interpolation	0.9475	1.2766	6.854	7.952	0/0	0/48
Direct DST	0.0000	0.0000	1.591	1.926	0/0	0/48

The complete audited report retains smaller-head ladders, intermediate POD ranks and time-step diagnostics. Failed sampled-quadrature certificates remain in the earlier attempt and are not promoted.

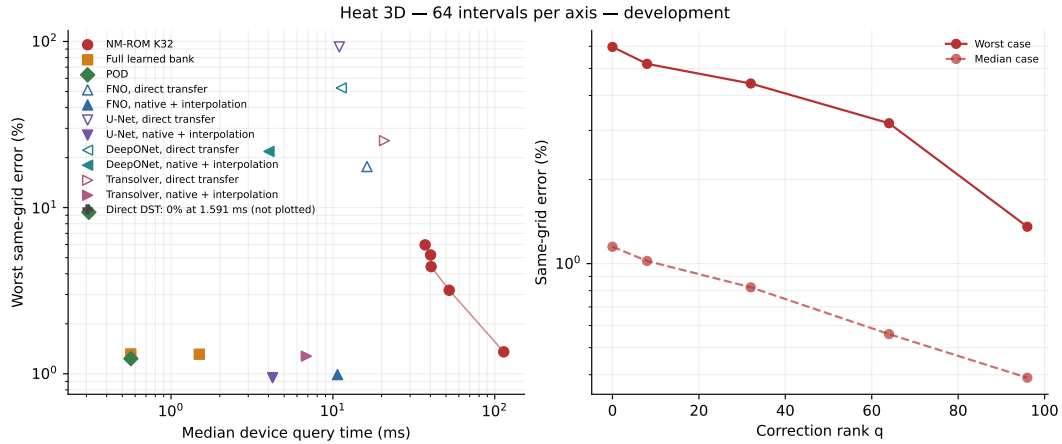


Figure I.2: Provisional paired development measurements from Table I.6. The right panel isolates the fixed-head correction ladder; the left retains classical controls and operator variants. Zero-error direct solves are identified separately rather than clipped onto the logarithmic error axis.

I.3 POISSON 3D: PROVISIONAL DEVELOPMENT PANEL

Provisional: Audited development data with all four operator families. Static weights are placed on the GPU during setup. New operator curves are still improving and longer common schedules are under evaluation. Direct transfer variants and quadrature certificates fail; preserved-domain FNO padding is a pending control. Every measured pair below shares one allocation. Finer-grid POD is rebuilt offline from the same training members; it is a mesh-adapted classical control, not frozen-weight transfer. The saved stopping records are internally consistent, but selected latent states were not retained for an independent weak-gradient audit.

Table I.7: 64 intervals per axis; 16 development cases per method. Relative field error against the same-grid discrete solution. Each accuracy/cost pair comes from the same invocation in job 3995104. NF/NS counts nonfinite/nonstationary cases; outliers count calls above 1.5 times the method median.

Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers/calls
NM-ROM K16, q0	0.2149	0.5726	3.390	3.965	0/0	0/48
NM-ROM K16, q16	0.1492	0.4580	3.435	4.269	0/0	0/48
NM-ROM K16, q32	0.1159	0.3215	3.301	4.113	0/0	0/48
NM-ROM K16, q64	0.0687	0.2001	3.465	4.269	0/0	0/48
NM-ROM K16, q96	0.0539	0.1598	3.566	4.219	0/0	0/48
Full bank, weak solve	0.0520	0.1444	0.944	1.049	0/0	0/48
Full bank, Galerkin	0.0530	0.1474	0.479	0.741	0/0	3/48
POD16	1.8304	5.2702	0.203	0.554	0/0	5/48
POD48	0.1484	0.5576	0.287	0.403	0/0	2/48
POD128	0.0066	0.0360	0.480	0.761	0/0	4/48
FNO, direct transfer	16.0680	17.1866	17.107	17.678	0/0	0/48
FNO, native + interpolation	0.2852	0.3656	11.197	11.617	0/0	0/48
U-Net, direct transfer	168.0982	219.8928	12.157	12.513	0/0	0/48
U-Net, native + interpolation	0.2828	0.3769	4.262	4.712	0/0	0/48
DeepONet, direct transfer	16.7244	26.8325	7.221	7.538	0/0	0/48
DeepONet, native + interpolation	2.1141	4.2024	3.016	3.261	0/0	0/48
Transolver, direct transfer	16.9674	22.0653	12.719	13.382	0/0	0/48
Transolver, native + interpolation	0.7201	1.0698	4.493	5.311	0/0	0/48
Direct DST	0.0000	0.0000	0.220	0.501	0/0	6/48

The complete audited report retains the K8 ladder, intermediate POD ranks, and failed sampled-quadrature diagnostics. They are not presented as missing experiments.

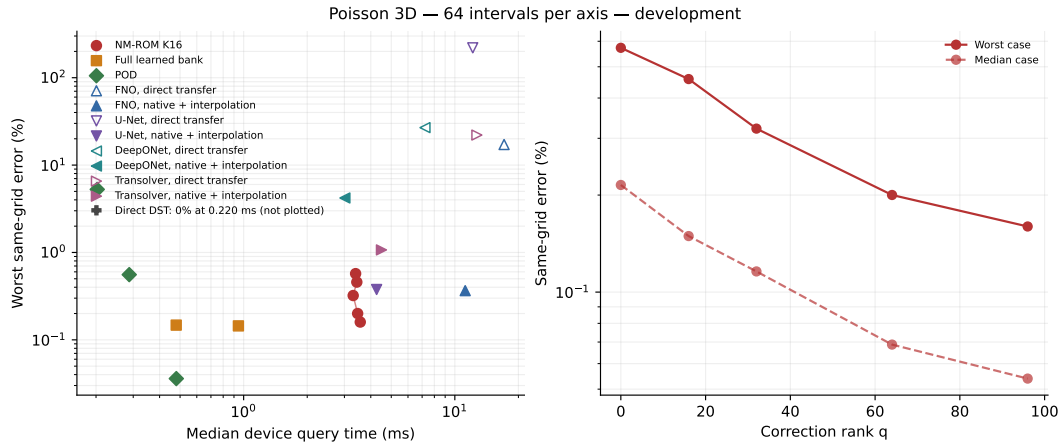


Figure I.3: Provisional paired development measurements from Table I.7. The right panel isolates the fixed-head correction ladder; the left retains classical controls and operator variants. Zero-error direct solves are identified separately rather than clipped onto the logarithmic error axis.

I.4 NAVIER-STOKES 3D: PROVISIONAL DEVELOPMENT PANEL

Provisional: Independently audited new paired development measurements with all four operator families. Errors use the initial vector-field norm and exclude time zero. Learned-bank and head accuracy targets fail. Saved latent histories support sampled independent weak-gradient checks. A strict cross-run frozen-field replay gate failed and remains failed; no numerical-equivalence or implementation-speed claim is made. Augmented training and larger-bank tuning remain in progress.

Table I.8: 32 periodic points per axis; 8 development cases per method. Initial-field-normalized evolved error; same-grid reference. Each accuracy/cost pair comes from the same invocation in job 3995695. NF/NS counts nonfinite/nonstationary cases; outliers count calls above 1.5 times the method median.

Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers/calls
NM-ROM K16, q0	28.5319	50.4758	779.231	781.497	0/0	0/24
NM-ROM K16, q16	28.2539	49.9583	810.785	812.390	0/0	0/24
NM-ROM K16, q32	27.8899	49.2381	970.197	973.237	0/0	0/24
NM-ROM K16, q64	27.0931	47.6297	1297.298	1302.637	0/0	0/24
NM-ROM K16, q128	25.2664	45.3348	1865.999	1873.513	0/0	0/24
Full bank, Galerkin R512	13.4503	24.2696	43.878	44.156	0/0	0/24
POD16, weak solve	87.7996	91.9857	25.084	26.989	0/0	0/24
POD32, weak solve	78.1203	85.5263	27.511	28.055	0/0	0/24
POD48, weak solve	67.3545	76.1070	35.081	35.300	0/0	0/24
POD128, Galerkin	34.5481	47.3334	5.609	5.832	0/0	0/24
POD256, Galerkin	24.8410	36.0594	11.294	11.345	0/0	0/24
POD512, Galerkin	15.2837	27.7537	43.895	44.105	0/0	0/24
FOM, dt 0.001	0.0031	0.0051	24.712	25.475	0/0	0/24
FOM, dt 0.002	0.0130	0.0213	12.810	13.163	0/0	0/24
FOM, dt 0.004	0.0524	0.0866	6.886	7.224	0/0	0/24
FOM, dt 0.008	0.2125	0.3598	3.893	4.227	0/0	0/24
FNO, raw	1.3422	1.7664	8.787	10.315	0/0	0/24
FNO, projected	1.0974	1.5121	8.970	12.449	0/0	1/24
U-Net, raw	3.4540	4.6468	1.982	2.112	0/0	0/24
U-Net, projected	3.0414	4.2830	2.094	2.212	0/0	0/24
DeepONet, raw	36.8223	45.3970	4.124	4.362	0/0	0/24
DeepONet, Leray projected	35.8894	44.8450	4.372	4.600	0/0	0/24
Transolver, raw	2.2017	2.9715	3.317	4.436	0/0	1/24
Transolver, Leray projected	1.3773	2.0366	3.398	4.553	0/0	1/24

All timed arms are included; the separate larger-bank snapshot-fit diagnostics remain in the complete archive. Leray projection enforces discrete incompressibility and its online cost is charged.

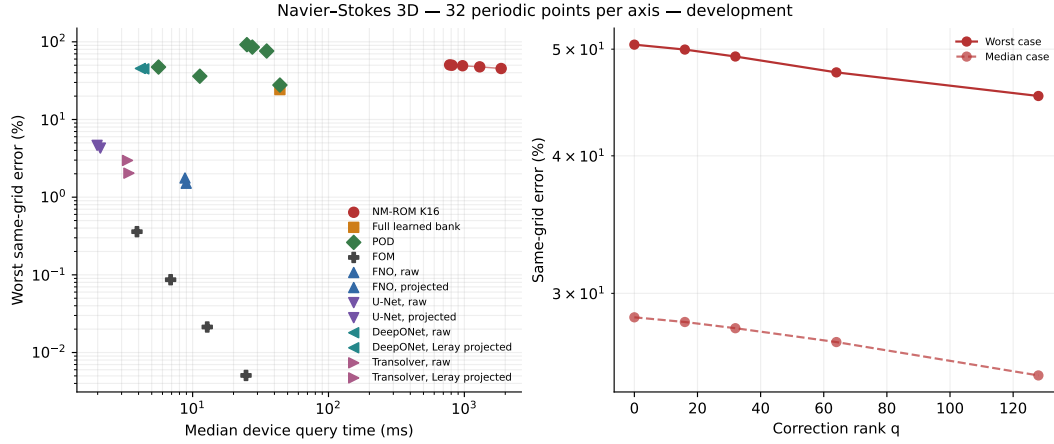


Figure I.4: Provisional paired development measurements from Table I.8. The right panel isolates the fixed-head correction ladder; the left retains classical controls and operator variants. Zero-error direct solves are identified separately rather than clipped onto the logarithmic error axis.

J GLOSSARY

Written for a reader who knows none of this project’s vocabulary.

Bank, head, latent code ($G, h_\theta, z; R, k$)

The fixed spatial functions the decoder combines (width R); the small network that maps k latent numbers to their coefficients; the k numbers solved for per query.

Correction rank q , rung, ladder, C_q

The number of fixed extra bank directions the solver may add to the head’s output; one value of q ; the sequence of rungs; the matrix of directions, a nested prefix chosen offline. $q = 0$ is the head alone, $q = R$ the linear model on the bank.

Test count M , weak tests, row scaling

How many sine test functions the residual is averaged against; the functions; the diagonal weights on those averaged equations. All three change which solution is selected.

m , EQ, rule, NNLS, fit states, reachable states

Quadrature node count; empirical quadrature, evaluating a sum over all nodes by a weighted sum over m of them; one fitted instance; the non-negative least-squares fit; the states used to fit a rule; states the solver actually visits, as opposed to stored snapshots.

ρ , primary and tight bars, confirmed, marginal

The rule’s relative error on the projected advection term on held-out reachable states; $\rho_{\max} \leq 0.116$ and ≤ 0.06 ; a construction every one of whose re-draws passes; one only some of whose re-draws pass. Never the NNLS fit residual, and never a single draw.

Dense

Evaluating the advection term at every interior node, exactly.

Evolved vs all-times metric, $t=0$ compression

Worst error over output times $t > 0$; over all output times including $t = 0$; the error at $t = 0$ alone, which is the decoder’s reconstruction of the supplied initial field and bounds the all-times metric from below.

Same-grid, reference, physical error

Error against the converged full-order solve on the same mesh (isolates the reduction); against a much finer solve (includes the mesh’s own discretisation error); the cell’s declared accuracy metric.

Three layers: bank floor, best-found, solved

Error of the best linear projection onto the bank (what nothing at inference can beat); the smallest error any point of the augmented manifold attains, found by a multistart oracle (an upper bound on the true minimum); the error the deployed iteration returns.

Stationary, budget exit, early-stopped, converged, strict

The solve met the normalised-gradient tolerance; it hit its iteration cap; any output produced under a

cap or stall; the pre-registered completion rule that also accepts an attained initial fit whose residual is at round-off; the earlier rule that does not.

Admissible

Eligible for a frontier or a ratio: converged under the completion rule as written, with stationarity 10^{-6} at every step or a residual-rule exit, and for a quadrature subject a rule that passes its bar. Arms run at the loose tolerance 10^{-3} are reported everywhere but are not admissible.

Non-dominated, frontier, error span, cost span

A point nothing else beats on both axes; the set of such points; ratio of largest to smallest error among a ladder’s non-dominated converged points; the same for cost. The tunability bar: monotone, at least three non-dominated points, at least $2\times$ on both spans, nothing early-stopped.

Same-allocation, same-job, cross-job

Two measurements inside one cluster allocation on one GPU, which may be divided; two from different allocations, which may not.

Development cohort, held-out, sealed cohort

Cases opened for diagnosis or selection; cases excluded from training, which may still be used for validation; cases opened once after choices are frozen. The Burgers sealed cohort has been opened and is reported in Table F.30.

Pre-registered, amended

Declared before the observations used to test a claim; changed later in a dated design amendment. Poisson’s deciding falsification interpretation and the wave tie band were amended after earlier observations; their literal and amended readings are reported separately.

Checkpoint recency

Whether the best validation checkpoint falls near the end of the training budget; a training-progress heuristic, not a convergence proof or a bound on attainable error.

POD-LSPG, k' , DST, CG, Newton

The classical linear reduced model on a snapshot basis of rank k' solved through the same weak objective; the direct discrete sine transform solve, exact for separable constant-coefficient operators on a rectangle; conjugate gradients; the full-order nonlinear iteration for Burgers.

FNO, U-Net, DeepONet, Transolver

Fourier, convolutional, branch-trunk and physics-attention solution-map models. Each is fixed once trained; its evaluation grid is a cost-accuracy control of its own (Table F.31), but one that does not change what the model can represent.

GPU median, p95, NF/NS, timing outliers

Median and ninety-fifth percentile of retained device query times; numbers of cases with nonfinite output or a failure of the recorded stationarity rule; calls above one and a half times their method’s median. These summaries are distinct from accuracy or convergence guarantees.

Direct transfer, native plus interpolation, projected operator

Evaluation of fixed weights on a changed mesh; evaluation on the training mesh followed by charged interpolation; an operator prediction followed by a charged divergence-free projection.

Checkpoint, frozen, incumbent

Saved network weights; unchanged for every result; the one Burgers checkpoint every cell shares.

Provisional, pending

A number that is real but rests on one draw, one seed or an unfinished replication; a placeholder for a run that has not landed.

K PAIRED COMPARISONS AGAINST ITERATIVE CG

These are collected development comparisons. Each row uses its own recorded job for both method and CG time. The comparator is the fastest retained passing CG setting whose displayed error does not exceed the method error; this finite-set selection is not a fixed denominator for every row. Poisson and heat use CG tolerance 10^{-2} in the main-text rows. At the finest wave mesh the uncorrected comparison uses $\Delta t = 0.005$ and tolerance 10^{-2} ; the corrected and linear-bank comparisons use $\Delta t = 0.0025$ and tolerance 10^{-6} . The raw setting names and all meshes are retained below.

Heat uses an earlier audited checkpoint. Its error uses the physical reference over all requested times; Poisson uses the same-grid steady reference. Wave error measures displacement only; the NM-ROM models fail their full-state target. GPU time includes initialization, the solve and requested resident-device outputs, excluding host transfers. The linear-bank rows are baseline measurements

and are not nonlinear NM-ROM speedups. These iterative comparisons supplement, rather than replace, the direct-transform controls in the other tables.

Table K.1: All collected CG comparisons. Errors are worst relative L^2 percentages, times are medians in milliseconds, and S is CG time divided by method time. Settings and source jobs identify every denominator.

Problem	N	Method	Error (%)	GPU ms	CG error (%)	CG ms	S	CG setting	Job
Poisson	256	NM-ROM, $q = 0$	3.1567	6.7031	0.1527	22.2898	3.33×	cg_0.01	3780692
Poisson	256	NM-ROM, $q = 256$	0.9689	7.1213	0.1527	22.2898	3.13×	cg_0.01	3780692
Poisson	256	Linear-bank ROM	0.7459	1.6928	0.1527	22.2898	13.17×	cg_0.01	3780692
Poisson	1024	NM-ROM, $q = 0$	3.1495	3.9406	0.0722	60.0170	15.23×	cg_0.01	3783813
Poisson	1024	NM-ROM, $q = 256$	0.9648	4.3188	0.0722	60.0170	13.90×	cg_0.01	3783813
Poisson	1024	Linear-bank ROM	0.7421	1.8300	0.0722	60.0170	32.80×	cg_0.01	3783813
Wave	64	NM-ROM, $q = 0$	6.3109	184.5351	2.7812	41.8936	0.23×	cgdt_0.005_tol_0.01	3780447
Wave	64	NM-ROM, $q = 32$	2.6815	2500.0601	1.5967	69.3737	0.03×	cgdt_0.005_tol_1e-06	3780447
Wave	64	Linear-bank ROM	2.6913	1.1685	1.5967	69.3737	59.37×	cgdt_0.005_tol_1e-06	3780447
Wave	256	NM-ROM, $q = 0$	6.1035	183.5155	0.4008	142.5111	0.78×	cg_1e-06	3783805
Wave	256	NM-ROM, $q = 32$	2.8017	2520.1766	0.4008	142.5111	0.06×	cg_1e-06	3783805
Wave	256	Linear-bank ROM	2.8115	1.4277	0.4008	142.5111	99.82×	cg_1e-06	3783805
Wave	1024	NM-ROM, $q = 0$	6.1066	192.4624	5.1413	583.0259	3.03×	cgdt_0.005_tol_0.01	3780450
Wave	1024	NM-ROM, $q = 32$	2.8094	2529.4450	0.4010	1043.3681	0.41×	cg_1e-06	3780450
Wave	1024	Linear-bank ROM	2.8192	4.5624	0.4010	1043.3681	228.69×	cg_1e-06	3780450
Heat	64	NM-ROM	4.5593	11.2268	1.5860	2.5038	0.22×	fom_cg_cn_tol1e2	3529772
Heat	64	Linear-bank ROM	1.6758	0.1083	1.5860	2.5038	23.12×	fom_cg_cn_tol1e2	3529772
Heat	256	NM-ROM	4.5557	11.3414	0.9255	6.9389	0.61×	fom_cg_cn_tol1e2	3529772
Heat	256	Linear-bank ROM	1.6758	0.1382	0.9255	6.9389	50.23×	fom_cg_cn_tol1e2	3529772
Heat	1024	NM-ROM	4.5555	12.2062	0.7707	59.1780	4.85×	fom_cg_cn_tol1e2	3529772
Heat	1024	Linear-bank ROM	1.6758	0.8381	0.7707	59.1780	70.61×	fom_cg_cn_tol1e2	3529772

Timing repetitions remain in the underlying records. Heat’s outlier counts are in the retained comparison JSON; the Poisson and wave summary inputs do not expose counts, so none are inferred from their medians. The hash-pinned snapshot and source-file hashes are recorded in tables/rewrite-provenance.json.

L DETAILED BURGERS COMPARISON AND CORRECTION SETTINGS

Table L.1: Complete Burgers comparison summary. The ratios in this historical diagnostic are explicitly *reduced time divided by FOM time*, the inverse of speedup S used in the main text. The evolved and all-times non-dominated counts are kept separate.

mesh	job	GPU	timed subjects	admissible reduced (\$5)	non-dom. (evolved)	non-dom. (all-times)	cheapest reduced / cheapest FOM	cheapest reduced / converged FFT	ladder vs-ref span
256 ²	3789570	NVIDIA A100 80GB PCIe	48	27	0	0	4.48×	0.446×	1.13×
512 ²	3805065	NVIDIA A100 80GB PCIe	48	23	0	0	2.92×	0.257×	1.36×
1024 ²	3789572	NVIDIA H200	30	16	3	0	2.27×	0.191×	1.38×

Table L.2: Fixed-test-space and scheduled correction studies. The fixed study holds M constant; the scheduled study changes it with q . Development timings and held-out errors originate in different jobs and are not combined into a held-out speedup. Quadrature status refers to the displayed rule and construction evidence.

q	M	dev evolved %	sealed evolved %	all %	vs ref %	device ms	EQ evolved %	EQ ms	EQ rule (status)
0	1088	1.2657	—	—	—	848.0	—	—	dense, job 3780177
64	1088	1.0593	—	—	—	1359.4	—	—	dense, job 3780177
128	1088	0.8711	—	—	—	1886.5	—	—	dense, job 3780177
256	1088	0.5194	—	—	—	4377.9	—	—	dense, job 3780177
0	64	1.8890	10.1120	2.5629	4.56	283.9	1.8891	58.6	confirmed (3 of 3 re-draws)
16	128	1.3985	1.4617	2.4806	4.11	360.8	1.4270	81.1	confirmed (3 of 3 re-draws)
32	192	1.2336	1.4458	2.3534	4.08	434.7	1.2493	97.8	confirmed (2 of 2 re-draws)
64	320	1.0843	1.3252	2.1489	4.10	621.1	1.0840	148.9	confirmed (2 of 2 re-draws)
128	576	0.8930	1.0964	1.8116	4.08	1186.9	0.8931	273.5	single-draw
256	1088	0.5194	0.6789	0.9053	4.04	3939.8	0.5129	746.0	single-draw

M MAIN THREE-DIMENSIONAL EXPERIMENT RECORDS

The main-paper tables are generated from the accepted snapshots listed in tables/main-experiments-provenance.json. Burgers uses the prospectively selected primary seed on the frozen final cohort; Poisson and Navier–Stokes use their later accepted development panels. Heat uses its audited development error/runtime panel while the paired CG

measurement is pending. The earlier three-dimensional supplement retains its original snapshot and is not the source of the new main comparison tables.

Within each column block, every method and its FOM come from the same allocation, with the recorded requested outputs. Poisson CG uses its efficient path without timed audit histories; separate replay checks verify residuals and stopping. No method’s field error is taken from a different invocation to improve its ratio. All repeated calls remain in the evidence, and the maximum field error retains the worst repetition. The fixed FOM denominator is a timing reference, not a claim that every method attains the same accuracy as the FOM.

For Burgers the main table uses the original, frozen primary seed; the second independent initialization is shown below. For NS, all learned methods and POD share the augmented training family, but their training budgets differ. Its bank/head is a warm training lineage, not independent seed replication. POD dimensions and correction ranks are stated in the main captions rather than treated as interchangeable.

Table M.1: Burgers3D final seed comparison. Errors are worst evolved same-grid percentages; times are median GPU milliseconds. Both seeds use the same reserved cases, without selection of the better seed.

Method	Seed 0 error (%)	Seed 0 ms	Seed 1 error (%)	Seed 1 ms
NM-ROM, uncorrected	16.263	219.161	13.245	219.218
NM-ROM, corrected	4.399	364.677	4.220	400.940
POD	2.561	310.989	2.561	311.710
FNO	2.378	6.072	2.268	6.083
U-Net	2.451	2.115	2.249	2.033
DeepONet	20.575	2.578	20.145	2.565
Transolver	2.017	3.489	5.246	3.509
FOM	2.312	8.337	2.312	8.528

Table M.2: Meaningful deployment controls. Training parameters and model sizes are fixed within each study and reported in the setup.

Control	Measured role
Correction rank q	Enlarges the trial space without retraining; can improve accuracy at greater solve cost.
Solver tolerance	Controls convergence; field-error gains plateau when representation error dominates.
Iteration limit	Bounds work; an early-stopped solve retains its numerical status.
EQ sample count	Selects a validated stored quadrature rule where EQ is used; chiefly changes residual cost.

N ADDITIONAL ANALYSIS OF THE CORRECTION SETTINGS

Rank against test count: the family at fixed M . The rank alone moves the error: hold the test count fixed, add correction directions, and the error falls at every step while the cost rises (Table 5). The ROM approximates the discrete system, so the same-grid error is what the knob controls. Against the fine reference the knob moves the error only from 4.56 to 4.04 % ($1.13\times$), because at 256^2 every rung’s reference error is $1.00\text{--}1.13\times$ the mesh’s own discretisation error of 4.03 %: the reduction adds little on top of the discretisation. And a same-job full-order setting at 15.7 ms reaches 2.47 % against the reference, more accurate than every reduced subject at a small fraction of the top rung’s cost; at 512^2 the ladder runs 3.76 to 2.76 % against the reference over a converged same-grid error of 2.70 %, all 39 reduced subjects above it and the 2 full-order settings below it coarser solves cancelling against the reference, not better ones; at 1024^2 the rungs run 3.86 to 2.80 % over 2.14 %. Correction rank chiefly changes reduction error here; discretisation limits the gain against the fine reference. At fixed $M = 256$ the ladder spans only $1.22\times$ and **fails** the pre-registered bar (Table F.5: monotone on evolved, every rung converged, error span $\geq 2\times$, and two generator clauses, ≥ 3 non-dominated points and cost span $\geq 2\times$). **At $M = 1088$ the ladder $q = 0, 64, 128, 256$, inside one job is monotone, every rung converged, spans $2.44\times$ in evolved error for $5.16\times$ in cost with**

4 **non-dominated points** and passes. $M = 1088$ is the pre-registered pure-rank ladder (the largest fixed M holding every rung to $q = 256$); the $q = 512$ extension did not converge and enters no span; M at $q = 256$ saturates at $M^* = 2176$.

Sealed cohort and seeds (Tables F.30, F.29). The sealed cohort was opened once, after every choice was frozen; on it all 4 checkpoints (the incumbent and 3 fresh seeds) give a ladder monotone on both metrics, 3 of them meeting the knob bar, with top-rung errors of 0.59–0.68 %. The one checkpoint-specific number is the incumbent’s uncorrected $q = 0$ rung, 10.1120 % sealed against 1.8890 % in development (the seeds: 2.15–3.43 %): one sealed case converged to a wrong branch, and the first correction rung removes it ($q = 16$: 1.46–1.87 % on every checkpoint). The pre-registered sealed-over-development ratio thus fails at $q = 0$ for the incumbent alone (5.35; worst seed-mean ratio 1.40); the sealed numbers are the headline and the development table records seed variability (the incumbent beats every seed there at $q = 16, 32, 64, 128$). Rungs not converged everywhere (seed2 at $q = 64$ (3 budget exits) sealed, the one bar failure; one seed at $q = 256$ in development) are reported as run, not tuned; the fixed- M ladder, one development-cohort checkpoint, is unchanged.

Quadrature and tolerance move cost. Quadrature is a cost lever at equal error ($4.77\times$ at $q = 0$, $4.81\times$ at $q = 128$) and the tolerance $10^{-6} \rightarrow 10^{-3}$ removes a further 27–23 % (Table F.10; those arms are not admissible). The earlier ladder’s rule set breaks upward at $q = 256$; the replication-selected set is monotone at both tolerances, $4.2\text{--}5.3\times$ cheaper than its dense twins, its $q = 128, 256$ rules single-draw (Table 5). Transferred to 512^2 (Table F.12) it stays monotone; its $q = 128$ and 256 transfers pass the primary bar there though single-draw at source, the ladder-rule transfers at $q \geq 64$ do not, and at $q = 256$ the transfer reaches 0.5510 % in $21\times$ less time than its twin. Solver knobs chiefly change cost in these accurate comparisons (Table F.19); changed numerical settings can also change error.

Validate on reachable states, then re-draw. A quadrature rule must be judged on the states the solver reaches, not on its fit to the states it was built from. The NNLS fit residual says nothing about held-out error (Figure F.1: a rule fitting to $1.5e - 04$ reaches $\rho_{\max} = 0.462$ held-out, $4.0\times$ the primary bar). **The pre-registered replication re-drew every construction** (Table F.25): $\rho_{\max}$ moves $1.4\text{--}9.6\times$ within one construction; the ladder’s rules are **confirmed at** $q = 0, 16, 32$ and **marginal at** $q = 64(2/6), 128(4/5), 256(1/5)$; one passing rule does not certify its construction; a re-draw does (Appendix G.2). One gate failed: same-source rule pairs transferred to 512^2 are not bitwise identical (each transfer refits its own draw), and one $q = 32$ source passes the primary bar in one transfer and only the secondary in the other, an $8.7\times$ spread in $\rho_{\max}$, unexplained; no falsification clause fired.

N.1 DETAILED MATCHED-DIMENSION COMPARISONS

At matched $k = 16$, the Burgers neural head outperforms the tested affine and POD comparators in achieved error, without a per-millisecond advantage. Comparisons use a training-fitted rank- k affine map, a quadratic map, the free bank and POD-LSPG in the same solver (Tables F.17, F.18). On Burgers the head reaches 2.5629 % against 56.9296 % (fitted affine map) and 61.6503 % (POD-16); on Poisson at 1024^2 it wins per dimension (6.0927 % against 17.2966 %) and loses per ms to POD-128 (4.3452 % at 5.930 vs 5.843 ms). In three layers (Table F.6) the reduction layer sits $6.5\times$ above the floor, the solver layer costs 0.018 pp on the all-times metric, where the $t = 0$ term dominates: the head binds.

N.2 DETAILED LINEAR AND NONLINEAR CONTROLS

On linear PDEs the full bank removes the nonlinear solve. On Poisson the top rung is the most accurate and cheapest. At 1024^2 (Table F.7) the rungs run $3.1495 \rightarrow 0.9648\%$ at flat cost, and $q = R$ lands on the floor at 0.7421 % as the *cheapest* point (4.42 vs 3.248 ms exact). Heat and the reflective wave show similar cost collapse, with a small wave accuracy exception (Appendix G.5). Against unpreconditioned CG at 10^{-6} , the previous comparator, reduced rungs are $3.7\text{--}23.4\times$ faster on Poisson in the same job; the transform is faster still by $1.3\text{--}4.0\times$. Both algorithms are retained as named comparators (Appendix G.6). **The collapse is confounded with a weak bank:** $3.4\text{--}4.0\times$ worse than a POD basis of equal rank on both cells. No reduced arm beats the direct solve.

Navier–Stokes collapses differently. On 2D incompressible flow the confirmatory ladder gate failed (Table F.36): at $K = 16$ and 32 the held-out oracle beats POD- K by only $1.19\times$ and $1.15\times$ against a bar of 2.0 (oracle values are upper bounds), and at $t = 0$ POD- 32 is more accurate (POD/oracle 0.82): linear wins where data lie in a low-dimensional subspace. Retraining the head on 128 – 512 trajectories moves the oracle 27.4938 to 20.6889 to 19.8279% (gains 25% then 4% ; “ambiguous” by the pre-registered rule, Table G.1; the bank saw all 512). A family of dimension 8 instead of 14 lowers every error 3.3 – $4.4\times$ yet leaves the head’s edge over POD- 16 at $1.39\times$. With $4\times$ the data (2048 trajectories), that ratio reaches 1.46 . Training and held-out errors use different normalisations; their ratio cannot establish a shift from generalisation to capacity limits. Across $K = 16$ and 32 , family dimension 14 and 8 , and 128 – 2048 trajectories, this head class never beats POD- K by the pre-registered $2.0\times$ on held-out decaying 2D Navier–Stokes (ratios 1.15 – 1.46). An exploratory ladder on that failed manifold (Table G.2) closes the manifold gap only at $q = R$, where the head no longer matters; POD-LSPG at matched dimension is more accurate at every rung and cheaper below the top, no neural rung non-dominated: the falsification clause’s named outcome. A trade between rungs needs a residual nonlinear in the coefficients and a manifold that beats POD held-out: Poisson, heat and waves lack the first, this cell the second, Burgers had both and still lost on cost. At ten-fold lower viscosity (Appendix G.7) linear floors collapse 12 – $19\times$ while the head degrades 3.3 – $4.5\times$, every neural rung beats every POD-LSPG rank, $q \geq 64$ is on the reduced-only frontier: the manifold’s edge appears where the Kolmogorov width is worst. Nothing reduced is on the full-order frontier, the fixed- M ladder spans $1.36\times$ and the mesh is under-resolved (discretisation error 19.3 – 20.9%): reduced-versus-reduced only.

On the L-shaped domain (Poisson, a linear residual; no fast transform) reduced models are cheaper, POD most of all. The direct solve’s cost climbs with the mesh, at 512^2 CG at tolerance 10^{-2} is the cheaper full-order arm (29.13 ms at 0.385% , more accurate than any reduced arm), and the reduced arms stay nearly flat (Table 6). Against the cheapest same-job full-order arm the head at $q = 64$ is $2.77\times$ and $6.07\times$ cheaper, POD- 128 $2.94\times$ and $7.23\times$; the head is non-dominated only on accuracy (6 – 19% more cost than POD- 128 , 15% less error). Nothing is claimed beyond 512^2 ; the banks’ reconstruction degrades from 6.8 to 16.7% on unswept validation sources (Table F.39).